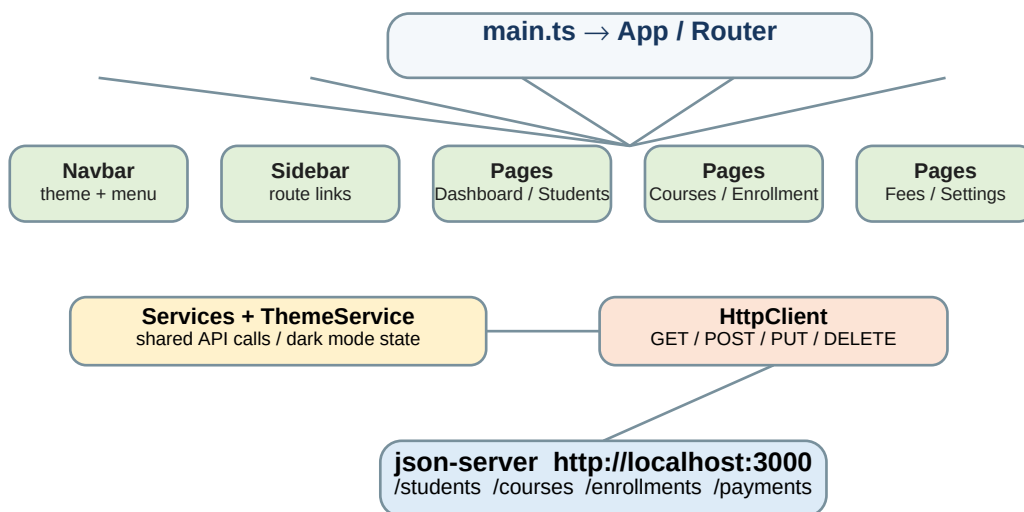


COURSE MANAGEMENT SYSTEM

Complete Project Documentation & Code Learning Guide

Simple Roman Urdu + English technical terms



Data flow: User action → Component → HTTP request → db.json → response → signal/form → template

Is guide ka maqsad: aap apne project ka har module, file, connection, code pattern aur data flow samajh kar viva / interview mein explain kar saken.

Generated from the uploaded project ZIP. Source files are described as they exist in that archive; the guide also flags important limitations where the UI looks complete but backend/security behavior is only simulated.

1. Is documentation ko kaise parhna hai

Har topic is order mein diya gaya hai: kya hai, kyun use hota hai, kaise kaam karta hai, phir project mein kahan use hua hai. Code appendix mein har source line ke saath us line ka simple purpose diya gaya hai. Blank lines aur braces bhi explain kiye gaye hain, kyun ke woh code structure ka hissa hain.

Important terms

Frontend: User ko nazar aane wala hissa: pages, buttons, forms, tables aur styling.

Angular component: UI ka aik chhota independent hissa; aam tor par TS + HTML + CSS files.

Template: HTML file jo batati hai ke screen par kya dikhana hai.

Service: Reusable class jahan shared logic/API calls rakh sakte hain.

Route: URL aur page component ke darmiyan mapping.

Signal: Angular ka reactive state container. Read: `value()`; update: `set(...)` ya `update(...)`.

Reactive Form: Form jiska fields aur validation TypeScript se manage hota hai.

HTTP request: Frontend ka backend ko data read/create/update/delete karne ka message.

json-server: Development tool jo db.json se simple REST API banata hai; production database nahi.

Observable / subscribe: Async data stream; subscribe karne par response milne par callback run hota hai.

localStorage: Browser mein key/value strings save karta hai; doosre device/user ko sync nahi karta.

CRUD: Create, Read, Update, Delete — records manage karne ke 4 basic operations.

Scope aur honest notes

Uploaded ZIP mein 7 main pages hain: Dashboard, Students, Courses, Enrollment, Fees, Settings, Admin Profile; saath Navbar, Sidebar aur services. Course page `CoursesService` use karta hai; Student service bhi maujood hai lekin Student component apni API calls direct `HttpClient` se karta hai. Settings ka password change sirf UI confirmation hai: backend/authentication se password update nahi hota. Logout `localStorage` clear karta hai, lekin is project mein real login/session security nazar nahi aati. Dashboard ke monthly chart values hard-coded demo values hain. These distinctions viva mein batana important hai.

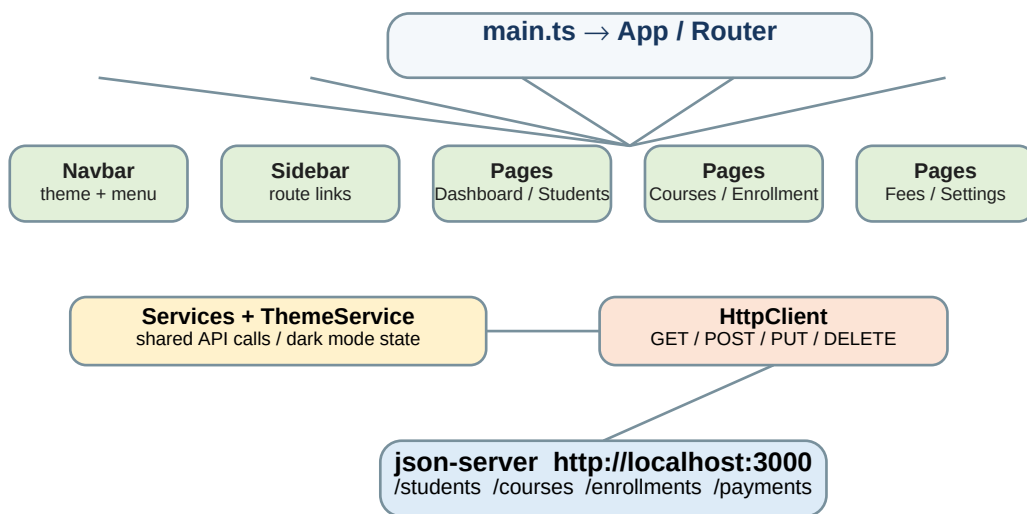
2. Project ka purpose aur modules

Yeh Course Management System student records, courses, enrollment aur fee payments ko aik UI se manage karne ke liye banaya gaya hai. Is version mein Angular frontend aur json-server development API istemal ho rahi hai.

Main modules

Module	Kaam	API / storage
Dashboard	Counts, total enrolled fee, pending fee; chart ka demo data	GET students/courses/enrollments/payments
Students	Student add, list, edit, delete; department selection	/students
Courses	Course add, list, edit, delete; seats/fee details	/courses via CoursesService
Enrollment	Student ko course se link karna; fee/date/status save	/enrollments + GET students/courses
Fees & Payments	Payment submit/edit/delete; paid/remaining calculation; status	/payments + /enrollments
Settings	Profile local save, password-form demo, accent color, JSON export, reset	localStorage + HTTP
Admin Profile	Admin info/photo/edit mode	localStorage
Navbar / Sidebar	Navigation, theme toggle, mobile sidebar event	Router + ThemeService

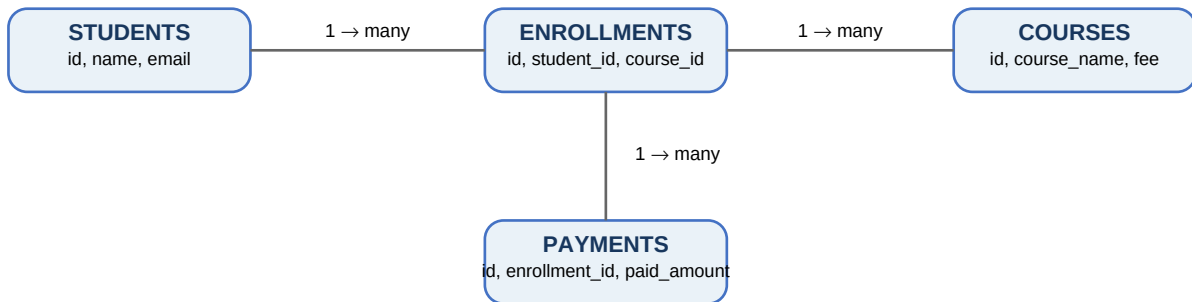
3. Architecture — files aik doosre se kaise connect hoti hain



Data flow: User action → Component → HTTP request → db.json → response → signal/form → template

Flow: main.ts app ko start karta hai → app.config.ts router aur HttpClient provide karta hai → app.routes.ts URL ke hisaab se page component select karta hai → component TS data/logic handle karta hai → HTML signal/form values display karti hai → CSS appearance control karti hai → HttpClient json-server ko request bhejta hai → response wapas signal mein set hota hai → template automatically refresh hoti hai.

4. Data model aur relationships



Student aur Course ka direct many-to-many relation enrollment ke zariye banta hai: aik student multiple courses le sakta hai aur aik course mein multiple students ho sakte hain. Enrollment aik joining record hai jo student_id aur course_id rakhta hai. Payment enrollment_id se batata hai ke payment kis enrollment ki fee ke liye hui.

db.json mein present collections

students — 3 record(s)

```
{
  "id": "1",
  "name": "Mahaz Naseem Abbasi",
  "father_name": "Muhammad Naseem",
  "email": "abbasimahaz562@gmail.com",
  "phone": "03480526933",
  "address": "Abbottabad",
  "department": "Computer Science"
}
```

courses — 5 record(s)

```
{
  "id": "1",
  "course_name": "Angular",
  "course_description": "Frontend framework",
  "course_duration": "3 months",
  "instructor": "Shahzad",
  "available_seats": "20",
  "course_fee": "10000"
}
```

enrollments — 2 record(s)

```
{
  "student_id": "1",
  "student_name": "Mahaz Naseem Abbasi",
  "course_id": "3",
  "course_name": "JavaScript",
  "course_fee": "8000",
  "enroll_date": "2026-10-06",
  "status": "Active",
  "id": "Z7JD8r352K8"
}
```

payments — 2 record(s)

```
{
  "enrollment_id": "Z7JD8r352K8",
  "student_name": "Mahaz Naseem Abbasi",
  "course_name": "JavaScript",
  "total_fee": "8000",
  "paid_amount": 5000,
  "payment_method": "Online",
  "payment_date": "2026-10-06",
  "id": "4XtfqMFxIsA"
}
```

\$schema — object record(s)

ID caution: JSON IDs kabhi numeric aur kabhi alphanumeric string ho sakti hain. ID ko Number(...) mein convert karna sirf numeric IDs ke liye sahi hai. Safer comparison: `String(a.id) === String(b.id)`.

Fee status rule

Unpaid: total paid 0 ya us se kam. Partial: paid amount total fee se kam ho lekin zero se zyada ho. Paid: paid amount total fee ke barabar ya zyada ho. Is project mein Fees table `getStatus(p.paid_amount, p.total_fee)` ko har single payment record ke liye call karta hai; agar enrollment par multiple partial payments hon, per-payment status ko total accumulated payments se compare karna logically incomplete ho sakta hai. Accurate enrollment-level status ke liye enrollment ki tamam payments sum karni chahiye.

5. Application start aur routing

src/main.ts — bootstrapApplication(App, appConfig) Angular app ko start karta hai. Error catch console mein startup error log karta hai.

src/app/app.ts — Navbar/sidebar state aur layout ko hold karta hai; RouterOutlet ke andar selected page render hota hai.

src/app/app.html — Layout components aur router outlet ko place karta hai.

src/app/app.config.ts — provideRouter(routes) navigation setup karta hai; provideHttpClient() HTTP requests enable karta hai.

src/app/app.routes.ts — URL paths ko Dashboard, Student, Course, Enrollment, Fees, Settings aur AdminProfile components se connect karta hai.

Route table from uploaded code

```
import { Routes } from '@angular/router';■
import { Student } from '../pages/student/student';■
import { Enrollment } from '../pages/enrollment/enrollment';■
import { Fees } from '../pages/fees/fees';■
import { Dashboard } from '../pages/dashboard/dashboard';■
import { Course } from '../pages/course/course';■
import { Setting } from '../pages/setting/setting';■
import { AdminProfile } from '../pages/admin-profile/admin-profile';■
■
export const routes: Routes = [■
  { path: '', redirectTo: 'dashboard', pathMatch: 'full' },■
  { path: 'dashboard', component: Dashboard },■
  { path: 'students', component: Student },■
  { path: 'courses', component: Course },■
  { path: 'enrollments', component: Enrollment },■
  { path: 'fees', component: Fees },■
  { path: 'settings', component: Setting },■
  { path: 'admin-profile', component: AdminProfile }■
];
```

Path '' default URL hai. redirectTo dashboard ka matlab root URL kholne par dashboard route par bhejna. pathMatch: 'full' batata hai ke empty path poori URL path se match hona chahiye.

src/app/pages/dashboard/dashboard.ts

Purpose: Dashboard par counts aur fee totals show karna.

Important state: totalStudents, totalCourses, totalEnrollments, totalFee, pendingFee signals hain. monthlyTarget/targetGrowth aur salesData chart ke display values hain.

Lifecycle/data flow: ngOnInit() → loadAllData() → GET /students, /courses, /enrollments, /payments → response se signals set.

Calculation: Enrollment fee total = enrollments ke course_fee ka sum. Pending = enrolled fees total – payments ke paid_amount ka sum.

Caution: Monthly chart salesData hard-coded hai; real monthly payment analytics se calculate nahi hota.

src/app/pages/student/student.ts

Purpose: Student CRUD.

Form: studentForm has name, father_name, email, phone, address, department; Validators.required sab fields ko required banata hai.

Add flow: openModal → user fills form → saveStudent validates → POST /students → success toast → closeModal → loadStudents.

Edit flow: editStudent(student) selected ID set karta hai, form setValue se values fill karta hai → saveStudent PUT /students/{id}.

Delete flow: deleteStudent(id) DELETE /students/{id}; response ke baad list reload.

Important note: Email field par sirf required validator hai, Validators.email nahi. Is liye malformed email bhi pass ho sakta hai.

src/app/pages/course/course.ts

Purpose: Course CRUD; CoursesService ke zariye backend requests.

Form: course_name, course_description, course_duration, instructor, available_seats, course_fee required hain.

Service connection: Component CoursesService inject karta hai. getCourses/addCourse/updateCourse/deleteCourse service mein HTTP calls karte hain.

Add/Edit decision: selectedCourseId null ho to add; otherwise update.

Seats dropdown: seatsOptions display options hain; selectSeats selected seat count form mein patch karta hai.

src/app/pages/enrollment/enrollment.ts

Purpose: Student aur course ko enrollment record mein connect karna.

Data loading: ngOnInit teen lists load karta hai: enrollments, students, courses.

Selection: selectStudent student_id/name set karta hai; selectCourse course_id/name/fee set karta hai.

Form: student_id, course_id, course_fee, enroll_date, status required; student_name/course_name extra display fields hain.

CRUD: POST /enrollments; PUT /enrollments/{id}; DELETE /enrollments/{id}.

Important note: selectedEnrollmentId currently typed number|null; agar backend string IDs use kare to types ko string|number|null banana safer hai.

src/app/pages/fees/fees.ts

Purpose: Enrollment ke against payment record manage karna.

Form: enrollment_id, total_fee, paid_amount, payment_method, payment_date required.

Selection/calculation: Enrollment select hone par total fee fill hoti hai; matching enrollment_id wali payments sum hoti hain; remaining = total – previouslyPaid – current entered payment.

Save: POST /payments or PUT /payments/{id}; success ke baad payments reload.

Delete: DELETE /payments/{id}.

ID warning: Current uploaded file ke IDs agar alphanumeric hain, selectedPaymentId:number|null aur strict equality p.enrollment_id === e.id bug de sakte hain. String comparison use karein.

Validation limitation: Validators.required numeric payment field par zero value ko required pass/fail behavior control value type ke mutabiq check karta hai; negative amount ya overpayment ko explicitly reject karne ka validator visible nahi.

src/app/pages/setting/setting.ts

Purpose: Settings page functionality.

src/app/pages/admin-profile/admin-profile.ts

Purpose: Admin profile details, edit mode aur photo upload/preview.

Storage: admin_profile aur admin_profile_picture localStorage keys use hoti hain.

Photo flow: File size/type validate → FileReader reads as data URL → signal and localStorage set.

Limits: 2MB limit checks file size; Base64 increases stored size. localStorage quotas apply.

Security: Profile values and photo are browser-local demo data, not authenticated server-side admin records.

src/app/layout/navbar/navbar.ts

Purpose: Top navigation, theme toggle, sidebar toggle event, logout UI.

Output: toggleSidebar = output<void>() emits an event parent component can listen to.

Theme: ThemeService.toggleTheme() flips shared isDarkMode signal.

Logout caution: localStorage.clear() clears all local storage keys for this origin; no server session revocation is implemented.

src/app/layout/sidebar/sidebar.ts

Purpose: Menu links and active navigation.

menuitems: Each item stores name, icon, route; HTML loops over it.

Output: linkClicked emits when a link is clicked so parent can close mobile sidebar.

Theme: Sidebar reads shared ThemeService.

6. Services aur shared concepts

CoursesService (src/app/services/courses.service.ts)

Service ka faida: API logic component se alag hota hai. Component ko endpoint aur HTTP details har jagah likhne ki zaroorat nahi. Course component getCourses(), addCourse(), updateCourse(), deleteCourse() call karta hai; service HttpClient ke zariye actual requests bhejti hai.

Students service (src/app/services/students.ts)

Yeh students ke GET/POST/PUT/DELETE methods provide karti hai, lekin uploaded Student component in methods ko use karne ke bajaye direct HttpClient use karta hai. Yeh duplicate approach hai; consistency ke liye future mein Student component ko Students service se connect kiya ja sakta hai.

ThemeService (src/app/services/theme.ts)

providedIn: 'root' se aik shared service app bhar available hoti hai. isDarkMode signal theme state rakhta hai. toggleTheme() signal ko opposite value se update karta hai. Navbar, Sidebar aur pages is service ko inject karte hain; template [class.dark] se CSS class attach karti hai.

Reactive form ka common flow

```
this.form = this.fb.group({
  name: ['', Validators.required]
});

if (this.form.invalid) {
  this.form.markAllAsTouched();
  return;
}

this.http.post(url, this.form.value).subscribe(...);
```

FormBuilder controls ka group banata hai. Validators rules lagata hai. form.invalid validation status deta hai. markAllAsTouched error messages visible karne mein help karta hai. form.value current values deta hai. subscribe async HTTP response ka wait/handler hai.

Signal syntax

```
count = signal<number>(0); // create
count(); // read
count.set(5); // replace value
count.update(v => v + 1); // old value se update
```

Signal ko normal variable ki tarah count nahi, count() se read karte hain. Agar template mein signal() use hota hai, value change par Angular UI update kar sakta hai.

7. Complete file map

Neeche important source files ka role diya gaya hai. CSS files appearance handle karti hain; HTML files screen structure; TS files state and logic; service files reusable API/theme logic; app configuration and routes app ko wire karti hain.

File	Role
src/app/app.config.ts	TypeScript logic/configuration
src/app/app.css	Visual styling / responsive layout
src/app/app.html	UI template
src/app/app.routes.ts	TypeScript logic/configuration
src/app/app.ts	TypeScript logic/configuration
src/app/layout/navbar/navbar.css	Visual styling / responsive layout
src/app/layout/navbar/navbar.html	UI template
src/app/layout/navbar/navbar.ts	TypeScript logic/configuration
src/app/layout/sidebar/sidebar.css	Visual styling / responsive layout
src/app/layout/sidebar/sidebar.html	UI template
src/app/layout/sidebar/sidebar.ts	TypeScript logic/configuration
src/app/pages/admin-profile/admin-profile.css	Visual styling / responsive layout
src/app/pages/admin-profile/admin-profile.html	UI template
src/app/pages/admin-profile/admin-profile.ts	TypeScript logic/configuration
src/app/pages/course/course.css	Visual styling / responsive layout
src/app/pages/course/course.html	UI template
src/app/pages/course/course.ts	TypeScript logic/configuration
src/app/pages/dashboard/dashboard.css	Visual styling / responsive layout
src/app/pages/dashboard/dashboard.html	UI template
src/app/pages/dashboard/dashboard.ts	TypeScript logic/configuration
src/app/pages/enrollment/enrollment.css	Visual styling / responsive layout
src/app/pages/enrollment/enrollment.html	UI template
src/app/pages/enrollment/enrollment.ts	TypeScript logic/configuration
src/app/pages/fees/fees.css	Visual styling / responsive layout
src/app/pages/fees/fees.html	UI template
src/app/pages/fees/fees.ts	TypeScript logic/configuration
src/app/pages/setting/setting.css	Visual styling / responsive layout
src/app/pages/setting/setting.html	UI template
src/app/pages/setting/setting.ts	TypeScript logic/configuration
src/app/pages/student/student.css	Visual styling / responsive layout
src/app/pages/student/student.html	UI template
src/app/pages/student/student.ts	TypeScript logic/configuration
src/app/services/courses.service.ts	TypeScript logic/configuration
src/app/services/students.ts	TypeScript logic/configuration
src/app/services/theme.ts	TypeScript logic/configuration
src/index.html	UI template
src/main.ts	TypeScript logic/configuration
src/styles.css	Visual styling / responsive layout

8. Line-by-line code appendix

Is appendix mein har source line original form mein aur uska simple role diya gaya hai. Explanations syntax pattern aur file context par based hain; CSS declarations ko styling purpose ke saath explain kiya gaya hai. Source ko change karne se pehle original line ko bhi dekhein.

src/app/app.config.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	<code>import { ApplicationConfig } from '@angular/core';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
2	<code>import { provideRouter } from '@angular/router';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
3	<code>import { provideHttpClient } from '@angular/common/http';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
4	<code>import { routes } from './app.routes';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
5		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
6	<code>export const appConfig: ApplicationConfig = {</code>	Value export karta hai taake doosri file import kar sake.
7	<code> providers: [</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
8	<code> provideRouter(routes),</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
9	<code> provideHttpClient() // ■ YE ZAROORI HAI</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
10	<code>]</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
11	<code>};</code>	Pichla block/function/object close karta hai.

src/app/app.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
2	<code>App Body</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
3	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
4	<code>.app-body {</code>	Code block start hota hai; andar related statements group hote hain.
5	<code> display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
6	<code> min-height: calc(100vh - 70px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
7	<code> position: relative;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
8	<code>}</code>	Pichla block/function/object close karta hai.
9		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
10	<code>.sidebar-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
11	<code> flex-shrink: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
12	<code> transition: transform 0.3s ease;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
13	<code>}</code>	Pichla block/function/object close karta hai.
14		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
15	<code>.main-content {</code>	Code block start hota hai; andar related statements group hote hain.

Line	Original code	Simple explanation
16	<code>flex: 1;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
17	<code>background: #f5f6fa;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
18	<code>transition: background 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
19	<code>overflow-y: auto;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
20	<code>min-width: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
21	<code>}</code>	Pichla block/function/object close karta hai.
22		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
23	<code>.main-content.dark {</code>	Code block start hota hai; andar related statements group hote hain.
24	<code>background: #0f0f1a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
25	<code>}</code>	Pichla block/function/object close karta hai.
26		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
27	<code>/* Overlay – hidden by default */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
28	<code>.sidebar-overlay {</code>	Code block start hota hai; andar related statements group hote hain.
29	<code>display: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
30	<code>}</code>	Pichla block/function/object close karta hai.
31		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
32	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
33	<code>Desktop (>768px)</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
34	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
35	<code>@media (min-width: 769px) {</code>	Responsive CSS rule: screen width/device condition par styles apply karta hai.
36	<code>.sidebar-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
37	<code>transform: translateX(0);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
38	<code>}</code>	Pichla block/function/object close karta hai.
39	<code>}</code>	Pichla block/function/object close karta hai.
40		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
41	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
42	<code>Mobile (<=768px)</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
43	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
44	<code>@media (max-width: 768px) {</code>	Responsive CSS rule: screen width/device condition par styles apply karta hai.
45		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
46	<code>/* Sidebar hidden by default (slide out) */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
47	<code>.sidebar-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
48	<code>position: fixed;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
49	<code>top: 70px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
50	<code>left: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
51	bottom: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
52	z-index: 999;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
53	transform: translateX(-100%);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
54	transition: transform 0.3s ease;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
55	}	Pichla block/function/object close karta hai.
56		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
57	/* Jab hamburger click ho → slide in */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
58	.sidebar-wrapper.open {	Code block start hota hai; andar related statements group hote hain.
59	transform: translateX(0);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
60	}	Pichla block/function/object close karta hai.
61		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
62	/* Overlay show */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
63	.sidebar-overlay {	Code block start hota hai; andar related statements group hote hain.
64	display: block;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
65	position: fixed;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
66	top: 70px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
67	left: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
68	right: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
69	bottom: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
70	background: rgba(0, 0, 0, 0.5);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
71	backdrop-filter: blur(2px);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
72	z-index: 998;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
73	animation: fadeIn 0.2s ease;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
74	}	Pichla block/function/object close karta hai.
75		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
76	@keyframes fadeIn {	Code block start hota hai; andar related statements group hote hain.
77	from { opacity: 0; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
78	to { opacity: 1; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
79	}	Pichla block/function/object close karta hai.
80		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
81	.main-content {	Code block start hota hai; andar related statements group hote hain.
82	width: 100%;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
83	margin-left: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
84	}	Pichla block/function/object close karta hai.
85	}	Pichla block/function/object close karta hai.

src/app/app.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Line	Original code	Simple explanation
1	<app-navbar (toggleSidebar)="toggleSidebar()"></app-navbar>	HTML element/component: page ka structure ya UI control banata hai.
2		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
3	<div class="app-body">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
4		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
5	<!-- Overlay (mobile) -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
6	@if (sidebarOpen()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
7	<div class="sidebar-overlay" (click)="closeSidebar()"></div>	User click par specified TypeScript method/expression chalti hai.
8	}	Pichla block/function/object close karta hai.
9		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
10	<!-- Sidebar -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
11	<div class="sidebar-wrapper" [class.open]="sidebarOpen()">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
12	<app-sidebar (linkClicked)="closeSidebar()"></app-sidebar>	HTML element/component: page ka structure ya UI control banata hai.
13	</div>	HTML element/component: page ka structure ya UI control banata hai.
14		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
15	<!-- Main Content -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
16	<main class="main-content" [class.dark]="themeService.isDarkMode()">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
17	<router-outlet></router-outlet>	HTML element/component: page ka structure ya UI control banata hai.
18	</main>	HTML element/component: page ka structure ya UI control banata hai.
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	</div>	HTML element/component: page ka structure ya UI control banata hai.

src/app/app.routes.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	import { Routes } from '@angular/router';	Import: doosri library/file se required class ya function is file mein available karta hai.
2	import { Student } from '../pages/student/student';	Import: doosri library/file se required class ya function is file mein available karta hai.
3	import { Enrollment } from '../pages/enrollment/enrollment';	Import: doosri library/file se required class ya function is file mein available karta hai.
4	import { Fees } from '../pages/fees/fees';	Import: doosri library/file se required class ya function is file mein available karta hai.
5	import { Dashboard } from '../pages/dashboard/dashboard';	Import: doosri library/file se required class ya function is file mein available karta hai.
6	import { Course } from '../pages/course/course';	Import: doosri library/file se required class ya function is file mein available karta hai.
7	import { Setting } from '../pages/setting/setting';	Import: doosri library/file se required class ya function is file mein available karta hai.
8	import { AdminProfile } from '../pages/admin-profile/admin-profile';	Import: doosri library/file se required class ya function is file mein available karta hai.
9		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
10	export const routes: Routes = [	Value export karta hai taake doosri file import kar sake.
11	{ path: '', redirectTo: 'dashboard', pathMatch: 'full' },	Navigation route define/import karta hai; URL ko relevant page component se connect karta hai.
12	{ path: 'dashboard', component: Dashboard },	Navigation route define/import karta hai; URL ko relevant page component se connect karta hai.
13	{ path: 'students', component: Student },	Navigation route define/import karta hai; URL ko relevant page component se connect karta hai.
14	{ path: 'courses', component: Course },	Navigation route define/import karta hai; URL ko relevant page component se connect karta hai.
15	{ path: 'enrollments', component: Enrollment },	Navigation route define/import karta hai; URL ko relevant page component se connect karta hai.
16	{ path: 'fees', component: Fees },	Navigation route define/import karta hai; URL ko relevant page component se connect karta hai.
17	{ path: 'settings', component: Setting},	Navigation route define/import karta hai; URL ko relevant page component se connect karta hai.
18	{ path: 'admin-profile', component: AdminProfile }	Navigation route define/import karta hai; URL ko relevant page component se connect karta hai.
19	];	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

src/app/app.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	import { Component, signal } from '@angular/core';	Import: doosri library/file se required class ya function is file mein available karta hai.
2	import { CommonModule } from '@angular/common';	Import: doosri library/file se required class ya function is file mein available karta hai.
3	import { RouterOutlet } from '@angular/router';	Import: doosri library/file se required class ya function is file mein available karta hai.
4	import { Navbar } from '../layout/navbar/navbar';	Import: doosri library/file se required class ya function is file mein available karta hai.
5	import { Sidebar } from '../layout/sidebar/sidebar';	Import: doosri library/file se required class ya function is file mein available karta hai.
6	import { ThemeService } from '../services/theme';	Import: doosri library/file se required class ya function is file mein available karta hai.
7		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
8	@Component({	Angular decorator: neech wali class ko Angular component declare karta hai.
9	selector: 'app-root',	Component ka HTML tag name; parent template mein isi selector se component lagta hai.
10	standalone: true,	Component standalone hai; NgModule mein declare karna zaroori nahi.
11	imports: [CommonModule, RouterOutlet, Navbar, Sidebar],	Template mein use hone wale Angular directives/forms/router features import karta hai.
12	templateUrl: './app.html',	Is component ki HTML template file ka path.
13	styleUrl: './app.css'	Is component ki CSS file ka path.
14	})	Pichla block/function/object close karta hai.
15	export class App {	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
16		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
17	// ■ Mobile sidebar state	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
18	sidebarOpen = signal<boolean>(false);	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	constructor(public themeService: ThemeService) {}	Shared ThemeService use hoti hai taake theme state/component access milay.
21		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
22	<code>toggleSidebar() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
23	<code>this.sidebarOpen.update(v => !v);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
24	<code>}</code>	Pichla block/function/object close karta hai.
25		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
26	<code>closeSidebar() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
27	<code>this.sidebarOpen.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
28	<code>}</code>	Pichla block/function/object close karta hai.
29	<code>}</code>	Pichla block/function/object close karta hai.

src/app/layout/navbar/navbar.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	<code>.navbar {</code>	Code block start hota hai; andar related statements group hote hain.
2	<code>height: 70px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
3	<code>background-color: white;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
4	<code>padding: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
5	<code>border-bottom: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
6	<code>transition: background 0.3s, border-color 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
7	<code>}</code>	Pichla block/function/object close karta hai.
8		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
9	<code>.navbar.dark {</code>	Code block start hota hai; andar related statements group hote hain.
10	<code>background-color: #0f0f1a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
11	<code>border-bottom-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
12	<code>}</code>	Pichla block/function/object close karta hai.
13		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
14	<code>.navbar.dark .form-control {</code>	Code block start hota hai; andar related statements group hote hain.
15	<code>background: #14141f;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
16	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
17	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
18	<code>}</code>	Pichla block/function/object close karta hai.
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	<code>.navbar.dark .form-control::placeholder {</code>	Code block start hota hai; andar related statements group hote hain.
21	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
22	<code>}</code>	Pichla block/function/object close karta hai.
23		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
24	<code>.navbar.dark .btn-outline-success {</code>	Code block start hota hai; andar related statements group hote hain.

Line	Original code	Simple explanation
25	color: #6ee7b7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
26	border-color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
27	}	Pichla block/function/object close karta hai.
28		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
29	.navbar.dark .btn.dropdown-toggle {	Code block start hota hai; andar related statements group hote hain.
30	color: #e4e4e7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
31	background: transparent;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
32	border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
33	}	Pichla block/function/object close karta hai.
34		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
35	.container-fluid {	Code block start hota hai; andar related statements group hote hain.
36	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
37	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
38	justify-content: space-between;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
39	}	Pichla block/function/object close karta hai.
40		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
41	.right-side {	Code block start hota hai; andar related statements group hote hain.
42	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
43	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
44	gap: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
45	}	Pichla block/function/object close karta hai.
46		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
47	.theme-toggle {	Code block start hota hai; andar related statements group hote hain.
48	width: 40px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
49	height: 40px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
50	border-radius: 50%;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
51	border: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
52	background: transparent;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
53	font-size: 18px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
54	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
55	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
56	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
57	justify-content: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
58	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
59	}	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
60		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
61	<code>.navbar.dark .theme-toggle {</code>	Code block start hota hai; andar related statements group hote hain.
62	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
63	<code>}</code>	Pichla block/function/object close karta hai.
64		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
65	<code>.theme-toggle:hover {</code>	Code block start hota hai; andar related statements group hote hain.
66	<code>background: #f3f4f6;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
67	<code>}</code>	Pichla block/function/object close karta hai.
68		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
69	<code>.navbar.dark .theme-toggle:hover {</code>	Code block start hota hai; andar related statements group hote hain.
70	<code>background: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
71	<code>}</code>	Pichla block/function/object close karta hai.
72	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
73	<code>Hamburger Button (Mobile Only)</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
74	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
75	<code>.hamburger-btn {</code>	Code block start hota hai; andar related statements group hote hain.
76	<code>display: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
77	<code>width: 42px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
78	<code>height: 42px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
79	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
80	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
81	<code>background: transparent;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
82	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
83	<code>font-size: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
84	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
85	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
86	<code>justify-content: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
87	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
88	<code>margin-right: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
89	<code>flex-shrink: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
90	<code>}</code>	Pichla block/function/object close karta hai.
91		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
92	<code>.navbar.dark .hamburger-btn {</code>	Code block start hota hai; andar related statements group hote hain.
93	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
94	<code>color: #9ca3af;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
95	}	Pichla block/function/object close karta hai.
96		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
97	.hamburger-btn:hover {	Code block start hota hai; andar related statements group hote hain.
98	background: #f3f4f6;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
99	color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
100	border-color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
101	}	Pichla block/function/object close karta hai.
102		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
103	.navbar.dark .hamburger-btn:hover {	Code block start hota hai; andar related statements group hote hain.
104	background: #1f1f2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
105	color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
106	}	Pichla block/function/object close karta hai.
107		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
108	/* ■ Mobile pe hi dikhao */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
109	@media (max-width: 768px) {	Responsive CSS rule: screen width/device condition par styles apply karta hai.
110	.hamburger-btn {	Code block start hota hai; andar related statements group hote hain.
111	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
112	}	Pichla block/function/object close karta hai.
113	}	Pichla block/function/object close karta hai.
114	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
115	Tablet & Mobile (<768px)	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
116	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
117	@media (max-width: 768px) {	Responsive CSS rule: screen width/device condition par styles apply karta hai.
118	.navbar {	Code block start hota hai; andar related statements group hote hain.
119	padding: 10px 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
120	height: 60px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
121	}	Pichla block/function/object close karta hai.
122		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
123	.container-fluid {	Code block start hota hai; andar related statements group hote hain.
124	gap: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
125	flex-wrap: nowrap; /* ■ koi wrap nahi */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
126	}	Pichla block/function/object close karta hai.
127		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
128	/* Hamburger chhota */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
129	.hamburger-btn {	Code block start hota hai; andar related statements group hote hain.
130	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
131	width: 38px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
132	height: 38px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
133	font-size: 18px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
134	margin-right: 4px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
135	}	Pichla block/function/object close karta hai.
136		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
137	/* Search form flexible */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
138	.container-fluid .d-flex {	Code block start hota hai; andar related statements group hote hain.
139	flex: 1;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
140	min-width: 0; /* ■ shrinks properly */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
141	gap: 6px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
142	}	Pichla block/function/object close karta hai.
143		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
144	/* Search input chhota */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
145	.form-control {	Code block start hota hai; andar related statements group hote hain.
146	padding: 8px 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
147	font-size: 13px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
148	min-width: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
149	flex: 1;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
150	}	Pichla block/function/object close karta hai.
151		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
152	/* Search button chhota */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
153	.btn-outline-success {	Code block start hota hai; andar related statements group hote hain.
154	padding: 8px 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
155	font-size: 13px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
156	white-space: nowrap;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
157	}	Pichla block/function/object close karta hai.
158		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
159	/* Theme toggle chhota */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
160	.theme-toggle {	Code block start hota hai; andar related statements group hote hain.
161	width: 36px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
162	height: 36px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
163	font-size: 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
164	flex-shrink: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
165	}	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
166		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
167	<code>/* Admin dropdown button chhota */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
168	<code>.dropdown-toggle {</code>	Code block start hota hai; andar related statements group hote hain.
169	<code>padding: 6px 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
170	<code>font-size: 13px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
171	<code>white-space: nowrap;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
172	<code>}</code>	Pichla block/function/object close karta hai.
173		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
174	<code>.right-side {</code>	Code block start hota hai; andar related statements group hote hain.
175	<code>gap: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
176	<code>flex-shrink: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
177	<code>}</code>	Pichla block/function/object close karta hai.
178	<code>}</code>	Pichla block/function/object close karta hai.
179		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
180	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
181	<code>Small Mobile (<500px)</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
182	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
183	<code>@media (max-width: 500px) {</code>	Responsive CSS rule: screen width/device condition par styles apply karta hai.
184	<code>.navbar {</code>	Code block start hota hai; andar related statements group hote hain.
185	<code>padding: 8px 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
186	<code>height: 56px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
187	<code>}</code>	Pichla block/function/object close karta hai.
188		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
189	<code>.hamburger-btn {</code>	Code block start hota hai; andar related statements group hote hain.
190	<code>width: 34px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
191	<code>height: 34px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
192	<code>font-size: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
193	<code>margin-right: 2px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
194	<code>}</code>	Pichla block/function/object close karta hai.
195		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
196	<code>/* Search button hide karo – sirf icon ya shortcut */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
197	<code>.btn-outline-success {</code>	Code block start hota hai; andar related statements group hote hain.
198	<code>display: none; /* ■ mobile pe hide */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
199	<code>}</code>	Pichla block/function/object close karta hai.
200		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
201	<code>/* Search input aur chhota */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
202	<code>.form-control {</code>	Code block start hota hai; andar related statements group hote hain.
203	<code>padding: 7px 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
204	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
205	<code>}</code>	Pichla block/function/object close karta hai.
206		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
207	<code>/* Theme toggle chhota */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
208	<code>.theme-toggle {</code>	Code block start hota hai; andar related statements group hote hain.
209	<code>width: 32px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
210	<code>height: 32px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
211	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
212	<code>}</code>	Pichla block/function/object close karta hai.
213		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
214	<code>/* Admin text hide karo – sirf icon */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
215	<code>.dropdown-toggle {</code>	Code block start hota hai; andar related statements group hote hain.
216	<code>padding: 4px 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
217	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
218	<code>}</code>	Pichla block/function/object close karta hai.
219		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
220	<code>.dropdown-toggle i {</code>	Code block start hota hai; andar related statements group hote hain.
221	<code>margin-right: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
222	<code>}</code>	Pichla block/function/object close karta hai.
223		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
224	<code>/* Admin ka naam hide karo (agar text hai) */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
225	<code>.dropdown-toggle span {</code>	Code block start hota hai; andar related statements group hote hain.
226	<code>display: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
227	<code>}</code>	Pichla block/function/object close karta hai.
228	<code>}</code>	Pichla block/function/object close karta hai.
229		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
230	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
231	<code>Very Small (<380px) – 320px tak</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
232	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
233	<code>@media (max-width: 380px) {</code>	Responsive CSS rule: screen width/device condition par styles apply karta hai.
234	<code>.navbar {</code>	Code block start hota hai; andar related statements group hote hain.
235	<code>padding: 6px 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
236	height: 52px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
237	}	Pichla block/function/object close karta hai.
238		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
239	.hamburger-btn {	Code block start hota hai; andar related statements group hote hain.
240	width: 30px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
241	height: 30px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
242	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
243	}	Pichla block/function/object close karta hai.
244		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
245	.form-control {	Code block start hota hai; andar related statements group hote hain.
246	padding: 6px 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
247	font-size: 11px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
248	}	Pichla block/function/object close karta hai.
249		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
250	.theme-toggle {	Code block start hota hai; andar related statements group hote hain.
251	width: 28px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
252	height: 28px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
253	font-size: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
254	}	Pichla block/function/object close karta hai.
255		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
256	.dropdown-toggle {	Code block start hota hai; andar related statements group hote hain.
257	padding: 4px 6px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
258	font-size: 11px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
259	}	Pichla block/function/object close karta hai.
260	}	Pichla block/function/object close karta hai.

src/app/layout/navbar/navbar.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Line	Original code	Simple explanation
1	<nav class="navbar" [class.dark]="themeService.isDarkMode()">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
2	<div class="container-fluid">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
3	<!-- ■ Sirf hamburger button add kiya -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
4	<button class="hamburger-btn" (click)="onToggleSidebar()" title="Menu">■</button>	User click par specified TypeScript method/expression chalti hai.
5		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
6	<!-- Search form (left side) -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
7	<form class="d-flex" role="search">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.

Line	Original code	Simple explanation
8	<code><input class="form-control me-2" type="search" placeholder="Search" aria-label="Search" /></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
9	<code><button class="btn btn-outline-success" type="submit">Search</button></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
10	<code></form></code>	HTML element/component: page ka structure ya UI control banata hai.
11		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
12	<code><!-- Right side: Theme toggle + Admin dropdown --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
13	<code><div class="right-side"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
14	<code><!-- ■ / ■■ Toggle Button --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
15	<code><button class="theme-toggle" (click)="toggleTheme()" title="Toggle Theme"></code>	User click par specified TypeScript method/expression chalti hai.
16	<code>{{ themeService.isDarkMode() ? '■■' : '■' }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
17	<code></button></code>	HTML element/component: page ka structure ya UI control banata hai.
18		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
19	<code><!-- Admin dropdown --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
20	<code><div class="dropdown"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
21	<code><button</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
22	<code>class="btn dropdown-toggle"</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
23	<code>type="button"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
24	<code>data-bs-toggle="dropdown"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
25	<code>aria-expanded="false"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
26	<code>></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
27	<code><i class="fa-solid fa-user me-2"></i>Admin</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
28	<code></button></code>	HTML element/component: page ka structure ya UI control banata hai.
29	<code><ul class="dropdown-menu dropdown-menu-end"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
30	<code></code>	HTML element/component: page ka structure ya UI control banata hai.
31	<code>■■ Settings</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
32	<code></code>	HTML element/component: page ka structure ya UI control banata hai.
33	<code></code>	HTML element/component: page ka structure ya UI control banata hai.
34	<code>■ Admin Profile</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
35	<code></code>	HTML element/component: page ka structure ya UI control banata hai.
36	<code><hr class="dropdown-divider" /></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
37	<code></code>	HTML element/component: page ka structure ya UI control banata hai.
38	<code></code>	User click par specified TypeScript method/expression chalti hai.
39	<code>>■ Logout</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
40	<code>></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
41	<code></code>	HTML element/component: page ka structure ya UI control banata hai.
42	<code></code>	HTML element/component: page ka structure ya UI control banata hai.
43	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
44	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
45	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
46	<code></nav></code>	HTML element/component: page ka structure ya UI control banata hai.

src/app/layout/navbar/navbar.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	<code>import { Component, signal, output } from '@angular/core';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
2	<code>import { CommonModule } from '@angular/common';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
3	<code>import { RouterLink } from '@angular/router';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
4	<code>import { ThemeService } from '../services/theme';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
5		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
6	<code>@Component({</code>	Angular decorator: neech wali class ko Angular component declare karta hai.
7	<code>imports: [CommonModule, RouterLink],</code>	Template mein use hone wale Angular directives/forms/router features import karta hai.
8	<code>selector: 'app-navbar',</code>	Component ka HTML tag name; parent template mein isi selector se component lagta hai.
9	<code>styleUrl: './navbar.css',</code>	Is component ki CSS file ka path.
10	<code>templateUrl: './navbar.html',</code>	Is component ki HTML template file ka path.
11	<code>})</code>	Pichla block/function/object close karta hai.
12	<code>export class Navbar {</code>	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
13		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
14	<code>// ■ Output event</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
15	<code>toggleSidebar = output<void>();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
16		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
17	<code>isAdminMenuOpen = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
18		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
19	<code>constructor(public themeService: ThemeService) {}</code>	Shared ThemeService use hoti hai taake theme state/component access milay.
20		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
21	<code>toggleTheme() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
22	<code>this.themeService.toggleTheme();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
23	<code>}</code>	Pichla block/function/object close karta hai.
24		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
25	<code>onToggleSidebar() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.

Line	Original code	Simple explanation
26	<code>this.toggleSidebar.emit();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
27	<code>}</code>	Pichla block/function/object close karta hai.
28		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
29	<code>logout() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
30	<code>if (confirm('Are you sure you want to logout?')) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
31	<code>localStorage.clear();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
32	<code>window.location.href = '/';</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
33	<code>}</code>	Pichla block/function/object close karta hai.
34	<code>}</code>	Pichla block/function/object close karta hai.
35	<code>}</code>	Pichla block/function/object close karta hai.

src/app/layout/sidebar/sidebar.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	<code>.sidebar {</code>	Code block start hota hai; andar related statements group hote hain.
2	<code>width: 260px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
3	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
4	<code>border-right: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
5	<code>padding: 24px 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
6	<code>transition: background 0.3s, border-color 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
7	<code>min-height: 100%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
8	<code>flex-shrink: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
9	<code>}</code>	Pichla block/function/object close karta hai.
10		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
11	<code>.sidebar.dark {</code>	Code block start hota hai; andar related statements group hote hain.
12	<code>background: #14141f;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
13	<code>border-right-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
14	<code>}</code>	Pichla block/function/object close karta hai.
15		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
16	<code>.sidebar-header {</code>	Code block start hota hai; andar related statements group hote hain.
17	<code>padding: 0 24px 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
18	<code>}</code>	Pichla block/function/object close karta hai.
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	<code>.sidebar-header h1 {</code>	Code block start hota hai; andar related statements group hote hain.
21	<code>font-size: 22px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
22	<code>font-weight: 700;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
23	line-height: 1.2;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
24	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
25	margin: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
26	}	Pichla block/function/object close karta hai.
27		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
28	.sidebar.dark .sidebar-header h1 {	Code block start hota hai; andar related statements group hote hain.
29	color: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
30	}	Pichla block/function/object close karta hai.
31		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
32	.sidebar-nav {	Code block start hota hai; andar related statements group hote hain.
33	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
34	flex-direction: column;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
35	gap: 4px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
36	padding: 0 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
37	}	Pichla block/function/object close karta hai.
38		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
39	.nav-item {	Code block start hota hai; andar related statements group hote hain.
40	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
41	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
42	gap: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
43	padding: 12px 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
44	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
45	color: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
46	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
47	font-size: 15px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
48	font-weight: 500;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
49	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
50	text-decoration: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
51	}	Pichla block/function/object close karta hai.
52		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
53	.sidebar.dark .nav-item {	Code block start hota hai; andar related statements group hote hain.
54	color: #9ca3af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
55	}	Pichla block/function/object close karta hai.
56		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
57	/* Hover – light mode */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.

Line	Original code	Simple explanation
58	<code>.nav-item:hover {</code>	Code block start hota hai; andar related statements group hote hain.
59	<code>background: #f3f4f6;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
60	<code>color: #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
61	<code>}</code>	Pichla block/function/object close karta hai.
62		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
63	<code>/* Hover – dark mode */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
64	<code>.sidebar.dark .nav-item:hover {</code>	Code block start hota hai; andar related statements group hote hain.
65	<code>background: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
66	<code>color: #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
67	<code>}</code>	Pichla block/function/object close karta hai.
68		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
69	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
70	<code>Active – Clean & Clear</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
71	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
72	<code>.nav-item.active {</code>	Code block start hota hai; andar related statements group hote hain.
73	<code>background: #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
74	<code>color: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
75	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
76	<code>box-shadow: 0 4px 14px rgba(16, 185, 129, 0.4);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
77	<code>}</code>	Pichla block/function/object close karta hai.
78		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
79	<code>/* Active – icon ko white karo */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
80	<code>.nav-item.active .nav-icon {</code>	Code block start hota hai; andar related statements group hote hain.
81	<code>filter: brightness(0) invert(1); /* emoji ko white banata hai */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
82	<code>color: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
83	<code>}</code>	Pichla block/function/object close karta hai.
84		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
85	<code>/* Active – text ko extra white + bold */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
86	<code>.nav-item.active .nav-text {</code>	Code block start hota hai; andar related statements group hote hain.
87	<code>color: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
88	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
89	<code>}</code>	Pichla block/function/object close karta hai.
90		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
91	<code>/* Icon base */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
92	<code>.nav-icon {</code>	Code block start hota hai; andar related statements group hote hain.

Lin e	Original code	Simple explanation
93	font-size: 18px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
94	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
95	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
96	justify-content: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
97	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
98	}	Pichla block/function/object close karta hai.
99		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
100	/* Text base */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
101	.nav-text {	Code block start hota hai; andar related statements group hote hain.
102	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
103	}	Pichla block/function/object close karta hai.
104	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
105	Mobile – Sidebar full height	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
106	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
107	@media (max-width: 768px) {	Responsive CSS rule: screen width/device condition par styles apply karta hai.
108	.sidebar {	Code block start hota hai; andar related statements group hote hain.
109	width: 260px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
110	height: 100%;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
111	overflow-y: auto;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
112	box-shadow: 4px 0 20px rgba(0, 0, 0, 0.3);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
113	}	Pichla block/function/object close karta hai.
114	}	Pichla block/function/object close karta hai.

src/app/layout/sidebar/sidebar.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Lin e	Original code	Simple explanation
1	<aside class="sidebar" [class.dark]="themeService.isDarkMode()">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
2		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
3	<div class="sidebar-header">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
4	Course Management System	HTML element/component: page ka structure ya UI control banata hai.
5	</div>	HTML element/component: page ka structure ya UI control banata hai.
6		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
7	<nav class="sidebar-nav">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
8	@for (item of menuItems; track item.name) {	Angular loop: array ke har item ke liye HTML repeat karta hai.
9	<a	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
10	class="nav-item"	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.

Line	Original code	Simple explanation
11	<code>[routerLink]="item.route"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
12	<code>routerLinkActive="active"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
13	<code>(click)="onLinkClick()"</code>	User click par specified TypeScript method/expression chalti hai.
14	<code>></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
15	<code>{{ item.icon }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
16	<code>{{ item.name }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
17	<code></code>	HTML element/component: page ka structure ya UI control banata hai.
18	<code>}</code>	Pichla block/function/object close karta hai.
19	<code></nav></code>	HTML element/component: page ka structure ya UI control banata hai.
20		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
21	<code></aside></code>	HTML element/component: page ka structure ya UI control banata hai.

src/app/layout/sidebar/sidebar.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	<code>import { Component, signal, output, input } from '@angular/core';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
2	<code>import { CommonModule } from '@angular/common';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
3	<code>import { RouterLink, RouterLinkActive } from '@angular/router';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
4	<code>import { ThemeService } from '../services/theme';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
5		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
6	<code>@Component({</code>	Angular decorator: neech wali class ko Angular component declare karta hai.
7	<code>selector: 'app-sidebar',</code>	Component ka HTML tag name; parent template mein isi selector se component lagta hai.
8	<code>standalone: true,</code>	Component standalone hai; NgModule mein declare karna zaroori nahi.
9	<code>imports: [CommonModule, RouterLink, RouterLinkActive],</code>	Template mein use hone wale Angular directives/forms/router features import karta hai.
10	<code>styleUrl: './sidebar.css',</code>	Is component ki CSS file ka path.
11	<code>templateUrl: './sidebar.html',</code>	Is component ki HTML template file ka path.
12	<code>})</code>	Pichla block/function/object close karta hai.
13	<code>export class Sidebar {</code>	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
14		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
15	<code>// ■ Output – mobile pe link click hone pe parent ko batane ke liye</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
16	<code>linkClicked = output<void>();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
17		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
18	<code>constructor(public themeService: ThemeService) {}</code>	Shared ThemeService use hoti hai taake theme state/component access milay.
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	<code>menuItems = [</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
21	<code>{ name: 'Dashboard', icon: '■', route: '/dashboard' },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
22	<code>{ name: 'Student', icon: '■■■', route: '/students' },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
23	<code>{ name: 'Courses', icon: '■', route: '/courses' },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
24	<code>{ name: 'Enrollment', icon: '■', route: '/enrollments' },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
25	<code>{ name: 'Fees', icon: '■', route: '/fees' },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
26	<code>{ name: 'Settings', icon: '■■■', route: '/settings' }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
27	<code>];</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
28		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
29	<code>onLinkClick() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
30	<code>this.linkClicked.emit();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
31	<code>}</code>	Pichla block/function/object close karta hai.
32	<code>}</code>	Pichla block/function/object close karta hai.

src/app/pages/admin-profile/admin-profile.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
2	<code>Admin Profile Page</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
3	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
4	<code>.admin-profile-page {</code>	Code block start hota hai; andar related statements group hote hain.
5	<code>padding: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
6	<code>background: #f5f6fa;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
7	<code>min-height: 100vh;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
8	<code>font-family: 'Inter', sans-serif;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
9	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
10	<code>transition: background 0.3s, color 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
11	<code>}</code>	Pichla block/function/object close karta hai.
12	<code>.admin-profile-page.dark { background: #0f0f1a; color: #e4e4e7; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
13		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
14	<code>.page-header {</code>	Code block start hota hai; andar related statements group hote hain.
15	<code>display: flex; justify-content: space-between;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
16	<code>align-items: center; margin-bottom: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
17	<code>}</code>	Pichla block/function/object close karta hai.
18	<code>.page-header h2 { font-size: 24px; font-weight: 700; margin: 0; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
19	<code>.admin-profile-page.dark .page-header h2 { color: #ffffff; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Lin e	Original code	Simple explanation
20		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
21	<code>.edit-toggle-btn {</code>	Code block start hota hai; andar related statements group hote hain.
22	<code>background: #465fff; color: white; border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
23	<code>padding: 10px 20px; border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
24	<code>font-size: 14px; font-weight: 600; cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
25	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
26	<code>box-shadow: 0 4px 12px rgba(70, 95, 255, 0.3);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
27	<code>}</code>	Pichla block/function/object close karta hai.
28	<code>.edit-toggle-btn:hover { background: #3548d4; transform: translateY(-1px); }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
29		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
30	<code>/* Layout */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
31	<code>.profile-container {</code>	Code block start hota hai; andar related statements group hote hain.
32	<code>display: grid;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
33	<code>grid-template-columns: 300px 1fr;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
34	<code>gap: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
35	<code>}</code>	Pichla block/function/object close karta hai.
36		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
37	<code>.profile-card {</code>	Code block start hota hai; andar related statements group hote hain.
38	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
39	<code>border-radius: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
40	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
41	<code>box-shadow: 0 2px 8px rgba(0, 0, 0, 0.04);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
42	<code>padding: 28px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
43	<code>transition: background 0.3s, border-color 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
44	<code>}</code>	Pichla block/function/object close karta hai.
45	<code>.admin-profile-page.dark .profile-card {</code>	Code block start hota hai; andar related statements group hote hain.
46	<code>background: #14141f;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
47	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
48	<code>}</code>	Pichla block/function/object close karta hai.
49		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
50	<code>/* Avatar Card */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
51	<code>.avatar-card {</code>	Code block start hota hai; andar related statements group hote hain.
52	<code>display: flex; flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
53	<code>align-items: center; text-align: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
54	<code>height: fit-content;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Lin e	Original code	Simple explanation
55	}	Pichla block/function/object close karta hai.
56		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
57	.avatar {	Code block start hota hai; andar related statements group hote hain.
58	width: 120px; height: 120px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
59	border-radius: 50%;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
60	background: linear-gradient(135deg, #10b981, #059669);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
61	color: white;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
62	font-size: 48px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
63	font-weight: 700;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
64	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
65	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
66	justify-content: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
67	margin-bottom: 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
68	box-shadow: 0 10px 30px rgba(16, 185, 129, 0.3);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
69	}	Pichla block/function/object close karta hai.
70		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
71	.avatar-card h3 {	Code block start hota hai; andar related statements group hote hain.
72	font-size: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
73	font-weight: 700;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
74	margin: 0 0 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
75	}	Pichla block/function/object close karta hai.
76	.admin-profile-page.dark .avatar-card h3 { color: #ffffff; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
77		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
78	.role-badge {	Code block start hota hai; andar related statements group hote hain.
79	display: inline-block;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
80	background: #dbeafe;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
81	color: #1e40af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
82	padding: 6px 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
83	border-radius: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
84	font-size: 13px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
85	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
86	margin-bottom: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
87	}	Pichla block/function/object close karta hai.
88	.admin-profile-page.dark .role-badge {	Code block start hota hai; andar related statements group hote hain.
89	background: #1e3a8a; color: #93c5fd;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
90	}	Pichla block/function/object close karta hai.
91		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
92	.admin-id {	Code block start hota hai; andar related statements group hote hain.
93	font-size: 13px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
94	color: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
95	margin: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
96	}	Pichla block/function/object close karta hai.
97		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
98	/* Info Card */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
99	.info-card h3 {	Code block start hota hai; andar related statements group hote hain.
100	font-size: 18px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
101	font-weight: 700;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
102	margin: 0 0 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
103	padding-bottom: 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
104	border-bottom: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
105	}	Pichla block/function/object close karta hai.
106	.admin-profile-page.dark .info-card h3 {	Code block start hota hai; andar related statements group hote hain.
107	color: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
108	border-bottom-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
109	}	Pichla block/function/object close karta hai.
110		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
111	.info-list {	Code block start hota hai; andar related statements group hote hain.
112	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
113	flex-direction: column;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
114	gap: 4px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
115	}	Pichla block/function/object close karta hai.
116		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
117	.info-row {	Code block start hota hai; andar related statements group hote hain.
118	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
119	justify-content: space-between;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
120	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
121	padding: 14px 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
122	border-bottom: 1px solid #f3f4f6;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
123	}	Pichla block/function/object close karta hai.
124	.admin-profile-page.dark .info-row { border-bottom-color: #1f1f2e; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
125	<code>.info-row:last-child { border-bottom: none; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
126		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
127	<code>.info-label {</code>	Code block start hota hai; andar related statements group hote hain.
128	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
129	<code>font-weight: 500;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
130	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
131	<code>}</code>	Pichla block/function/object close karta hai.
132	<code>.admin-profile-page.dark .info-label { color: #9ca3af; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
133		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
134	<code>.info-value {</code>	Code block start hota hai; andar related statements group hote hain.
135	<code>font-size: 15px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
136	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
137	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
138	<code>}</code>	Pichla block/function/object close karta hai.
139	<code>.admin-profile-page.dark .info-value { color: #ffffff; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
140		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
141	<code>/* Edit Form */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
142	<code>.edit-form {</code>	Code block start hota hai; andar related statements group hote hain.
143	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
144	<code>flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
145	<code>gap: 18px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
146	<code>}</code>	Pichla block/function/object close karta hai.
147		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
148	<code>.form-group {</code>	Code block start hota hai; andar related statements group hote hain.
149	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
150	<code>flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
151	<code>gap: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
152	<code>}</code>	Pichla block/function/object close karta hai.
153		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
154	<code>.form-group label {</code>	Code block start hota hai; andar related statements group hote hain.
155	<code>font-size: 13px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
156	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
157	<code>color: #374151;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
158	<code>}</code>	Pichla block/function/object close karta hai.
159	<code>.admin-profile-page.dark .form-group label { color: #d1d5db; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
160		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
161	<code>.form-group input {</code>	Code block start hota hai; andar related statements group hote hain.
162	<code>padding: 12px 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
163	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
164	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
165	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
166	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
167	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
168	<code>outline: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
169	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
170	<code>}</code>	Pichla block/function/object close karta hai.
171	<code>.admin-profile-page.dark .form-group input {</code>	Code block start hota hai; andar related statements group hote hain.
172	<code>background: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
173	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
174	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
175	<code>}</code>	Pichla block/function/object close karta hai.
176	<code>.form-group input:focus {</code>	Code block start hota hai; andar related statements group hote hain.
177	<code>border-color: #465fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
178	<code>box-shadow: 0 0 3px rgba(70, 95, 255, 0.15);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
179	<code>}</code>	Pichla block/function/object close karta hai.
180	<code>.form-group input.error {</code>	Code block start hota hai; andar related statements group hote hain.
181	<code>border-color: #ef4444 !important;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
182	<code>box-shadow: 0 0 3px rgba(239, 68, 68, 0.15) !important;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
183	<code>}</code>	Pichla block/function/object close karta hai.
184		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
185	<code>.error-msg {</code>	Code block start hota hai; andar related statements group hote hain.
186	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
187	<code>color: #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
188	<code>font-weight: 500;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
189	<code>}</code>	Pichla block/function/object close karta hai.
190		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
191	<code>.edit-actions {</code>	Code block start hota hai; andar related statements group hote hain.
192	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
193	<code>gap: 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
194	<code>justify-content: flex-end;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
195	margin-top: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
196	padding-top: 18px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
197	border-top: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
198	}	Pichla block/function/object close karta hai.
199	.admin-profile-page.dark .edit-actions { border-top-color: #2a2a3a; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
200		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
201	.save-btn {	Code block start hota hai; andar related statements group hote hain.
202	background: #10b981; color: white; border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
203	padding: 10px 24px; border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
204	font-size: 14px; font-weight: 600; cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
205	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
206	}	Pichla block/function/object close karta hai.
207	.save-btn:hover { background: #059669; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
208		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
209	.cancel-btn {	Code block start hota hai; andar related statements group hote hain.
210	background: #6b7280; color: white; border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
211	padding: 10px 20px; border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
212	font-size: 14px; font-weight: 600; cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
213	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
214	}	Pichla block/function/object close karta hai.
215	.cancel-btn:hover { background: #4b5563; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
216		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
217	/* Toast */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
218	.toast-popup {	Code block start hota hai; andar related statements group hote hain.
219	position: fixed; top: 90px; right: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
220	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
221	border-left: 4px solid #ef4444;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
222	border-radius: 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
223	padding: 14px 18px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
224	display: flex; align-items: center; gap: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
225	box-shadow: 0 10px 30px rgba(0, 0, 0, 0.15);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
226	z-index: 2000;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
227	min-width: 280px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
228	}	Pichla block/function/object close karta hai.
229	.toast-popup.success { border-left-color: #10b981; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
230	<code>.admin-profile-page.dark .toast-popup { background: #1f1f2e; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
231		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
232	<code>.toast-text { flex: 1; font-size: 14px; font-weight: 500; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
233	<code>.admin-profile-page.dark .toast-text { color: #e4e4e7; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
234	<code>.toast-close {</code>	Code block start hota hai; andar related statements group hote hain.
235	<code>background: transparent; border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
236	<code>color: #9ca3af; cursor: pointer; padding: 2px 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
237	<code>}</code>	Pichla block/function/object close karta hai.
238	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
239	<code>Profile Picture</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
240	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
241	<code>.avatar-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
242	<code>position: relative;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
243	<code>display: inline-block;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
244	<code>margin-bottom: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
245	<code>}</code>	Pichla block/function/object close karta hai.
246		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
247	<code>.avatar-img {</code>	Code block start hota hai; andar related statements group hote hain.
248	<code>width: 120px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
249	<code>height: 120px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
250	<code>border-radius: 50%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
251	<code>object-fit: cover;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
252	<code>border: 4px solid #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
253	<code>box-shadow: 0 10px 30px rgba(16, 185, 129, 0.3);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
254	<code>}</code>	Pichla block/function/object close karta hai.
255		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
256	<code>.avatar {</code>	Code block start hota hai; andar related statements group hote hain.
257	<code>width: 120px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
258	<code>height: 120px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
259	<code>border-radius: 50%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
260	<code>background: linear-gradient(135deg, #10b981, #059669);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
261	<code>color: white;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
262	<code>font-size: 48px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
263	<code>font-weight: 700;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
264	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
265	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
266	<code>justify-content: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
267	<code>box-shadow: 0 10px 30px rgba(16, 185, 129, 0.3);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
268	<code>border: 4px solid #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
269	<code>}</code>	Pichla block/function/object close karta hai.
270		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
271	<code>/* Camera Button */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
272	<code>.camera-btn {</code>	Code block start hota hai; andar related statements group hote hain.
273	<code>position: absolute;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
274	<code>bottom: 5px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
275	<code>right: 5px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
276	<code>width: 38px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
277	<code>height: 38px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
278	<code>border-radius: 50%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
279	<code>background: #465fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
280	<code>color: white;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
281	<code>border: 3px solid #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
282	<code>font-size: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
283	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
284	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
285	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
286	<code>justify-content: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
287	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
288	<code>box-shadow: 0 4px 12px rgba(70, 95, 255, 0.4);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
289	<code>}</code>	Pichla block/function/object close karta hai.
290	<code>.admin-profile-page.dark .camera-btn {</code>	Code block start hota hai; andar related statements group hote hain.
291	<code>border-color: #14141f;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
292	<code>}</code>	Pichla block/function/object close karta hai.
293	<code>.camera-btn:hover {</code>	Code block start hota hai; andar related statements group hote hain.
294	<code>background: #3548d4;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
295	<code>transform: scale(1.1);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
296	<code>}</code>	Pichla block/function/object close karta hai.
297		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
298	<code>/* Photo Actions */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.

Line	Original code	Simple explanation
299	<code>.photo-actions {</code>	Code block start hota hai; andar related statements group hote hain.
300	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
301	<code>gap: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
302	<code>justify-content: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
303	<code>margin-bottom: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
304	<code>flex-wrap: wrap;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
305	<code>}</code>	Pichla block/function/object close karta hai.
306		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
307	<code>.change-photo-btn {</code>	Code block start hota hai; andar related statements group hote hain.
308	<code>background: #465fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
309	<code>color: white;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
310	<code>border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
311	<code>padding: 8px 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
312	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
313	<code>font-size: 13px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
314	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
315	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
316	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
317	<code>}</code>	Pichla block/function/object close karta hai.
318	<code>.change-photo-btn:hover {</code>	Code block start hota hai; andar related statements group hote hain.
319	<code>background: #3548d4;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
320	<code>transform: translateY(-1px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
321	<code>}</code>	Pichla block/function/object close karta hai.
322		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
323	<code>.remove-photo-btn {</code>	Code block start hota hai; andar related statements group hote hain.
324	<code>background: transparent;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
325	<code>color: #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
326	<code>border: 1px solid #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
327	<code>padding: 8px 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
328	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
329	<code>font-size: 13px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
330	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
331	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
332	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
333	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
334	<code>.remove-photo-btn:hover {</code>	Code block start hota hai; andar related statements group hote hain.
335	<code>background: #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
336	<code>color: white;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
337	<code>}</code>	Pichla block/function/object close karta hai.
338		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
339	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
340	<code>Full Image Preview</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
341	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
342	<code>.image-preview-overlay {</code>	Code block start hota hai; andar related statements group hote hain.
343	<code>position: fixed;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
344	<code>inset: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
345	<code>background: rgba(0, 0, 0, 0.9);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
346	<code>backdrop-filter: blur(8px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
347	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
348	<code>flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
349	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
350	<code>justify-content: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
351	<code>z-index: 9999;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
352	<code>padding: 40px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
353	<code>animation: fadeIn 0.2s ease;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
354	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
355	<code>}</code>	Pichla block/function/object close karta hai.
356		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
357	<code>@keyframes fadeIn {</code>	Code block start hota hai; andar related statements group hote hain.
358	<code>from { opacity: 0; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
359	<code>to { opacity: 1; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
360	<code>}</code>	Pichla block/function/object close karta hai.
361		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
362	<code>.preview-image {</code>	Code block start hota hai; andar related statements group hote hain.
363	<code>max-width: 90vw;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
364	<code>max-height: 80vh;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
365	<code>border-radius: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
366	<code>box-shadow: 0 20px 80px rgba(0, 0, 0, 0.8);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
367	<code>border: 4px solid #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
368	animation: zoomIn 0.3s ease;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
369	cursor: default;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
370	}	Pichla block/function/object close karta hai.
371		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
372	@keyframes zoomIn {	Code block start hota hai; andar related statements group hote hain.
373	from { transform: scale(0.7); opacity: 0; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
374	to { transform: scale(1); opacity: 1; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
375	}	Pichla block/function/object close karta hai.
376		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
377	.preview-close-btn {	Code block start hota hai; andar related statements group hote hain.
378	position: absolute;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
379	top: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
380	right: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
381	width: 44px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
382	height: 44px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
383	border-radius: 50%;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
384	background: rgba(255, 255, 255, 0.15);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
385	color: white;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
386	border: 2px solid rgba(255, 255, 255, 0.3);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
387	font-size: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
388	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
389	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
390	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
391	justify-content: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
392	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
393	z-index: 10000;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
394	}	Pichla block/function/object close karta hai.
395		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
396	.preview-close-btn:hover {	Code block start hota hai; andar related statements group hote hain.
397	background: #ef4444;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
398	border-color: #ef4444;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
399	transform: rotate(90deg) scale(1.1);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
400	}	Pichla block/function/object close karta hai.
401		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
402	.preview-hint {	Code block start hota hai; andar related statements group hote hain.

Line	Original code	Simple explanation
403	margin-top: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
404	color: rgba(255, 255, 255, 0.6);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
405	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
406	font-weight: 500;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
407	letter-spacing: 0.3px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
408	}	Pichla block/function/object close karta hai.
409		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
410	/* Avatar pe hover effect */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
411	.avatar-img:hover {	Code block start hota hai; andar related statements group hote hain.
412	transform: scale(1.05);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
413	box-shadow: 0 15px 40px rgba(16, 185, 129, 0.5);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
414	transition: 0.3s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
415	}	Pichla block/function/object close karta hai.
416		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
417	/* Responsive */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
418	@media (max-width: 768px) {	Responsive CSS rule: screen width/device condition par styles apply karta hai.
419	.profile-container { grid-template-columns: 1fr; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
420	.admin-profile-page { padding: 16px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
421	.page-header { flex-direction: column; gap: 12px; align-items: stretch; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
422	.info-row { flex-direction: column; align-items: flex-start; gap: 4px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
423	}	Pichla block/function/object close karta hai.

src/app/pages/admin-profile/admin-profile.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Line	Original code	Simple explanation
1	<div class="admin-profile-page" [class.dark]="themeService.isDarkMode()">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
2		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
3	@if (showToast()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
4	<div class="toast-popup" [class.success]="toastType() === 'success'">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
5	{{ toastMessage() }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
6	<button class="toast-close" (click)="showToast.set(false)">X</button>	User click par specified TypeScript method/expression chalti hai.
7	</div>	HTML element/component: page ka structure ya UI control banata hai.
8	}	Pichla block/function/object close karta hai.
9		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
10	<div class="page-header">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.

Lin e	Original code	Simple explanation
11	<code><h2>Admin Profile</h2></code>	HTML element/component: page ka structure ya UI control banata hai.
12	<code><button class="edit-toggle-btn" (click)="toggleEdit()"></code>	User click par specified TypeScript method/expression chalti hai.
13	<code>{{ isEditing() ? 'X Cancel' : 'Edit Profile' }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
14	<code></button></code>	HTML element/component: page ka structure ya UI control banata hai.
15	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
16		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
17	<code><div class="profile-container"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
18		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
19	<code><!-- Left: Avatar Card --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
20	<code><div class="profile-card avatar-card"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
21		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
22	<code><!-- Hidden file input --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
23	<code><input</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
24	<code>type="file"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
25	<code>id="photoInput"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
26	<code>accept="image/*"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
27	<code>(change)="onPhotoSelected(\$event)"</code>	Input/select ki value change par handler chalta hai.
28	<code>style="display: none;"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
29	<code>/></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
30		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
31	<code><!-- Avatar with photo or letter --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
32	<code><div class="avatar-wrapper"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
33	<code>@if (profilePicture()) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
34	<code><img</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
35	<code>[src]="profilePicture()"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
36	<code>alt="Profile"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
37	<code>class="avatar-img"</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
38	<code>(click)="openImagePreview()"</code>	User click par specified TypeScript method/expression chalti hai.
39	<code>style="cursor: pointer;"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
40	<code>title="Click to view full image"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
41	<code>/></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
42	<code>} @else {</code>	Alternative branch: pehli condition false ho to yeh code chalta hai.
43	<code><div class="avatar"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
44	<code>{{ profile().name.charAt(0) }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.

Line	Original code	Simple explanation
45	</div>	HTML element/component: page ka structure ya UI control banata hai.
46	}	Pichla block/function/object close karta hai.
47		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
48	<button class="camera-btn" (click)="triggerFileInput()" title="Change Photo">	User click par specified TypeScript method/expression chalti hai.
49	■	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
50	</button>	HTML element/component: page ka structure ya UI control banata hai.
51	</div>	HTML element/component: page ka structure ya UI control banata hai.
52		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
53	<!-- ■ Buttons -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
54	<div class="photo-actions">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
55	<button class="change-photo-btn" (click)="triggerFileInput()">	User click par specified TypeScript method/expression chalti hai.
56	{{ profilePicture() ? 'Change Photo' : 'Upload Photo' }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
57	</button>	HTML element/component: page ka structure ya UI control banata hai.
58		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
59	@if (profilePicture()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
60	<button class="remove-photo-btn" (click)="removePhoto()">	User click par specified TypeScript method/expression chalti hai.
61	Remove	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
62	</button>	HTML element/component: page ka structure ya UI control banata hai.
63	}	Pichla block/function/object close karta hai.
64	</div>	HTML element/component: page ka structure ya UI control banata hai.
65		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
66	{{ profile().name }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
67	{{ profile().role }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
68	<p class="admin-id">ID: {{ profile().id }}</p>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
69	</div>	HTML element/component: page ka structure ya UI control banata hai.
70		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
71	<!-- Right: Info Card -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
72	<div class="profile-card info-card">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
73	Account Information	HTML element/component: page ka structure ya UI control banata hai.
74		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
75	<!-- VIEW MODE -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
76	@if (!isEditing()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
77	<div class="info-list">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
78	<div class="info-row">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.

Line	Original code	Simple explanation
79	<code>■ Name</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
80	<code>{{ profile().name }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
81	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
82	<code><div class="info-row"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
83	<code>■ Email</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
84	<code>{{ profile().email }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
85	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
86	<code><div class="info-row"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
87	<code>■ Phone</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
88	<code>{{ profile().phone }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
89	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
90	<code><div class="info-row"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
91	<code>■ Role</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
92	<code>{{ profile().role }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
93	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
94	<code><div class="info-row"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
95	<code>■ Joined</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
96	<code>{{ profile().joinedDate }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
97	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
98	<code><div class="info-row"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
99	<code>■ Last Login</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
100	<code>{{ profile().lastLogin }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
101	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
102	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
103	<code>}</code>	Pichla block/function/object close karta hai.
104		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
105	<code><!-- EDIT MODE --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
106	<code>@if (isEditing()) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
107	<code><form [formGroup]="profileForm" (ngSubmit)="saveProfile()" class="edit-form"></code>	Form submit par specified handler chalata hai.
108		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
109	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
110	<code><label>Name *</label></code>	HTML element/component: page ka structure ya UI control banata hai.
111	<code><input formControlName="name" [class.error]="isInvalid('name')" /></code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
112	<code>@if (isInvalid('name')) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.

Line	Original code	Simple explanation
113	<code>■ Name required</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
114	<code>}</code>	Pichla block/function/object close karta hai.
115	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
116		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
117	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
118	<code><label>Email *</label></code>	HTML element/component: page ka structure ya UI control banata hai.
119	<code><input type="email" formControlName="email" [class.error]="invalid('email')" /></code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
120	<code>@if (invalid('email')) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
121	<code>■ Valid email required</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
122	<code>}</code>	Pichla block/function/object close karta hai.
123	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
124		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
125	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
126	<code><label>Phone *</label></code>	HTML element/component: page ka structure ya UI control banata hai.
127	<code><input formControlName="phone" [class.error]="invalid('phone')" /></code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
128	<code>@if (invalid('phone')) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
129	<code>■ Phone required</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
130	<code>}</code>	Pichla block/function/object close karta hai.
131	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
132		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
133	<code><div class="edit-actions"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
134	<code><button type="button" class="cancel-btn" (click)="toggleEdit()">Cancel</button></code>	User click par specified TypeScript method/expression chalti hai.
135	<code><button type="submit" class="save-btn">■ Save Changes</button></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
136	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
137		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
138	<code></form></code>	HTML element/component: page ka structure ya UI control banata hai.
139	<code>}</code>	Pichla block/function/object close karta hai.
140	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
141		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
142	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
143		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
144	<code><!-- ■ Full Image Preview Modal --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
145	<code>@if (showImagePreview()) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
146	<code><div class="image-preview-overlay" (click)="closeImagePreview()"></code>	User click par specified TypeScript method/expression chalti hai.
147	<code><button class="preview-close-btn" (click)="closeImagePreview()">X</button></code>	User click par specified TypeScript method/expression chalti hai.

Line	Original code	Simple explanation
148	<img	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
149	[src]="profilePicture()"	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
150	alt="Full Preview"	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
151	class="preview-image"	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
152	(click)="\$event.stopPropagation()"	User click par specified TypeScript method/expression chalti hai.
153	/>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
154	<p class="preview-hint">Click anywhere or press ESC to close</p>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
155	</div>	HTML element/component: page ka structure ya UI control banata hai.
156	}	Pichla block/function/object close karta hai.
157	</div>	HTML element/component: page ka structure ya UI control banata hai.

src/app/pages/admin-profile/admin-profile.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	import { Component, OnInit, signal, inject } from '@angular/core';	Import: doosri library/file se required class ya function is file mein available karta hai.
2	import { CommonModule } from '@angular/common';	Import: doosri library/file se required class ya function is file mein available karta hai.
3	import { FormBuilder, ReactiveFormsModule, Validators } from '@angular/forms';	Import: doosri library/file se required class ya function is file mein available karta hai.
4	import { ThemeService } from '../../../services/theme';	Import: doosri library/file se required class ya function is file mein available karta hai.
5		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
6	@Component({	Angular decorator: neech wali class ko Angular component declare karta hai.
7	selector: 'app-admin-profile',	Component ka HTML tag name; parent template mein isi selector se component lagta hai.
8	standalone: true,	Component standalone hai; NgModule mein declare karna zaroori nahi.
9	imports: [CommonModule, ReactiveFormsModule],	Template mein use hone wale Angular directives/forms/router features import karta hai.
10	templateUrl: './admin-profile.html',	Is component ki HTML template file ka path.
11	styleUrl: './admin-profile.css'	Is component ki CSS file ka path.
12	})	Pichla block/function/object close karta hai.
13	export class AdminProfile implements OnInit {	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
14		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
15		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
16	private fb = inject(FormBuilder);	FormBuilder leta hai taake reactive form aur controls bana sake.
17	public themeService = inject(ThemeService);	Shared ThemeService use hoti hai taake theme state/component access milay.
18		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
19	// ■ Open full image	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
20	openImagePreview() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
21	if (this.profilePicture()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.

Lin e	Original code	Simple explanation
22	<code>this.showImagePreview.set(true);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
23	<code>}</code>	Pichla block/function/object close karta hai.
24	<code>}</code>	Pichla block/function/object close karta hai.
25		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
26	<code>// ■ Close preview</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
27	<code>closeImagePreview() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
28	<code>this.showImagePreview.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
29	<code>}</code>	Pichla block/function/object close karta hai.
30	<code>// ■ Full image preview</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
31	<code>showImagePreview = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/update().
32	<code>// Toast</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
33	<code>showToast = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/update().
34	<code>toastMessage = signal<string>('');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/update().
35	<code>toastType = signal<'error' 'success'>('error');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/update().
36		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
37	<code>// Edit mode</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
38	<code>isEditing = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/update().
39		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
40	<code>// ■ Profile picture</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
41	<code>profilePicture = signal<string>('');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/update().
42		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
43		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
44	<code>// Profile data</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
45	<code>profile = signal<any>({</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/update().
46	<code>id: 'A1',</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
47	<code>name: 'Shahzad Naseem',</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
48	<code>email: 'admin@cms.com',</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
49	<code>phone: '03001234567',</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
50	<code>role: 'Super Admin',</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
51	<code>joinedDate: '2026-01-01',</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
52	<code>lastLogin: new Date().toISOString().slice(0, 16).replace('T', ' ');</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
53	<code>});</code>	Pichla block/function/object close karta hai.
54		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
55	<code>profileForm = this.fb.group({</code>	Reactive form group banata hai; fields, current values aur validation ek jagah manage hotay hain.
56	<code>name: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.

Line	Original code	Simple explanation
57	email: ['', [Validators.required, Validators.email]],	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
58	phone: ['', Validators.required]	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
59	});	Pichla block/function/object close karta hai.
60		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
61	ngOnInit() {	Component initialize hone ke baad Angular is method ko automatically call karta hai.
62	// Load from localStorage	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
63	const saved = localStorage.getItem('admin_profile');	Browser ke localStorage se saved string read karta hai.
64	if (saved) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
65	const data = JSON.parse(saved);	JSON text ko JavaScript object/value mein convert karta hai.
66	this.profile.update(p => ({ ...p, ...data }));	Arrow function: callback/function ko short syntax mein define karta hai.
67	}	Pichla block/function/object close karta hai.
68		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
69	// ■ Load profile picture	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
70	const savedPic = localStorage.getItem('admin_profile_picture');	Browser ke localStorage se saved string read karta hai.
71	if (savedPic) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
72	this.profilePicture.set(savedPic);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
73	}	Pichla block/function/object close karta hai.
74		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
75	// Fill form	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
76	this.profileForm.patchValue({	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
77	name: this.profile().name,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
78	email: this.profile().email,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
79	phone: this.profile().phone	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
80	});	Pichla block/function/object close karta hai.
81	// ■ Escape key se close	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
82	document.addEventListener('keydown', (e) => {	Arrow function: callback/function ko short syntax mein define karta hai.
83	if (e.key === 'Escape') {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
84	this.showImagePreview.set(false);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
85	}	Pichla block/function/object close karta hai.
86	});	Pichla block/function/object close karta hai.
87	}	Pichla block/function/object close karta hai.
88		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
89	showToastMsg(msg: string, type: 'error' 'success' = 'error') {	Function/method define karta hai; call hone par is block ka kaam hota hai.
90	this.toastMessage.set(msg);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
91	this.toastType.set(type);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
92	this.showToast.set(true);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
93	setTimeout(() => this.showToast.set(false), 3000);	Arrow function: callback/function ko short syntax mein define karta hai.

Line	Original code	Simple explanation
94	}	Pichla block/function/object close karta hai.
95		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
96	// ■ Photo upload	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
97	onPhotoSelected(event: Event) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
98	const input = event.target as HTMLInputElement;	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
99	const file = input.files?.[0];	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
100		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
101	if (!file) return;	Condition: andar ka code sirf tab chalta hai jab condition true ho.
102		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
103	// Size check (max 2MB)	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
104	if (file.size > 2 * 1024 * 1024) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
105	this.showToastMsg('■ Image must be less than 2MB', 'error');	User ko success/error ka short notification dikhata hai.
106	return;	Current function ko yahin stop karta hai.
107	}	Pichla block/function/object close karta hai.
108		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
109	// Type check	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
110	if (!file.type.startsWith('image/')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
111	this.showToastMsg('■ Please select an image file', 'error');	User ko success/error ka short notification dikhata hai.
112	return;	Current function ko yahin stop karta hai.
113	}	Pichla block/function/object close karta hai.
114		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
115	const reader = new FileReader();	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
116	reader.onload = () => {	Arrow function: callback/function ko short syntax mein define karta hai.
117	const base64 = reader.result as string;	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
118	this.profilePicture.set(base64);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
119	localStorage.setItem('admin_profile_picture', base64);	Browser localStorage mein string save karta hai; browser-side persistence.
120	this.showToastMsg('■ Profile picture updated!', 'success');	User ko success/error ka short notification dikhata hai.
121	};	Pichla block/function/object close karta hai.
122	reader.readAsDataURL(file);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
123	}	Pichla block/function/object close karta hai.
124		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
125	// ■ Remove photo	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
126	removePhoto() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
127	if (!confirm('Remove profile picture?')) return;	Condition: andar ka code sirf tab chalta hai jab condition true ho.
128	this.profilePicture.set('');	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
129	localStorage.removeItem('admin_profile_picture');	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
130	this.showToastMsg('■ Profile picture removed', 'success');	User ko success/error ka short notification dikhata hai.

Line	Original code	Simple explanation
131	}	Pichla block/function/object close karta hai.
132		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
133	// ■ Trigger file input	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
134	triggerFileInput() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
135	const input = document.getElementById('photoInput') as HTMLInputElement;	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
136	input?.click();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
137	}	Pichla block/function/object close karta hai.
138		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
139	toggleEdit() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
140	this.isEditing.update(v => !v);	Arrow function: callback/function ko short syntax mein define karta hai.
141	if (!this.isEditing()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
142	this.profileForm.patchValue({	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
143	name: this.profile().name,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
144	email: this.profile().email,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
145	phone: this.profile().phone	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
146	});	Pichla block/function/object close karta hai.
147	}	Pichla block/function/object close karta hai.
148	}	Pichla block/function/object close karta hai.
149		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
150	saveProfile() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
151	if (this.profileForm.invalid) {	Check karta hai ke form/field validation pass kar raha hai ya nahi.
152	this.profileForm.markAllAsTouched();	Validation messages dikhane ke liye sab controls ko touched mark karta hai.
153	this.showToastMsg('■ Please fill all fields correctly', 'error');	User ko success/error ka short notification dikhata hai.
154	return;	Current function ko yahin stop karta hai.
155	}	Pichla block/function/object close karta hai.
156		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
157	this.profile.update(p => ({ ...p, ...this.profileForm.value }));	Arrow function: callback/function ko short syntax mein define karta hai.
158	localStorage.setItem('admin_profile', JSON.stringify(this.profileForm.value));	Browser localStorage mein string save karta hai; browser-side persistence.
159		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
160	this.showToastMsg('■ Profile updated successfully!', 'success');	User ko success/error ka short notification dikhata hai.
161	this.isEditing.set(false);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
162	}	Pichla block/function/object close karta hai.
163		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
164	isValid(field: string): boolean {	Code block start hota hai; andar related statements group hote hain.
165	const c = this.profileForm.get(field);	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
166	return !(c && c.invalid && (c.dirty c.touched));	Check karta hai ke form/field validation pass kar raha hai ya nahi.
167	}	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
168	}	Pichla block/function/object close karta hai.

src/app/pages/course/course.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
2	Courses Page – Light (default)	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
3	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
4	.courses-page {	Code block start hota hai; andar related statements group hote hain.
5	padding: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
6	background: #f5f6fa;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
7	min-height: 100vh;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
8	font-family: 'Inter', sans-serif;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
9	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
10	transition: background 0.3s, color 0.3s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
11	}	Pichla block/function/object close karta hai.
12		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
13	.courses-page.dark {	Code block start hota hai; andar related statements group hote hain.
14	background: #0f0f1a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
15	color: #e4e4e7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
16	}	Pichla block/function/object close karta hai.
17		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
18	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
19	Page Header	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
20	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
21	.page-header {	Code block start hota hai; andar related statements group hote hain.
22	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
23	justify-content: space-between;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
24	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
25	margin-bottom: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
26	}	Pichla block/function/object close karta hai.
27		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
28	.page-header h2 {	Code block start hota hai; andar related statements group hote hain.
29	font-size: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
30	font-weight: 700;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
31	margin: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
32	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
33	}	Pichla block/function/object close karta hai.
34		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
35	.courses-page.dark .page-header h2 {	Code block start hota hai; andar related statements group hote hain.
36	color: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
37	}	Pichla block/function/object close karta hai.
38		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
39	.add-new-btn {	Code block start hota hai; andar related statements group hote hain.
40	background: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
41	color: #fff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
42	border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
43	padding: 10px 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
44	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
45	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
46	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
47	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
48	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
49	box-shadow: 0 4px 12px rgba(16, 185, 129, 0.3);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
50	}	Pichla block/function/object close karta hai.
51		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
52	.add-new-btn:hover {	Code block start hota hai; andar related statements group hote hain.
53	background: #059669;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
54	transform: translateY(-1px);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
55	}	Pichla block/function/object close karta hai.
56		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
57	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
58	Form	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
59	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
60	.course-form {	Code block start hota hai; andar related statements group hote hain.
61	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
62	padding: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
63	border-radius: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
64	border: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
65	box-shadow: 0 2px 8px rgba(0, 0, 0, 0.04);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
66	display: grid;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
67	grid-template-columns: repeat(auto-fit, minmax(180px, 1fr));	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
68	gap: 15px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
69	margin-bottom: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
70	transition: background 0.3s, border-color 0.3s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
71	}	Pichla block/function/object close karta hai.
72		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
73	.courses-page.dark .course-form {	Code block start hota hai; andar related statements group hote hain.
74	background: #14141f;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
75	border-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
76	}	Pichla block/function/object close karta hai.
77		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
78	.course-form input {	Code block start hota hai; andar related statements group hote hain.
79	padding: 10px 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
80	border: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
81	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
82	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
83	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
84	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
85	outline: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
86	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
87	}	Pichla block/function/object close karta hai.
88		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
89	.courses-page.dark .course-form input {	Code block start hota hai; andar related statements group hote hain.
90	background: #1f1f2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
91	border-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
92	color: #e4e4e7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
93	}	Pichla block/function/object close karta hai.
94		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
95	.courses-page.dark .course-form input::placeholder {	Code block start hota hai; andar related statements group hote hain.
96	color: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
97	}	Pichla block/function/object close karta hai.
98		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
99	.course-form input:focus {	Code block start hota hai; andar related statements group hote hain.
100	border-color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
101	box-shadow: 0 0 0 3px rgba(16, 185, 129, 0.15);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
102	}	Pichla block/function/object close karta hai.
103		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
104	.form-actions {	Code block start hota hai; andar related statements group hote hain.
105	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
106	gap: 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
107	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
108	grid-column: 1 / -1;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
109	justify-content: flex-end;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
110	}	Pichla block/function/object close karta hai.
111		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
112	.save-btn {	Code block start hota hai; andar related statements group hote hain.
113	background: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
114	color: #fff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
115	border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
116	padding: 10px 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
117	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
118	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
119	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
120	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
121	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
122	}	Pichla block/function/object close karta hai.
123		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
124	.save-btn:hover {	Code block start hota hai; andar related statements group hote hain.
125	background: #059669;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
126	transform: translateY(-1px);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
127	}	Pichla block/function/object close karta hai.
128		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
129	.cancel-btn {	Code block start hota hai; andar related statements group hote hain.
130	background: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
131	color: #fff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
132	border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
133	padding: 10px 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
134	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
135	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
136	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
137	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
138	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
139	}	Pichla block/function/object close karta hai.
140		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
141	.cancel-btn:hover {	Code block start hota hai; andar related statements group hote hain.
142	background: #4b5563;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
143	}	Pichla block/function/object close karta hai.
144		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
145	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
146	Table	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
147	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
148	.table-wrapper {	Code block start hota hai; andar related statements group hote hain.
149	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
150	border-radius: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
151	border: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
152	box-shadow: 0 2px 8px rgba(0, 0, 0, 0.04);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
153	overflow: hidden;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
154	transition: background 0.3s, border-color 0.3s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
155	}	Pichla block/function/object close karta hai.
156		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
157	.courses-page.dark .table-wrapper {	Code block start hota hai; andar related statements group hote hain.
158	background: #14141f;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
159	border-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
160	}	Pichla block/function/object close karta hai.
161		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
162	table {	Code block start hota hai; andar related statements group hote hain.
163	width: 100%;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
164	border-collapse: collapse;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
165	}	Pichla block/function/object close karta hai.
166		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
167	thead {	Code block start hota hai; andar related statements group hote hain.
168	background: #f9fafb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
169	transition: background 0.3s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
170	}	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
171		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
172	<code>.courses-page.dark thead {</code>	Code block start hota hai; andar related statements group hote hain.
173	<code>background: #1a1a28;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
174	<code>}</code>	Pichla block/function/object close karta hai.
175		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
176	<code>th {</code>	Code block start hota hai; andar related statements group hote hain.
177	<code>text-align: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
178	<code>padding: 14px 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
179	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
180	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
181	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
182	<code>text-transform: uppercase;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
183	<code>letter-spacing: 0.5px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
184	<code>border-bottom: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
185	<code>}</code>	Pichla block/function/object close karta hai.
186		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
187	<code>.courses-page.dark th {</code>	Code block start hota hai; andar related statements group hote hain.
188	<code>color: #9ca3af;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
189	<code>border-bottom-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
190	<code>}</code>	Pichla block/function/object close karta hai.
191		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
192	<code>td {</code>	Code block start hota hai; andar related statements group hote hain.
193	<code>padding: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
194	<code>text-align: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
195	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
196	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
197	<code>border-bottom: 1px solid #f3f4f6;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
198	<code>transition: background 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
199	<code>}</code>	Pichla block/function/object close karta hai.
200		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
201	<code>.courses-page.dark td {</code>	Code block start hota hai; andar related statements group hote hain.
202	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
203	<code>border-bottom-color: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
204	<code>}</code>	Pichla block/function/object close karta hai.
205		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
206	<code>tbody tr:last-child td {</code>	Code block start hota hai; andar related statements group hote hain.
207	<code>border-bottom: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
208	<code>}</code>	Pichla block/function/object close karta hai.
209		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
210	<code>tbody tr:hover td {</code>	Code block start hota hai; andar related statements group hote hain.
211	<code>background: #f9fafb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
212	<code>}</code>	Pichla block/function/object close karta hai.
213		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
214	<code>.courses-page.dark tbody tr:hover td {</code>	Code block start hota hai; andar related statements group hote hain.
215	<code>background: #1a1a28;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
216	<code>}</code>	Pichla block/function/object close karta hai.
217		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
218	<code>.name-cell {</code>	Code block start hota hai; andar related statements group hote hain.
219	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
220	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
221	<code>}</code>	Pichla block/function/object close karta hai.
222		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
223	<code>.courses-page.dark .name-cell {</code>	Code block start hota hai; andar related statements group hote hain.
224	<code>color: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
225	<code>}</code>	Pichla block/function/object close karta hai.
226		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
227	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
228	<code>Action Buttons</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
229	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
230	<code>.actions {</code>	Code block start hota hai; andar related statements group hote hain.
231	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
232	<code>gap: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
233	<code>justify-content: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
234	<code>}</code>	Pichla block/function/object close karta hai.
235		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
236	<code>.edit-btn {</code>	Code block start hota hai; andar related statements group hote hain.
237	<code>background: #465fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
238	<code>color: #fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
239	<code>border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
240	<code>padding: 7px 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Lin e	Original code	Simple explanation
241	<code>border-radius: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
242	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
243	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
244	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
245	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
246	<code>}</code>	Pichla block/function/object close karta hai.
247		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
248	<code>.edit-btn:hover {</code>	Code block start hota hai; andar related statements group hote hain.
249	<code>background: #3548d4;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
250	<code>transform: translateY(-1px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
251	<code>}</code>	Pichla block/function/object close karta hai.
252		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
253	<code>.delete-btn {</code>	Code block start hota hai; andar related statements group hote hain.
254	<code>background: #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
255	<code>color: #fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
256	<code>border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
257	<code>padding: 7px 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
258	<code>border-radius: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
259	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
260	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
261	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
262	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
263	<code>}</code>	Pichla block/function/object close karta hai.
264		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
265	<code>.delete-btn:hover {</code>	Code block start hota hai; andar related statements group hote hain.
266	<code>background: #dc2626;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
267	<code>transform: translateY(-1px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
268	<code>}</code>	Pichla block/function/object close karta hai.
269		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
270	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
271	<code>Available Seats — Custom Dropdown</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
272	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
273	<code>.seats-dropdown {</code>	Code block start hota hai; andar related statements group hote hain.
274	<code>position: relative;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
275	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
276		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
277	<code>.seats-btn {</code>	Code block start hota hai; andar related statements group hote hain.
278	<code>width: 100%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
279	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
280	<code>justify-content: space-between;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
281	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
282	<code>padding: 10px 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
283	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
284	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
285	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
286	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
287	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
288	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
289	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
290	<code>text-align: left;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
291	<code>}</code>	Pichla block/function/object close karta hai.
292		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
293	<code>.courses-page.dark .seats-btn {</code>	Code block start hota hai; andar related statements group hote hain.
294	<code>background: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
295	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
296	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
297	<code>}</code>	Pichla block/function/object close karta hai.
298		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
299	<code>.seats-btn:hover,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
300	<code>.seats-btn.active {</code>	Code block start hota hai; andar related statements group hote hain.
301	<code>border-color: #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
302	<code>box-shadow: 0 0 0 3px rgba(16, 185, 129, 0.15);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
303	<code>}</code>	Pichla block/function/object close karta hai.
304		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
305	<code>.seats-btn .arrow {</code>	Code block start hota hai; andar related statements group hote hain.
306	<code>font-size: 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
307	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
308	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
309	<code>}</code>	Pichla block/function/object close karta hai.
310		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
311	<code>.seats-btn.active .arrow {</code>	Code block start hota hai; andar related statements group hote hain.
312	<code>transform: rotate(180deg);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
313	<code>}</code>	Pichla block/function/object close karta hai.
314		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
315	<code>/* Dropdown Menu */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
316	<code>.seats-menu {</code>	Code block start hota hai; andar related statements group hote hain.
317	<code>position: absolute;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
318	<code>top: calc(100% + 6px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
319	<code>left: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
320	<code>right: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
321	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
322	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
323	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
324	<code>box-shadow: 0 10px 30px rgba(0, 0, 0, 0.1);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
325	<code>max-height: 220px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
326	<code>overflow-y: auto;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
327	<code>z-index: 50;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
328	<code>animation: fadeIn 0.15s ease;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
329	<code>}</code>	Pichla block/function/object close karta hai.
330		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
331	<code>.courses-page.dark .seats-menu {</code>	Code block start hota hai; andar related statements group hote hain.
332	<code>background: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
333	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
334	<code>box-shadow: 0 10px 30px rgba(0, 0, 0, 0.5);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
335	<code>}</code>	Pichla block/function/object close karta hai.
336		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
337	<code>@keyframes fadeIn {</code>	Code block start hota hai; andar related statements group hote hain.
338	<code>from { opacity: 0; transform: translateY(-4px); }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
339	<code>to { opacity: 1; transform: translateY(0); }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
340	<code>}</code>	Pichla block/function/object close karta hai.
341		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
342	<code>.seats-item {</code>	Code block start hota hai; andar related statements group hote hain.
343	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
344	<code>justify-content: space-between;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
345	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
346	padding: 10px 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
347	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
348	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
349	transition: 0.15s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
350	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
351	}	Pichla block/function/object close karta hai.
352		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
353	.courses-page.dark .seats-item {	Code block start hota hai; andar related statements group hote hain.
354	color: #e4e4e7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
355	}	Pichla block/function/object close karta hai.
356		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
357	.seats-item:hover {	Code block start hota hai; andar related statements group hote hain.
358	background: #f3f4f6;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
359	color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
360	}	Pichla block/function/object close karta hai.
361		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
362	.courses-page.dark .seats-item:hover {	Code block start hota hai; andar related statements group hote hain.
363	background: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
364	color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
365	}	Pichla block/function/object close karta hai.
366		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
367	.seats-code {	Code block start hota hai; andar related statements group hote hain.
368	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
369	}	Pichla block/function/object close karta hai.
370		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
371	.seats-count {	Code block start hota hai; andar related statements group hote hain.
372	font-size: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
373	color: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
374	background: #f3f4f6;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
375	padding: 3px 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
376	border-radius: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
377	}	Pichla block/function/object close karta hai.
378		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
379	.courses-page.dark .seats-count {	Code block start hota hai; andar related statements group hote hain.
380	background: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
381	color: #9ca3af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
382	}	Pichla block/function/object close karta hai.
383		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
384	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
385	Empty State	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
386	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
387	.empty-state {	Code block start hota hai; andar related statements group hote hain.
388	padding: 60px 20px !important;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
389	}	Pichla block/function/object close karta hai.
390		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
391	.empty-content {	Code block start hota hai; andar related statements group hote hain.
392	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
393	flex-direction: column;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
394	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
395	gap: 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
396	}	Pichla block/function/object close karta hai.
397		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
398	.empty-icon {	Code block start hota hai; andar related statements group hote hain.
399	font-size: 48px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
400	margin-bottom: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
401	opacity: 0.6;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
402	}	Pichla block/function/object close karta hai.
403		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
404	.empty-content h3 {	Code block start hota hai; andar related statements group hote hain.
405	font-size: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
406	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
407	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
408	margin: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
409	}	Pichla block/function/object close karta hai.
410		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
411	.courses-page.dark .empty-content h3 {	Code block start hota hai; andar related statements group hote hain.
412	color: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
413	}	Pichla block/function/object close karta hai.
414		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
415	.empty-content p {	Code block start hota hai; andar related statements group hote hain.

Line	Original code	Simple explanation
416	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
417	color: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
418	margin: 0 0 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
419	}	Pichla block/function/object close karta hai.
420		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
421	.courses-page.dark .empty-content p {	Code block start hota hai; andar related statements group hote hain.
422	color: #9ca3af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
423	}	Pichla block/function/object close karta hai.
424		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
425	.add-first-btn {	Code block start hota hai; andar related statements group hote hain.
426	background: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
427	color: #fff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
428	border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
429	padding: 10px 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
430	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
431	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
432	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
433	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
434	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
435	box-shadow: 0 4px 12px rgba(16, 185, 129, 0.3);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
436	}	Pichla block/function/object close karta hai.
437		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
438	.add-first-btn:hover {	Code block start hota hai; andar related statements group hote hain.
439	background: #059669;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
440	transform: translateY(-2px);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
441	}	Pichla block/function/object close karta hai.
442		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
443	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
444	Modal Popup	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
445	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
446	.modal-overlay {	Code block start hota hai; andar related statements group hote hain.
447	position: fixed;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
448	top: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
449	left: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
450	right: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
451	<code>bottom: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
452	<code>background: rgba(0, 0, 0, 0.6);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
453	<code>backdrop-filter: blur(4px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
454	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
455	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
456	<code>justify-content: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
457	<code>z-index: 1000;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
458	<code>padding: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
459	<code>animation: fadeIn 0.2s ease;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
460	<code>}</code>	Pichla block/function/object close karta hai.
461		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
462	<code>.modal-box {</code>	Code block start hota hai; andar related statements group hote hain.
463	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
464	<code>border-radius: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
465	<code>padding: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
466	<code>max-width: 700px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
467	<code>width: 100%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
468	<code>max-height: 90vh;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
469	<code>overflow-y: auto;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
470	<code>box-shadow: 0 20px 60px rgba(0, 0, 0, 0.3);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
471	<code>animation: slideUp 0.3s ease;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
472	<code>}</code>	Pichla block/function/object close karta hai.
473		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
474	<code>.courses-page.dark .modal-box {</code>	Code block start hota hai; andar related statements group hote hain.
475	<code>background: #14141f;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
476	<code>box-shadow: 0 20px 60px rgba(0, 0, 0, 0.8);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
477	<code>}</code>	Pichla block/function/object close karta hai.
478		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
479	<code>@keyframes slideUp {</code>	Code block start hota hai; andar related statements group hote hain.
480	<code>from { opacity: 0; transform: translateY(20px); }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
481	<code>to { opacity: 1; transform: translateY(0); }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
482	<code>}</code>	Pichla block/function/object close karta hai.
483		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
484	<code>.modal-header {</code>	Code block start hota hai; andar related statements group hote hain.
485	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
486	<code>justify-content: space-between;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
487	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
488	<code>padding: 20px 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
489	<code>border-bottom: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
490	<code>}</code>	Pichla block/function/object close karta hai.
491		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
492	<code>.courses-page.dark .modal-header {</code>	Code block start hota hai; andar related statements group hote hain.
493	<code>border-bottom-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
494	<code>}</code>	Pichla block/function/object close karta hai.
495		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
496	<code>.modal-header h3 {</code>	Code block start hota hai; andar related statements group hote hain.
497	<code>font-size: 18px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
498	<code>font-weight: 700;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
499	<code>margin: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
500	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
501	<code>}</code>	Pichla block/function/object close karta hai.
502		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
503	<code>.courses-page.dark .modal-header h3 {</code>	Code block start hota hai; andar related statements group hote hain.
504	<code>color: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
505	<code>}</code>	Pichla block/function/object close karta hai.
506		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
507	<code>.modal-close {</code>	Code block start hota hai; andar related statements group hote hain.
508	<code>background: transparent;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
509	<code>border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
510	<code>font-size: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
511	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
512	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
513	<code>padding: 4px 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
514	<code>border-radius: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
515	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
516	<code>}</code>	Pichla block/function/object close karta hai.
517		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
518	<code>.modal-close:hover {</code>	Code block start hota hai; andar related statements group hote hain.
519	<code>background: #f3f4f6;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
520	<code>color: #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
521	}	Pichla block/function/object close karta hai.
522		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
523	.courses-page.dark .modal-close:hover {	Code block start hota hai; andar related statements group hote hain.
524	background: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
525	}	Pichla block/function/object close karta hai.
526		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
527	/* Modal Form */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
528	.modal-form {	Code block start hota hai; andar related statements group hote hain.
529	padding: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
530	display: grid;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
531	grid-template-columns: 1fr 1fr;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
532	gap: 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
533	}	Pichla block/function/object close karta hai.
534		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
535	.modal-form .form-group {	Code block start hota hai; andar related statements group hote hain.
536	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
537	flex-direction: column;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
538	gap: 6px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
539	}	Pichla block/function/object close karta hai.
540		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
541	.modal-form .form-group:nth-child(5) {	Code block start hota hai; andar related statements group hote hain.
542	grid-column: 1 / -1; /* Available seats full width */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
543	}	Pichla block/function/object close karta hai.
544		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
545	.modal-form .form-group label {	Code block start hota hai; andar related statements group hote hain.
546	font-size: 13px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
547	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
548	color: #374151;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
549	}	Pichla block/function/object close karta hai.
550		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
551	.courses-page.dark .modal-form .form-group label {	Code block start hota hai; andar related statements group hote hain.
552	color: #d1d5db;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
553	}	Pichla block/function/object close karta hai.
554		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
555	.modal-form input {	Code block start hota hai; andar related statements group hote hain.
556	padding: 10px 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
557	border: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
558	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
559	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
560	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
561	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
562	outline: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
563	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
564	}	Pichla block/function/object close karta hai.
565		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
566	.courses-page.dark .modal-form input {	Code block start hota hai; andar related statements group hote hain.
567	background: #1f1f2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
568	border-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
569	color: #e4e4e7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
570	}	Pichla block/function/object close karta hai.
571		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
572	.modal-form input:focus {	Code block start hota hai; andar related statements group hote hain.
573	border-color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
574	box-shadow: 0 0 0 3px rgba(16, 185, 129, 0.15);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
575	}	Pichla block/function/object close karta hai.
576		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
577	.modal-actions {	Code block start hota hai; andar related statements group hote hain.
578	grid-column: 1 / -1;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
579	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
580	gap: 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
581	justify-content: flex-end;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
582	margin-top: 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
583	padding-top: 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
584	border-top: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
585	}	Pichla block/function/object close karta hai.
586		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
587	.courses-page.dark .modal-actions {	Code block start hota hai; andar related statements group hote hain.
588	border-top-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
589	}	Pichla block/function/object close karta hai.
590		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
591		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
592	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
593	<code>Responsive</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
594	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
595	<code>@media (max-width: 600px) {</code>	Responsive CSS rule: screen width/device condition par styles apply karta hai.
596	<code>.modal-form {</code>	Code block start hota hai; andar related statements group hote hain.
597	<code>grid-template-columns: 1fr;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
598	<code>}</code>	Pichla block/function/object close karta hai.
599	<code>}</code>	Pichla block/function/object close karta hai.
600	<code>@media (max-width: 768px) {</code>	Responsive CSS rule: screen width/device condition par styles apply karta hai.
601	<code>.courses-page {</code>	Code block start hota hai; andar related statements group hote hain.
602	<code>padding: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
603	<code>}</code>	Pichla block/function/object close karta hai.
604		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
605	<code>.page-header {</code>	Code block start hota hai; andar related statements group hote hain.
606	<code>flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
607	<code>gap: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
608	<code>align-items: stretch;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
609	<code>}</code>	Pichla block/function/object close karta hai.
610		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
611	<code>.course-form {</code>	Code block start hota hai; andar related statements group hote hain.
612	<code>grid-template-columns: 1fr;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
613	<code>}</code>	Pichla block/function/object close karta hai.
614		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
615	<code>.table-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
616	<code>overflow-x: auto;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
617	<code>}</code>	Pichla block/function/object close karta hai.
618		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
619	<code>table {</code>	Code block start hota hai; andar related statements group hote hain.
620	<code>min-width: 700px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
621	<code>}</code>	Pichla block/function/object close karta hai.
622	<code>}</code>	Pichla block/function/object close karta hai.

src/app/pages/course/course.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Line	Original code	Simple explanation
1	<code><div class="courses-page" [class.dark]="themeService.isDarkMode()"></code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
2		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
3	<code><!-- Page Header --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
4	<code><div class="page-header"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
5	<code><h2>Courses</h2></code>	HTML element/component: page ka structure ya UI control banata hai.
6		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
7	<code><!-- ■ Button sirf tab dikhe jab courses hain --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
8	<code>@if (courses().length > 0) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
9	<code><button class="add-new-btn" (click)="openModal()"></code>	User click par specified TypeScript method/expression chalti hai.
10	<code>+ Add Course</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
11	<code></button></code>	HTML element/component: page ka structure ya UI control banata hai.
12	<code>}</code>	Pichla block/function/object close karta hai.
13	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
14		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
15	<code><!-- Table --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
16	<code><div class="table-wrapper"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
17	<code><table></code>	HTML element/component: page ka structure ya UI control banata hai.
18	<code><thead></code>	HTML element/component: page ka structure ya UI control banata hai.
19	<code><tr></code>	HTML element/component: page ka structure ya UI control banata hai.
20	<code><th>Course Name</th></code>	HTML element/component: page ka structure ya UI control banata hai.
21	<code><th>Course Description</th></code>	HTML element/component: page ka structure ya UI control banata hai.
22	<code><th>Course Duration</th></code>	HTML element/component: page ka structure ya UI control banata hai.
23	<code><th>Available Seats</th></code>	HTML element/component: page ka structure ya UI control banata hai.
24	<code><th>Instructor</th></code>	HTML element/component: page ka structure ya UI control banata hai.
25	<code><th>Actions</th></code>	HTML element/component: page ka structure ya UI control banata hai.
26	<code></tr></code>	HTML element/component: page ka structure ya UI control banata hai.
27	<code></thead></code>	HTML element/component: page ka structure ya UI control banata hai.
28	<code><tbody></code>	HTML element/component: page ka structure ya UI control banata hai.
29	<code>@for (course of courses(); track course.id) {</code>	Angular loop: array ke har item ke liye HTML repeat karta hai.
30	<code><tr></code>	HTML element/component: page ka structure ya UI control banata hai.
31	<code><td class="name-cell">{{ course.course_name }}</td></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
32	<code><td>{{ course.course_description }}</td></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
33	<code><td>{{ course.course_duration }}</td></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
34	<code><td>{{ course.available_seats }}</td></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
35	<code><td>{{ course.instructor }}</td></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
36	<code><td></code>	HTML element/component: page ka structure ya UI control banata hai.

Line	Original code	Simple explanation
37	<code><div class="actions"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
38	<code><button class="edit-btn" (click)="editCourse(course)">Edit</button></code>	User click par specified TypeScript method/expression chalti hai.
39	<code><button class="delete-btn" (click)="deleteCourse(course.id)">Delete</button></code>	User click par specified TypeScript method/expression chalti hai.
40	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
41	<code></td></code>	HTML element/component: page ka structure ya UI control banata hai.
42	<code></tr></code>	HTML element/component: page ka structure ya UI control banata hai.
43	<code>}</code>	Pichla block/function/object close karta hai.
44		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
45	<code><!-- ■ Empty State --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
46	<code>@if (courses().length === 0) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
47	<code><tr></code>	HTML element/component: page ka structure ya UI control banata hai.
48	<code><td colspan="6" class="empty-state"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
49	<code><div class="empty-content"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
50	<code><div class="empty-icon">■</div></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
51	<code><h3>No Courses Yet</h3></code>	HTML element/component: page ka structure ya UI control banata hai.
52	<code><p>Click the button below to add your first course</p></code>	HTML element/component: page ka structure ya UI control banata hai.
53	<code><button class="add-first-btn" (click)="openModal()"></code>	User click par specified TypeScript method/expression chalti hai.
54	<code>+ Add First Course</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
55	<code></button></code>	HTML element/component: page ka structure ya UI control banata hai.
56	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
57	<code></td></code>	HTML element/component: page ka structure ya UI control banata hai.
58	<code></tr></code>	HTML element/component: page ka structure ya UI control banata hai.
59	<code>}</code>	Pichla block/function/object close karta hai.
60	<code></tbody></code>	HTML element/component: page ka structure ya UI control banata hai.
61	<code></table></code>	HTML element/component: page ka structure ya UI control banata hai.
62	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
63		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
64	<code><!-- ■ Modal Popup --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
65	<code>@if (showModal()) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
66	<code><div class="modal-overlay" (click)="closeModal()"></code>	User click par specified TypeScript method/expression chalti hai.
67	<code><div class="modal-box" (click)="\$event.stopPropagation()"></code>	User click par specified TypeScript method/expression chalti hai.
68		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
69	<code><div class="modal-header"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
70	<code><h3>{{ selectedCourseId() === null ? 'Add New Course' : 'Update Course' }}</h3></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.

Line	Original code	Simple explanation
71	<code><button class="modal-close" (click)="closeModal()">X</button></code>	User click par specified TypeScript method/expression chalti hai.
72	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
73		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
74	<code><form [formGroup]="courseForm" (ngSubmit)="saveCourse()" class="modal-form"></code>	Form submit par specified handler chalata hai.
75		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
76	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
77	<code><label>Course Name</label></code>	HTML element/component: page ka structure ya UI control banata hai.
78	<code><input placeholder="Enter course name" formControlName="course_name" /></code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
79	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
80		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
81	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
82	<code><label>Course Description</label></code>	HTML element/component: page ka structure ya UI control banata hai.
83	<code><input placeholder="Enter description" formControlName="course_description" /></code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
84	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
85		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
86	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
87	<code><label>Course Duration</label></code>	HTML element/component: page ka structure ya UI control banata hai.
88	<code><input placeholder="e.g. 3 months" formControlName="course_duration" /></code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
89	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
90		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
91	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
92	<code><label>Instructor</label></code>	HTML element/component: page ka structure ya UI control banata hai.
93	<code><input placeholder="Enter instructor" formControlName="instructor" /></code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
94	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
95		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
96	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
97	<code><label>Available Seats</label></code>	HTML element/component: page ka structure ya UI control banata hai.
98	<code><div class="seats-dropdown"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
99	<code><button type="button" class="seats-btn"</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
100	<code>(click)="toggleSeatsDropdown()"</code>	User click par specified TypeScript method/expression chalti hai.
101	<code>[class.active]="showSeatsDropdown()"></code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
102	<code>{{ selectedSeatsLabel() 'Select Course Seats' }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
103	<code>▼</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.

Line	Original code	Simple explanation
104	</button>	HTML element/component: page ka structure ya UI control banata hai.
105		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
106	@if (showSeatsDropdown()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
107	<div class="seats-menu">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
108	@for (option of seatsOptions; track option.code) {	Angular loop: array ke har item ke liye HTML repeat karta hai.
109	<div class="seats-item" (click)="selectSeats(option)">	User click par specified TypeScript method/expression chalti hai.
110	{{ option.code }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
111	{{ option.seats }} seats	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
112	</div>	HTML element/component: page ka structure ya UI control banata hai.
113	}	Pichla block/function/object close karta hai.
114	</div>	HTML element/component: page ka structure ya UI control banata hai.
115	}	Pichla block/function/object close karta hai.
116	</div>	HTML element/component: page ka structure ya UI control banata hai.
117	</div>	HTML element/component: page ka structure ya UI control banata hai.
118	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
119	<label>Course Fee *</label>	HTML element/component: page ka structure ya UI control banata hai.
120	<input	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
121	type="number"	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
122	placeholder="e.g. 10000"	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
123	formControlName="course_fee"	HTML input ko TS reactive form ke same-named control se connect karta hai.
124	[class.error]="invalid('course_fee')"	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
125	/>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
126	@if (invalid('course_fee')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
127	■ Course fee is required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
128	}	Pichla block/function/object close karta hai.
129	</div>	HTML element/component: page ka structure ya UI control banata hai.
130		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
131	<div class="modal-actions">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
132	<button type="button" class="cancel-btn" (click)="closeModal()">Cancel</button>	User click par specified TypeScript method/expression chalti hai.
133	<button type="submit" class="save-btn">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
134	{{ selectedCourseId() === null ? 'Add Course' : 'Update Course' }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
135	</button>	HTML element/component: page ka structure ya UI control banata hai.
136	</div>	HTML element/component: page ka structure ya UI control banata hai.
137		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
138		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
139	</form>	HTML element/component: page ka structure ya UI control banata hai.
140	</div>	HTML element/component: page ka structure ya UI control banata hai.
141	</div>	HTML element/component: page ka structure ya UI control banata hai.
142	}	Pichla block/function/object close karta hai.
143	<!-- ■ Toast Notification -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
144	@if (showToast()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
145	<div class="toast-popup">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
146	■	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
147	{{ toastMessage() }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
148	<button class="toast-close" (click)="showToast.set(false)">X</button>	User click par specified TypeScript method/expression chalti hai.
149	</div>	HTML element/component: page ka structure ya UI control banata hai.
150		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
151		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
152	}	Pichla block/function/object close karta hai.
153		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
154		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
155	</div>	HTML element/component: page ka structure ya UI control banata hai.

src/app/pages/course/course.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	import { CommonModule } from '@angular/common';	Import: doosri library/file se required class ya function is file mein available karta hai.
2	import { Component, OnInit, inject, signal } from '@angular/core';	Import: doosri library/file se required class ya function is file mein available karta hai.
3	import { FormBuilder, ReactiveFormsModule, Validators } from '@angular/forms';	Import: doosri library/file se required class ya function is file mein available karta hai.
4	import { CoursesService } from '../../../services/courses.service';	Import: doosri library/file se required class ya function is file mein available karta hai.
5	import { ThemeService } from '../../../services/theme';	Import: doosri library/file se required class ya function is file mein available karta hai.
6		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
7	@Component({	Angular decorator: neech wali class ko Angular component declare karta hai.
8	selector: 'app-course',	Component ka HTML tag name; parent template mein isi selector se component lagta hai.
9	standalone: true,	Component standalone hai; NgModule mein declare karna zaroori nahi.
10	imports: [CommonModule, ReactiveFormsModule],	Template mein use hone wale Angular directives/forms/router features import karta hai.
11	templateUrl: './course.html',	Is component ki HTML template file ka path.
12	styleUrl: './course.css',	Is component ki CSS file ka path.
13	})	Pichla block/function/object close karta hai.
14	export class Course implements OnInit {	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
15		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
16	<code>showModal = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
17		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
18	<code>private fb = inject(FormBuilder);</code>	FormBuilder leta hai taake reactive form aur controls bana sake.
19	<code>private courseService = inject(CoursesService);</code>	Class property/method ko class ke andar se access karne ke liye mark karta hai.
20	<code>public themeService = inject(ThemeService);</code>	Shared ThemeService use hoti hai taake theme state/component access milay.
21		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
22	<code>courses = signal<any[]>([]);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
23	<code>selectedCourseId = signal<number null>(null);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
24		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
25	<code>showSeatsDropdown = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
26	<code>selectedSeatsLabel = signal<string>('');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
27		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
28	<code>// ■ Toast signals</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
29	<code>showToast = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
30	<code>toastMessage = signal<string>('');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
31	<code>toastType = signal<'error' 'success'>('error');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
32		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
33	<code>seatsOptions = [</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
34	<code>{ code: 'CS', seats: 20 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
35	<code>{ code: 'MLT', seats: 15 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
36	<code>{ code: 'Angular', seats: 20 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
37	<code>{ code: 'React', seats: 15 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
38	<code>{ code: 'Python', seats: 25 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
39	<code>{ code: 'Java', seats: 22 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
40	<code>{ code: 'Node', seats: 18 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
41	<code>{ code: 'UI/UX', seats: 12 }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
42	<code>];</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
43		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
44	<code>// ■ Sab fields required hain</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
45	<code>courseForm = this.fb.group({</code>	Reactive form group banata hai; fields, current values aur validation ek jagah manage hotay hain.
46	<code>course_name: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
47	<code>course_description: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
48	<code>course_duration: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.

Line	Original code	Simple explanation
49	<code>instructor: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
50	<code>available_seats: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
51	<code>course_fee: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
52	<code>});</code>	Pichla block/function/object close karta hai.
53		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
54	<code>ngOnInit() {</code>	Component initialize hone ke baad Angular is method ko automatically call karta hai.
55	<code> this.loadCourses();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
56	<code>}</code>	Pichla block/function/object close karta hai.
57		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
58	<code>// ■ Toast helper</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
59	<code>showToastMsg(msg: string, type: 'error' 'success' = 'error') {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
60	<code> this.toastMessage.set(msg);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
61	<code> this.toastType.set(type);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
62	<code> this.showToast.set(true);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
63	<code> setTimeout(() => this.showToast.set(false), 3000);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
64	<code>}</code>	Pichla block/function/object close karta hai.
65		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
66	<code>openModal() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
67	<code> this.selectedCourseId.set(null);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
68	<code> this.courseForm.reset();</code>	Form ko clear/default values par reset karta hai.
69	<code> this.selectedSeatsLabel.set('');</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
70	<code> this.showModal.set(true);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
71	<code>}</code>	Pichla block/function/object close karta hai.
72		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
73	<code>closeModal() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
74	<code> this.showModal.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
75	<code> this.selectedCourseId.set(null);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
76	<code> this.courseForm.reset();</code>	Form ko clear/default values par reset karta hai.
77	<code> this.selectedSeatsLabel.set('');</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
78	<code> this.showSeatsDropdown.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
79	<code>}</code>	Pichla block/function/object close karta hai.
80		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
81	<code>toggleSeatsDropdown() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
82	<code> this.showSeatsDropdown.update(v => !v);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
83	<code>}</code>	Pichla block/function/object close karta hai.
84		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
85	<code>selectSeats(option: { code: string; seats: number }) {</code>	Code block start hota hai; andar related statements group hote hain.
86	<code> this.selectedSeatsLabel.set(`\${option.code} – \${option.seats} seats`);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
87	<code> this.courseForm.patchValue({</code>	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
88	<code> available_seats: option.seats.toString()</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
89	<code> });</code>	Pichla block/function/object close karta hai.
90	<code> this.showSeatsDropdown.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
91	<code>}</code>	Pichla block/function/object close karta hai.
92		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
93	<code>loadCourses() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
94	<code> this.courseService.getCourses().subscribe({</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
95	<code> next: (data) => this.courses.set(data),</code>	Successful response ka handler; received data par UI/state update hoti hai.
96	<code> error: (err) => console.error('Load error:', err),</code>	Request fail ho to error handler chalta hai; debugging/error feedback ke liye.
97	<code> });</code>	Pichla block/function/object close karta hai.
98	<code>}</code>	Pichla block/function/object close karta hai.
99		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
100	<code>// ■ Updated saveCourse with validation + toast</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
101	<code>saveCourse() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
102	<code> if (this.courseForm.invalid) {</code>	Check karta hai ke form/field validation pass kar raha hai ya nahi.
103	<code> this.courseForm.markAllAsTouched();</code>	Validation messages dikhane ke liye sab controls ko touched mark karta hai.
104	<code> this.showToastMsg('■ Please fill all required fields', 'error');</code>	User ko success/error ka short notification dikhata hai.
105	<code> return;</code>	Current function ko yahin stop karta hai.
106	<code> }</code>	Pichla block/function/object close karta hai.
107		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
108	<code> if (this.selectedCourseId() === null) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
109	<code> this.courseService.addCourse(this.courseForm.value).subscribe(() => {</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
110	<code> this.showToastMsg('■ Course added successfully!', 'success');</code>	User ko success/error ka short notification dikhata hai.
111	<code> this.closeModal();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
112	<code> this.loadCourses();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
113	<code> });</code>	Pichla block/function/object close karta hai.
114	<code> } else {</code>	Alternative branch: pehli condition false ho to yeh code chalta hai.
115	<code> this.courseService</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
116	<code> .updateCourse(this.selectedCourseId()!, this.courseForm.value)</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
117	<code> .subscribe(() => {</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
118	<code> this.showToastMsg('■ Course updated successfully!', 'success');</code>	User ko success/error ka short notification dikhata hai.
119	<code> this.closeModal();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
120	<code> this.loadCourses();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
121	<code> });</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
122	}	Pichla block/function/object close karta hai.
123	}	Pichla block/function/object close karta hai.
124		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
125	editCourse(course: any) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
126	this.selectedCourseId.set(course.id);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
127	this.courseForm.setValue({	Form ke tamam controls ko values deta hai; required controls miss hon to error ho sakta hai.
128	course_name: course.course_name,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
129	course_description: course.course_description,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
130	course_duration: course.course_duration,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
131	instructor: course.instructor,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
132	available_seats: course.available_seats,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
133	course_fee: course.course_fee,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
134	});	Pichla block/function/object close karta hai.
135	this.selectedSeatsLabel.set(`Seats: \${course.available_seats}`);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
136	this.showModal.set(true);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
137	}	Pichla block/function/object close karta hai.
138		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
139	deleteCourse(id: number) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
140	this.courseService.deleteCourse(id).subscribe(() => {	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
141	this.showToastMsg('■ Course deleted', 'success');	User ko success/error ka short notification dikhati hai.
142	this.loadCourses();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
143	});	Pichla block/function/object close karta hai.
144	}	Pichla block/function/object close karta hai.
145		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
146	// ■ Field invalid check	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
147	isValid(field: string): boolean {	Code block start hota hai; andar related statements group hote hain.
148	const control = this.courseForm.get(field);	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
149	return !(control && control.invalid && (control.dirty control.touched));	Check karta hai ke form/field validation pass kar raha hai ya nahi.
150	}	Pichla block/function/object close karta hai.
151	}	Pichla block/function/object close karta hai.

src/app/pages/dashboard/dashboard.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
2	Dashboard (Default = LIGHT)	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
3	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
4	<code>.dashboard {</code>	Code block start hota hai; andar related statements group hote hain.
5	<code>padding: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
6	<code>background: #f5f6fa;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
7	<code>min-height: 100vh;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
8	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
9	<code>flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
10	<code>gap: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
11	<code>font-family: 'Inter', sans-serif;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
12	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
13	<code>box-sizing: border-box;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
14	<code>transition: background 0.3s, color 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
15	<code>}</code>	Pichla block/function/object close karta hai.
16		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
17	<code>.dashboard.dark {</code>	Code block start hota hai; andar related statements group hote hain.
18	<code>background: #0f0f1a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
19	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
20	<code>}</code>	Pichla block/function/object close karta hai.
21		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
22	<code>.top-row {</code>	Code block start hota hai; andar related statements group hote hain.
23	<code>display: grid;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
24	<code>grid-template-columns: 1fr 1.3fr;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
25	<code>gap: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
26	<code>}</code>	Pichla block/function/object close karta hai.
27		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
28	<code>.metrics-column {</code>	Code block start hota hai; andar related statements group hote hain.
29	<code>display: grid;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
30	<code>grid-template-columns: 1fr 1fr;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
31	<code>gap: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
32	<code>}</code>	Pichla block/function/object close karta hai.
33		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
34	<code>/* Metric Cards */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
35	<code>.metric-card {</code>	Code block start hota hai; andar related statements group hote hain.
36	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
37	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
38	<code>border-radius: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
39	padding: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
40	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
41	flex-direction: column;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
42	gap: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
43	transition: 0.3s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
44	}	Pichla block/function/object close karta hai.
45		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
46	.dashboard.dark .metric-card {	Code block start hota hai; andar related statements group hote hain.
47	background: #14141f;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
48	border-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
49	}	Pichla block/function/object close karta hai.
50		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
51	.metric-card:hover {	Code block start hota hai; andar related statements group hote hain.
52	transform: translateY(-4px);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
53	border-color: #465fff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
54	box-shadow: 0 10px 30px rgba(70, 95, 255, 0.15);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
55	}	Pichla block/function/object close karta hai.
56		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
57	.metric-icon {	Code block start hota hai; andar related statements group hote hain.
58	width: 44px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
59	height: 44px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
60	border-radius: 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
61	background: #eef2ff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
62	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
63	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
64	justify-content: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
65	font-size: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
66	}	Pichla block/function/object close karta hai.
67		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
68	.dashboard.dark .metric-icon {	Code block start hota hai; andar related statements group hote hain.
69	background: #1e1e3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
70	}	Pichla block/function/object close karta hai.
71		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
72	.metric-label {	Code block start hota hai; andar related statements group hote hain.
73	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
74	color: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
75	}	Pichla block/function/object close karta hai.
76		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
77	.dashboard.dark .metric-label {	Code block start hota hai; andar related statements group hote hain.
78	color: #9ca3af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
79	}	Pichla block/function/object close karta hai.
80		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
81	.metric-bottom {	Code block start hota hai; andar related statements group hote hain.
82	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
83	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
84	justify-content: space-between;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
85	gap: 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
86	}	Pichla block/function/object close karta hai.
87		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
88	.metric-value {	Code block start hota hai; andar related statements group hote hain.
89	font-size: 28px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
90	font-weight: 700;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
91	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
92	letter-spacing: -0.5px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
93	}	Pichla block/function/object close karta hai.
94		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
95	.dashboard.dark .metric-value {	Code block start hota hai; andar related statements group hote hain.
96	color: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
97	}	Pichla block/function/object close karta hai.
98		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
99	.badge {	Code block start hota hai; andar related statements group hote hain.
100	font-size: 11px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
101	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
102	padding: 4px 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
103	border-radius: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
104	white-space: nowrap;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
105	}	Pichla block/function/object close karta hai.
106		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
107	.badge.positive {	Code block start hota hai; andar related statements group hote hain.
108	background: #d1fae5;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
109	color: #065f46;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
110	}	Pichla block/function/object close karta hai.
111		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
112	.dashboard.dark .badge.positive {	Code block start hota hai; andar related statements group hote hain.
113	background: #064e3b;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
114	color: #6ee7b7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
115	}	Pichla block/function/object close karta hai.
116		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
117	.badge.negative {	Code block start hota hai; andar related statements group hote hain.
118	background: #fee2e2;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
119	color: #991b1b;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
120	}	Pichla block/function/object close karta hai.
121		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
122	.dashboard.dark .badge.negative {	Code block start hota hai; andar related statements group hote hain.
123	background: #7f1d1d;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
124	color: #fca5a5;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
125	}	Pichla block/function/object close karta hai.
126		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
127	/* Target Card */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
128	.target-card {	Code block start hota hai; andar related statements group hote hain.
129	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
130	border: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
131	border-radius: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
132	padding: 22px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
133	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
134	flex-direction: column;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
135	gap: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
136	}	Pichla block/function/object close karta hai.
137		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
138	.dashboard.dark .target-card {	Code block start hota hai; andar related statements group hote hain.
139	background: #14141f;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
140	border-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
141	}	Pichla block/function/object close karta hai.
142		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
143	.card-header {	Code block start hota hai; andar related statements group hote hain.

Line	Original code	Simple explanation
144	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
145	justify-content: space-between;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
146	align-items: flex-start;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
147	}	Pichla block/function/object close karta hai.
148		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
149	.card-header h3 {	Code block start hota hai; andar related statements group hote hain.
150	font-size: 18px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
151	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
152	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
153	margin: 0 0 4px 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
154	}	Pichla block/function/object close karta hai.
155		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
156	.dashboard.dark .card-header h3 {	Code block start hota hai; andar related statements group hote hain.
157	color: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
158	}	Pichla block/function/object close karta hai.
159		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
160	.card-header p {	Code block start hota hai; andar related statements group hote hain.
161	font-size: 13px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
162	color: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
163	margin: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
164	}	Pichla block/function/object close karta hai.
165		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
166	.dashboard.dark .card-header p {	Code block start hota hai; andar related statements group hote hain.
167	color: #9ca3af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
168	}	Pichla block/function/object close karta hai.
169		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
170	.more-btn {	Code block start hota hai; andar related statements group hote hain.
171	background: transparent;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
172	border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
173	font-size: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
174	color: #9ca3af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
175	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
176	padding: 0 6px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
177	}	Pichla block/function/object close karta hai.
178		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
179	<code>/* Gauge */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
180	<code>.gauge-wrap {</code>	Code block start hota hai; andar related statements group hote hain.
181	<code>position: relative;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
182	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
183	<code>justify-content: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
184	<code>margin: 6px 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
185	<code>}</code>	Pichla block/function/object close karta hai.
186		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
187	<code>.gauge-svg {</code>	Code block start hota hai; andar related statements group hote hain.
188	<code>width: 240px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
189	<code>height: 130px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
190	<code>}</code>	Pichla block/function/object close karta hai.
191		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
192	<code>.gauge-bg {</code>	Code block start hota hai; andar related statements group hote hain.
193	<code>stroke: #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
194	<code>}</code>	Pichla block/function/object close karta hai.
195		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
196	<code>.dashboard.dark .gauge-bg {</code>	Code block start hota hai; andar related statements group hote hain.
197	<code>stroke: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
198	<code>}</code>	Pichla block/function/object close karta hai.
199		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
200	<code>.gauge-text {</code>	Code block start hota hai; andar related statements group hote hain.
201	<code>position: absolute;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
202	<code>bottom: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
203	<code>left: 50%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
204	<code>transform: translateX(-50%);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
205	<code>text-align: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
206	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
207	<code>flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
208	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
209	<code>gap: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
210	<code>}</code>	Pichla block/function/object close karta hai.
211		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
212	<code>.gauge-value {</code>	Code block start hota hai; andar related statements group hote hain.
213	<code>font-size: 30px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
214	font-weight: 700;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
215	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
216	line-height: 1;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
217	}	Pichla block/function/object close karta hai.
218		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
219	.dashboard.dark .gauge-value {	Code block start hota hai; andar related statements group hote hain.
220	color: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
221	}	Pichla block/function/object close karta hai.
222		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
223	.gauge-badge {	Code block start hota hai; andar related statements group hote hain.
224	background: #d1fae5;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
225	color: #065f46;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
226	font-size: 11px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
227	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
228	padding: 3px 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
229	border-radius: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
230	}	Pichla block/function/object close karta hai.
231		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
232	.dashboard.dark .gauge-badge {	Code block start hota hai; andar related statements group hote hain.
233	background: #064e3b;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
234	color: #6ee7b7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
235	}	Pichla block/function/object close karta hai.
236		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
237	.target-desc {	Code block start hota hai; andar related statements group hote hain.
238	font-size: 13px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
239	color: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
240	text-align: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
241	line-height: 1.5;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
242	margin: 6px 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
243	}	Pichla block/function/object close karta hai.
244		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
245	.dashboard.dark .target-desc {	Code block start hota hai; andar related statements group hote hain.
246	color: #9ca3af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
247	}	Pichla block/function/object close karta hai.
248		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
249	<code>.target-stats {</code>	Code block start hota hai; andar related statements group hote hain.
250	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
251	<code>justify-content: space-around;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
252	<code>border-top: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
253	<code>padding-top: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
254	<code>}</code>	Pichla block/function/object close karta hai.
255		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
256	<code>.dashboard.dark .target-stats {</code>	Code block start hota hai; andar related statements group hote hain.
257	<code>border-top-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
258	<code>}</code>	Pichla block/function/object close karta hai.
259		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
260	<code>.stat {</code>	Code block start hota hai; andar related statements group hote hain.
261	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
262	<code>flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
263	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
264	<code>gap: 4px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
265	<code>}</code>	Pichla block/function/object close karta hai.
266		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
267	<code>.stat span {</code>	Code block start hota hai; andar related statements group hote hain.
268	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
269	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
270	<code>}</code>	Pichla block/function/object close karta hai.
271		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
272	<code>.dashboard.dark .stat span {</code>	Code block start hota hai; andar related statements group hote hain.
273	<code>color: #9ca3af;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
274	<code>}</code>	Pichla block/function/object close karta hai.
275		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
276	<code>.stat strong {</code>	Code block start hota hai; andar related statements group hote hain.
277	<code>font-size: 15px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
278	<code>font-weight: 700;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
279	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
280	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
281	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
282	<code>gap: 4px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
283	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
284		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
285	<code>.dashboard.dark .stat strong {</code>	Code block start hota hai; andar related statements group hote hain.
286	<code>color: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
287	<code>}</code>	Pichla block/function/object close karta hai.
288		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
289	<code>.up { color: #10b981; font-style: normal; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
290	<code>.down { color: #ef4444; font-style: normal; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
291		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
292	<code>/* Sales Chart */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
293	<code>.sales-card {</code>	Code block start hota hai; andar related statements group hote hain.
294	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
295	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
296	<code>border-radius: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
297	<code>padding: 22px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
298	<code>}</code>	Pichla block/function/object close karta hai.
299		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
300	<code>.dashboard.dark .sales-card {</code>	Code block start hota hai; andar related statements group hote hain.
301	<code>background: #14141f;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
302	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
303	<code>}</code>	Pichla block/function/object close karta hai.
304		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
305	<code>.bar-chart {</code>	Code block start hota hai; andar related statements group hote hain.
306	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
307	<code>gap: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
308	<code>margin-top: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
309	<code>height: 260px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
310	<code>}</code>	Pichla block/function/object close karta hai.
311		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
312	<code>.y-axis {</code>	Code block start hota hai; andar related statements group hote hain.
313	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
314	<code>flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
315	<code>justify-content: space-between;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
316	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
317	<code>color: #9ca3af;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
318	<code>padding-bottom: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
319	<code>text-align: right;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
320	<code>min-width: 30px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
321	<code>}</code>	Pichla block/function/object close karta hai.
322		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
323	<code>.bars-area {</code>	Code block start hota hai; andar related statements group hote hain.
324	<code>flex: 1;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
325	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
326	<code>align-items: flex-end;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
327	<code>justify-content: space-around;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
328	<code>gap: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
329	<code>border-bottom: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
330	<code>position: relative;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
331	<code>}</code>	Pichla block/function/object close karta hai.
332		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
333	<code>.dashboard.dark .bars-area {</code>	Code block start hota hai; andar related statements group hote hain.
334	<code>border-bottom-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
335	<code>}</code>	Pichla block/function/object close karta hai.
336		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
337	<code>.bar-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
338	<code>flex: 1;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
339	<code>height: 100%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
340	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
341	<code>flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
342	<code>justify-content: flex-end;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
343	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
344	<code>position: relative;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
345	<code>}</code>	Pichla block/function/object close karta hai.
346		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
347	<code>.bar {</code>	Code block start hota hai; andar related statements group hote hain.
348	<code>width: 65%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
349	<code>background: linear-gradient(180deg, #465fff, #3548d4);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
350	<code>border-radius: 5px 5px 0 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
351	<code>transition: 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
352	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
353	<code>min-height: 4px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
354	}	Pichla block/function/object close karta hai.
355		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
356	.bar:hover {	Code block start hota hai; andar related statements group hote hain.
357	background: linear-gradient(180deg, #7592ff, #465fff);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
358	transform: scaleY(1.03);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
359	transform-origin: bottom;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
360	}	Pichla block/function/object close karta hai.
361		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
362	.bar-label {	Code block start hota hai; andar related statements group hote hain.
363	font-size: 11px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
364	color: #9ca3af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
365	position: absolute;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
366	bottom: -22px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
367	}	Pichla block/function/object close karta hai.
368		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
369	/* Responsive */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
370	@media (max-width: 1000px) {	Responsive CSS rule: screen width/device condition par styles apply karta hai.
371	.top-row { grid-template-columns: 1fr; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
372	}	Pichla block/function/object close karta hai.
373		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
374	@media (max-width: 600px) {	Responsive CSS rule: screen width/device condition par styles apply karta hai.
375	.metrics-column { grid-template-columns: 1fr; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
376	.dashboard { padding: 16px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
377	.bar-chart { height: 200px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
378	.gauge-svg { width: 200px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
379	}	Pichla block/function/object close karta hai.

src/app/pages/dashboard/dashboard.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Line	Original code	Simple explanation
1	<div class="dashboard" [class.dark]="themeService.isDarkMode()">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
2		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
3	<!-- Top Row -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
4	<div class="top-row">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
5		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
6	<!-- Metric Cards -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.

Line	Original code	Simple explanation
7	<code><div class="metrics-column"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
8		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
9	<code><div class="metric-card"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
10	<code><div class="metric-icon">■</div></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
11	<code><div class="metric-label">Total Students</div></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
12	<code><div class="metric-bottom"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
13	<code>{{ totalStudents() }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
14	<code>↑ 11.01%</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
15	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
16	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
17		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
18	<code><div class="metric-card"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
19	<code><div class="metric-icon">■</div></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
20	<code><div class="metric-label">Total Courses</div></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
21	<code><div class="metric-bottom"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
22	<code>{{ totalCourses() }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
23	<code>↓ 9.05%</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
24	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
25	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
26		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
27	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
28		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
29	<code><!-- Enrollment Target --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
30	<code><div class="target-card"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
31	<code><div class="card-header"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
32	<code><div></code>	HTML element/component: page ka structure ya UI control banata hai.
33	<code><h3>Enrollment Target</h3></code>	HTML element/component: page ka structure ya UI control banata hai.
34	<code><p>Target you've set for each month</p></code>	HTML element/component: page ka structure ya UI control banata hai.
35	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
36	<code><button class="more-btn">■</button></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
37	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
38		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
39	<code><div class="gauge-wrap"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.

Line	Original code	Simple explanation
40	<code><svg viewBox="0 0 220 130" class="gauge-svg"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
41	<code><path d="M 20 110 A 90 90 0 0 1 200 110"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
42	<code>fill="none" stroke="#2a2a3a" stroke-width="14" stroke-linecap="round"/></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
43	<code><path d="M 20 110 A 90 90 0 0 1 200 110"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
44	<code>fill="none" stroke="#465fff" stroke-width="14" stroke-linecap="round"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
45	<code>[attr.stroke-dasharray]="283"</code>	Property/attribute binding: HTML property ko TS value se connect karta hai.
46	<code>[attr.stroke-dashoffset]="283 - (283 * monthlyTarget() / 100)" /></code>	Property/attribute binding: HTML property ko TS value se connect karta hai.
47	<code></svg></code>	HTML element/component: page ka structure ya UI control banata hai.
48	<code><div class="gauge-text"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
49	<code><div class="gauge-value">{{ monthlyTarget() }}%</div></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
50	<code><div class="gauge-badge">+{{ targetGrowth() }}%</div></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
51	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
52	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
53		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
54	<code><p class="target-desc"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
55	<code>{{ totalEnrollments() }} students enrolled so far. Keep up your good work!</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
56	<code></p></code>	HTML element/component: page ka structure ya UI control banata hai.
57		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
58	<code><div class="target-stats"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
59	<code><div class="stat"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
60	<code>Enrolled</code>	HTML element/component: page ka structure ya UI control banata hai.
61	<code>{{ totalEnrollments() }} <em class="up">↑</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
62	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
63	<code><div class="stat"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
64	<code>Courses</code>	HTML element/component: page ka structure ya UI control banata hai.
65	<code>{{ totalCourses() }} <em class="up">↑</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
66	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
67	<code><div class="stat"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
68	<code>Fees</code>	HTML element/component: page ka structure ya UI control banata hai.
69	<code>Rs {{ totalFee() }} <em class="up">↑</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
70	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
71	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
72	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.

Line	Original code	Simple explanation
73		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
74	</div>	HTML element/component: page ka structure ya UI control banata hai.
75		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
76	<!-- Monthly Fees Chart -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
77	<div class="sales-card">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
78	<div class="card-header">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
79	Monthly Fees Collected	HTML element/component: page ka structure ya UI control banata hai.
80	<button class="more-btn">■</button>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
81	</div>	HTML element/component: page ka structure ya UI control banata hai.
82		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
83	<div class="bar-chart">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
84	<div class="y-axis">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
85	400	HTML element/component: page ka structure ya UI control banata hai.
86	300	HTML element/component: page ka structure ya UI control banata hai.
87	200	HTML element/component: page ka structure ya UI control banata hai.
88	100	HTML element/component: page ka structure ya UI control banata hai.
89	0	HTML element/component: page ka structure ya UI control banata hai.
90	</div>	HTML element/component: page ka structure ya UI control banata hai.
91	<div class="bars-area">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
92	@for (item of salesData; track item.month) {	Angular loop: array ke har item ke liye HTML repeat karta hai.
93	<div class="bar-wrapper">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
94	<div class="bar" [style.height.%]="getBarHeight(item.value)"></div>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
95	{{ item.month }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
96	</div>	HTML element/component: page ka structure ya UI control banata hai.
97	}	Pichla block/function/object close karta hai.
98	</div>	HTML element/component: page ka structure ya UI control banata hai.
99	</div>	HTML element/component: page ka structure ya UI control banata hai.
100	</div>	HTML element/component: page ka structure ya UI control banata hai.
101		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
102	</div>	HTML element/component: page ka structure ya UI control banata hai.

src/app/pages/dashboard/dashboard.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	import { Component, OnInit, signal, inject } from '@angular/core';	Import: doosri library/file se required class ya function is file mein available karta hai.

Line	Original code	Simple explanation
2	<code>import { CommonModule } from '@angular/common';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
3	<code>import { HttpClient } from '@angular/common/http';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
4	<code>import { ThemeService } from '../services/theme';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
5		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
6	<code>@Component({</code>	Angular decorator: neech wali class ko Angular component declare karta hai.
7	<code>selector: 'app-dashboard',</code>	Component ka HTML tag name; parent template mein isi selector se component lagta hai.
8	<code>standalone: true,</code>	Component standalone hai; NgModule mein declare karna zaroori nahi.
9	<code>imports: [CommonModule],</code>	Template mein use hone wale Angular directives/forms/router features import karta hai.
10	<code>templateUrl: './dashboard.html',</code>	Is component ki HTML template file ka path.
11	<code>styleUrl: './dashboard.css',</code>	Is component ki CSS file ka path.
12	<code>})</code>	Pichla block/function/object close karta hai.
13	<code>export class Dashboard implements OnInit {</code>	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
14		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
15	<code>private http = inject(HttpClient);</code>	Angular se HttpClient service leta hai; API request bhejne ke liye.
16	<code>public themeService = inject(ThemeService);</code>	Shared ThemeService use hoti hai taake theme state/component access milay.
17		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
18	<code>// ■ Signals – real data se fill honge</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
19	<code>totalStudents = signal<number>(0);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set().update().
20	<code>totalCourses = signal<number>(0);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set().update().
21	<code>totalEnrollments = signal<number>(0);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set().update().
22	<code>totalFee = signal<number>(0);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set().update().
23	<code>pendingFee = signal<number>(0);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set().update().
24		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
25	<code>// Enrollment Target (percentage)</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
26	<code>monthlyTarget = signal<number>(72);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set().update().
27	<code>targetGrowth = signal<number>(10);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set().update().
28		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
29	<code>// Monthly fees chart</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
30	<code>salesData = [</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
31	<code>{ month: 'Jan', value: 160 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
32	<code>{ month: 'Feb', value: 350 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
33	<code>{ month: 'Mar', value: 200 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
34	<code>{ month: 'Apr', value: 280 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
35	<code>{ month: 'May', value: 180 },</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
36	{ month: 'Jun', value: 190 },	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
37	{ month: 'Jul', value: 270 },	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
38	{ month: 'Aug', value: 100 },	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
39	{ month: 'Sep', value: 210 },	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
40	{ month: 'Oct', value: 380 },	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
41	{ month: 'Nov', value: 260 },	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
42	{ month: 'Dec', value: 110 }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
43	];	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
44		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
45	maxSales = 400;	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
46		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
47	ngOnInit() {	Component initialize hone ke baad Angular is method ko automatically call karta hai.
48	console.log('■ Dashboard ngOnInit');	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
49	this.loadAllData();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
50	}	Pichla block/function/object close karta hai.
51		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
52	// ■ Sab data ek saath load karo	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
53	loadAllData() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
54	// Students	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
55	this.http.get<any[]>('http://localhost:3000/students')	HTTP GET request: backend se data read karta hai.
56	.subscribe({	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
57	next: (data) => {	Successful response ka handler; received data par UI/state update hoti hai.
58	console.log('■ Students:', data.length);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
59	this.totalStudents.set(data.length);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
60	},	Pichla block/function/object close karta hai.
61	error: (err) => console.error('■ Students error:', err)	Request fail ho to error handler chalta hai; debugging/error feedback ke liye.
62	});	Pichla block/function/object close karta hai.
63		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
64	// Courses	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
65	this.http.get<any[]>('http://localhost:3000/courses')	HTTP GET request: backend se data read karta hai.
66	.subscribe({	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
67	next: (data) => {	Successful response ka handler; received data par UI/state update hoti hai.
68	console.log('■ Courses:', data.length);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
69	this.totalCourses.set(data.length);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
70	},	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
71	<code>error: (err) => console.error('■ Courses error:', err)</code>	Request fail ho to error handler chalta hai; debugging/error feedback ke liye.
72	<code>});</code>	Pichla block/function/object close karta hai.
73		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
74	<code>// Enrollments</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
75	<code>this.http.get<any[]>('http://localhost:3000/enrollments')</code>	HTTP GET request: backend se data read karta hai.
76	<code>.subscribe({</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
77	<code>next: (data) => {</code>	Successful response ka handler; received data par UI/state update hoti hai.
78	<code>console.log('■ Enrollments:', data.length);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
79	<code>this.totalEnrollments.set(data.length);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
80		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
81	<code>// Total enrolled fee calculate</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
82	<code>const total = data.reduce((sum, e) => sum + Number(e.course_fee 0), 0);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
83	<code>this.totalFee.set(total);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
84	<code>console.log('■ Total enrolled fee:', total);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
85	<code>},</code>	Pichla block/function/object close karta hai.
86	<code>error: (err) => console.error('■ Enrollments error:', err)</code>	Request fail ho to error handler chalta hai; debugging/error feedback ke liye.
87	<code>});</code>	Pichla block/function/object close karta hai.
88		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
89	<code>// Payments – Pending calculate</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
90	<code>this.http.get<any[]>('http://localhost:3000/payments')</code>	HTTP GET request: backend se data read karta hai.
91	<code>.subscribe({</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
92	<code>next: (payments) => {</code>	Successful response ka handler; received data par UI/state update hoti hai.
93	<code>// Enrollments + Payments dono chahiye pending ke liye</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
94	<code>this.http.get<any[]>('http://localhost:3000/enrollments')</code>	HTTP GET request: backend se data read karta hai.
95	<code>.subscribe(enrollments => {</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
96	<code>const totalEnrolled = enrollments.reduce((sum, e) => sum + Number(e.course_fee 0), 0);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
97	<code>const totalPaid = payments.reduce((sum, p) => sum + Number(p.paid_amount 0), 0);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
98	<code>const pending = totalEnrolled - totalPaid;</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
99	<code>this.pendingFee.set(pending);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
100	<code>console.log('■ Total paid:', totalPaid, ' Pending:', pending);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
101	<code>});</code>	Pichla block/function/object close karta hai.
102	<code>},</code>	Pichla block/function/object close karta hai.
103	<code>error: (err) => console.error('■ Payments error:', err)</code>	Request fail ho to error handler chalta hai; debugging/error feedback ke liye.
104	<code>});</code>	Pichla block/function/object close karta hai.
105	<code>}</code>	Pichla block/function/object close karta hai.
106		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
107	<code>getBarHeight(value: number): number {</code>	Code block start hota hai; andar related statements group hote hain.

Line	Original code	Simple explanation
108	<code>return (value / this.maxSales) * 100;</code>	Function ka result return karta hai.
109	<code>}</code>	Pichla block/function/object close karta hai.
110	<code>}</code>	Pichla block/function/object close karta hai.

src/app/pages/enrollment/enrollment.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	<code>.enrollments-page {</code>	Code block start hota hai; andar related statements group hote hain.
2	<code>padding: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
3	<code>background: #f5f6fa;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
4	<code>min-height: 100vh;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
5	<code>font-family: 'Inter', sans-serif;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
6	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
7	<code>transition: background 0.3s, color 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
8	<code>}</code>	Pichla block/function/object close karta hai.
9	<code>.enrollments-page.dark { background: #0f0f1a; color: #e4e4e7; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
10		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
11	<code>.page-header {</code>	Code block start hota hai; andar related statements group hote hain.
12	<code>display: flex; justify-content: space-between;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
13	<code>align-items: center; margin-bottom: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
14	<code>}</code>	Pichla block/function/object close karta hai.
15	<code>.page-header h2 { font-size: 24px; font-weight: 700; margin: 0; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
16	<code>.enrollments-page.dark .page-header h2 { color: #ffffff; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
17		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
18	<code>.add-new-btn {</code>	Code block start hota hai; andar related statements group hote hain.
19	<code>background: #059669; color: #fff; border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
20	<code>padding: 10px 20px; border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
21	<code>font-size: 14px; font-weight: 600; cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
22	<code>box-shadow: 0 4px 12px rgba(70, 95, 255, 0.3);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
23	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
24	<code>}</code>	Pichla block/function/object close karta hai.
25	<code>.add-new-btn:hover { background: #059669; transform: translateY(-1px); }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
26		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
27	<code>.table-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
28	<code>background: #ffffff; border-radius: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
29	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
30	<code>box-shadow: 0 2px 8px rgba(0, 0, 0, 0.04);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
31	<code>overflow: hidden;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
32	<code>}</code>	Pichla block/function/object close karta hai.
33	<code>.enrollments-page.dark .table-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
34	<code>background: #14141f; border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
35	<code>}</code>	Pichla block/function/object close karta hai.
36		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
37	<code>table { width: 100%; border-collapse: collapse; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
38	<code>thead { background: #f9fafb; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
39	<code>.enrollments-page.dark thead { background: #1a1a28; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
40		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
41	<code>th {</code>	Code block start hota hai; andar related statements group hote hain.
42	<code>text-align: center; padding: 14px 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
43	<code>font-size: 12px; font-weight: 600; color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
44	<code>text-transform: uppercase; letter-spacing: 0.5px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
45	<code>border-bottom: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
46	<code>}</code>	Pichla block/function/object close karta hai.
47	<code>.enrollments-page.dark th { color: #9ca3af; border-bottom-color: #2a2a3a; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
48		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
49	<code>td {</code>	Code block start hota hai; andar related statements group hote hain.
50	<code>padding: 16px; text-align: center; font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
51	<code>border-bottom: 1px solid #f3f4f6;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
52	<code>}</code>	Pichla block/function/object close karta hai.
53	<code>.enrollments-page.dark td {</code>	Code block start hota hai; andar related statements group hote hain.
54	<code>color: #e4e4e7; border-bottom-color: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
55	<code>}</code>	Pichla block/function/object close karta hai.
56	<code>tbody tr:last-child td { border-bottom: none; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
57	<code>tbody tr:hover td { background: #f9fafb; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
58	<code>.enrollments-page.dark tbody tr:hover td { background: #1a1a28; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
59	<code>.name-cell { font-weight: 600; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
60		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
61	<code>.status-badge {</code>	Code block start hota hai; andar related statements group hote hain.
62	<code>display: inline-block; padding: 4px 12px; border-radius: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
63	<code>font-size: 12px; font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
64	<code>background: #fee2e2; color: #991b1b;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
65	}	Pichla block/function/object close karta hai.
66	<code>.enrollments-page.dark .status-badge { background: #7f1d1d; color: #fca5a5; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
67	<code>.status-badge.active { background: #d1fae5; color: #065f46; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
68	<code>.enrollments-page.dark .status-badge.active { background: #064e3b; color: #6ee7b7; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
69		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
70	<code>.actions { display: flex; gap: 6px; justify-content: center; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
71	<code>.edit-btn { background: #465fff; color: #fff; border: none; padding: 7px 14px; border-radius: 6px; font-size: 12px; font-weight: 600; cursor: pointer; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
72	<code>.edit-btn:hover { background: #3548d4; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
73	<code>.delete-btn { background: #ef4444; color: #fff; border: none; padding: 7px 14px; border-radius: 6px; font-size: 12px; font-weight: 600; cursor: pointer; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
74	<code>.delete-btn:hover { background: #dc2626; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
75		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
76	<code>.empty-state { padding: 60px 20px !important; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
77	<code>.empty-content { display: flex; flex-direction: column; align-items: center; gap: 10px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
78	<code>.empty-icon { font-size: 48px; opacity: 0.6; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
79	<code>.empty-content h3 { font-size: 20px; font-weight: 600; margin: 0; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
80	<code>.empty-content p { font-size: 14px; color: #6b7280; margin: 0 0 16px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
81	<code>.add-first-btn {</code>	Code block start hota hai; andar related statements group hote hain.
82	<code>background: #059669; color: #fff; border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
83	<code>padding: 10px 24px; border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
84	<code>font-weight: 600; cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
85	}	Pichla block/function/object close karta hai.
86		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
87	<code>/* Modal */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
88	<code>.modal-overlay {</code>	Code block start hota hai; andar related statements group hote hain.
89	<code>position: fixed; inset: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
90	<code>background: rgba(0, 0, 0, 0.6);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
91	<code>backdrop-filter: blur(4px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
92	<code>display: flex; align-items: center; justify-content: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
93	<code>z-index: 1000; padding: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
94	}	Pichla block/function/object close karta hai.
95	<code>.modal-box {</code>	Code block start hota hai; andar related statements group hote hain.
96	<code>background: #ffffff; border-radius: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
97	<code>max-width: 700px; width: 100%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
98	<code>max-height: 90vh; overflow-y: auto;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
99	<code>box-shadow: 0 20px 60px rgba(0, 0, 0, 0.3);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
100	<code>}</code>	Pichla block/function/object close karta hai.
101	<code>.enrollments-page.dark .modal-box { background: #14141f; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
102		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
103	<code>.modal-header {</code>	Code block start hota hai; andar related statements group hote hain.
104	<code>display: flex; justify-content: space-between; align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
105	<code>padding: 20px 24px; border-bottom: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
106	<code>}</code>	Pichla block/function/object close karta hai.
107	<code>.enrollments-page.dark .modal-header { border-bottom-color: #2a2a3a; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
108	<code>.modal-header h3 { font-size: 18px; font-weight: 700; margin: 0; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
109	<code>.modal-close {</code>	Code block start hota hai; andar related statements group hote hain.
110	<code>background: transparent; border: none; font-size: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
111	<code>color: #6b7280; cursor: pointer; padding: 4px 10px; border-radius: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
112	<code>}</code>	Pichla block/function/object close karta hai.
113		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
114	<code>.modal-form {</code>	Code block start hota hai; andar related statements group hote hain.
115	<code>padding: 24px; display: grid;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
116	<code>grid-template-columns: 1fr 1fr; gap: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
117	<code>}</code>	Pichla block/function/object close karta hai.
118	<code>.form-group { display: flex; flex-direction: column; gap: 6px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
119	<code>.form-group.full-width { grid-column: 1 / -1; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
120	<code>.form-group label { font-size: 13px; font-weight: 600; color: #374151; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
121	<code>.enrollments-page.dark .form-group label { color: #d1d5db; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
122		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
123	<code>.form-group input, .form-group select {</code>	Code block start hota hai; andar related statements group hote hain.
124	<code>padding: 10px 14px; border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
125	<code>border-radius: 8px; background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
126	<code>color: #1a1a2e; font-size: 14px; outline: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
127	<code>}</code>	Pichla block/function/object close karta hai.
128	<code>.enrollments-page.dark .form-group input,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
129	<code>.enrollments-page.dark .form-group select {</code>	Code block start hota hai; andar related statements group hote hain.
130	<code>background: #1f1f2e; border-color: #2a2a3a; color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
131	<code>}</code>	Pichla block/function/object close karta hai.
132	<code>.form-group input[readonly] {</code>	Code block start hota hai; andar related statements group hote hain.
133	<code>background: #f9fafb; color: #6b7280; cursor: not-allowed;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
134	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
135		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
136	<code>/* Custom Dropdown */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
137	<code>.custom-dropdown { position: relative; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
138	<code>.dropdown-btn {</code>	Code block start hota hai; andar related statements group hote hain.
139	<code>width: 100%; display: flex; justify-content: space-between;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
140	<code>align-items: center; padding: 10px 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
141	<code>border: 1px solid #e5e7eb; border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
142	<code>background: #ffffff; color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
143	<code>font-size: 14px; cursor: pointer; text-align: left;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
144	<code>}</code>	Pichla block/function/object close karta hai.
145	<code>.enrollments-page.dark .dropdown-btn {</code>	Code block start hota hai; andar related statements group hote hain.
146	<code>background: #1f1f2e; border-color: #2a2a3a; color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
147	<code>}</code>	Pichla block/function/object close karta hai.
148	<code>.dropdown-btn:hover, .dropdown-btn.active {</code>	Code block start hota hai; andar related statements group hote hain.
149	<code>border-color: #465fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
150	<code>box-shadow: 0 0 3px rgba(70, 95, 255, 0.15);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
151	<code>}</code>	Pichla block/function/object close karta hai.
152	<code>.dropdown-btn.error {</code>	Code block start hota hai; andar related statements group hote hain.
153	<code>border-color: #ef4444 !important;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
154	<code>box-shadow: 0 0 3px rgba(239, 68, 68, 0.15) !important;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
155	<code>}</code>	Pichla block/function/object close karta hai.
156	<code>.dropdown-btn .arrow { font-size: 10px; color: #6b7280; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
157	<code>.dropdown-menu {</code>	Code block start hota hai; andar related statements group hote hain.
158	<code>position: absolute; top: calc(100% + 6px); left: 0; right: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
159	<code>background: #ffffff; border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
160	<code>border-radius: 8px; box-shadow: 0 10px 30px rgba(0, 0, 0, 0.1);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
161	<code>max-height: 200px; overflow-y: auto; z-index: 50;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
162	<code>}</code>	Pichla block/function/object close karta hai.
163	<code>.enrollments-page.dark .dropdown-menu { background: #1f1f2e; border-color: #2a2a3a; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
164	<code>.dropdown-item {</code>	Code block start hota hai; andar related statements group hote hain.
165	<code>padding: 10px 14px; cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
166	<code>font-size: 14px; color: #1a1a2e; transition: 0.15s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
167	<code>}</code>	Pichla block/function/object close karta hai.
168	<code>.enrollments-page.dark .dropdown-item { color: #e4e4e7; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
169	<code>.dropdown-item:hover { background: #f3f4f6; color: #465fff; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
170	<code>.enrollments-page.dark .dropdown-item:hover { background: #2a2a3a; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
171		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
172	<code>.error-msg { font-size: 12px; color: #ef4444; margin-top: 4px; font-weight: 500; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
173		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
174	<code>.modal-actions {</code>	Code block start hota hai; andar related statements group hote hain.
175	<code>grid-column: 1 / -1; display: flex; gap: 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
176	<code>justify-content: flex-end; margin-top: 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
177	<code>padding-top: 16px; border-top: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
178	<code>}</code>	Pichla block/function/object close karta hai.
179	<code>.enrollments-page.dark .modal-actions { border-top-color: #2a2a3a; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
180	<code>.save-btn { background: #059669; color: #fff; border: none; padding: 10px 24px; border-radius: 8px; font-weight: 600; cursor: pointer; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
181	<code>.save-btn:hover { background: #047857; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
182	<code>.cancel-btn { background: #6b7280; color: #fff; border: none; padding: 10px 20px; border-radius: 8px; font-weight: 600; cursor: pointer; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
183	<code>.cancel-btn:hover { background: #4b5563; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
184		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
185	<code>/* Toast */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
186	<code>.toast-popup {</code>	Code block start hota hai; andar related statements group hote hain.
187	<code>position: fixed; top: 90px; right: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
188	<code>background: #ffffff; border-left: 4px solid #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
189	<code>border-radius: 10px; padding: 14px 18px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
190	<code>display: flex; align-items: center; gap: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
191	<code>box-shadow: 0 10px 30px rgba(0, 0, 0, 0.15);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
192	<code>z-index: 2000; min-width: 280px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
193	<code>}</code>	Pichla block/function/object close karta hai.
194	<code>.toast-popup.success { border-left-color: #10b981; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
195	<code>.enrollments-page.dark .toast-popup { background: #1f1f2e; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
196	<code>.toast-text { flex: 1; font-size: 14px; font-weight: 500; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
197	<code>.toast-close { background: transparent; border: none; color: #9ca3af; cursor: pointer; padding: 2px 8px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
198		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
199	<code>.form-select {</code>	Code block start hota hai; andar related statements group hote hain.
200	<code>width: 100%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
201	<code>padding: 10px 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
202		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
203	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
204	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
205		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
206	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
207	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
208		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
209	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
210	<code>outline: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
211		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
212	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
213	<code>}</code>	Pichla block/function/object close karta hai.
214		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
215	<code>.form-select:focus {</code>	Code block start hota hai; andar related statements group hote hain.
216	<code>border-color: #465fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
217		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
218	<code>box-shadow: 0 0 0 3px rgba(70, 95, 255, 0.15);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
219	<code>}</code>	Pichla block/function/object close karta hai.
220		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
221	<code>.form-select.error {</code>	Code block start hota hai; andar related statements group hote hain.
222	<code>border-color: #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
223	<code>}</code>	Pichla block/function/object close karta hai.
224		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
225	<code>.enrollments-page.dark .form-select {</code>	Code block start hota hai; andar related statements group hote hain.
226	<code>background: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
227	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
228	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
229	<code>}</code>	Pichla block/function/object close karta hai.
230		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
231	<code>.form-select option {</code>	Code block start hota hai; andar related statements group hote hain.
232	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
233	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
234	<code>}</code>	Pichla block/function/object close karta hai.
235		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
236	<code>.enrollments-page.dark .form-select option {</code>	Code block start hota hai; andar related statements group hote hain.
237	<code>background: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
238	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
239	}	Pichla block/function/object close karta hai.
240		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
241	@media (max-width: 768px) {	Responsive CSS rule: screen width/device condition par styles apply karta hai.
242	.enrollments-page { padding: 16px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
243	.page-header { flex-direction: column; gap: 12px; align-items: stretch; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
244	.table-wrapper { overflow-x: auto; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
245	table { min-width: 800px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
246	.modal-form { grid-template-columns: 1fr; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
247	}	Pichla block/function/object close karta hai.

src/app/pages/enrollment/enrollment.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Line	Original code	Simple explanation
1	<div class="enrollments-page" [class.dark]="themeService.isDarkMode()">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
2		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
3	@if (showToast()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
4	<div class="toast-popup" [class.success]="toastType() === 'success'">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
5	{{ toastMessage() }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
6	<button class="toast-close" (click)="showToast.set(false)">X</button>	User click par specified TypeScript method/expression chalti hai.
7	</div>	HTML element/component: page ka structure ya UI control banata hai.
8	}	Pichla block/function/object close karta hai.
9		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
10	<div class="page-header">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
11	Enrollments	HTML element/component: page ka structure ya UI control banata hai.
12	<button class="add-new-btn" (click)="openModal()">+ New Enrollment</button>	User click par specified TypeScript method/expression chalti hai.
13	</div>	HTML element/component: page ka structure ya UI control banata hai.
14		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
15	<div class="table-wrapper">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
16	<table>	HTML element/component: page ka structure ya UI control banata hai.
17	<thead>	HTML element/component: page ka structure ya UI control banata hai.
18	<tr>	HTML element/component: page ka structure ya UI control banata hai.
19	<th>Enrollment ID</th>	HTML element/component: page ka structure ya UI control banata hai.
20	<th>Student</th>	HTML element/component: page ka structure ya UI control banata hai.
21	<th>Course</th>	HTML element/component: page ka structure ya UI control banata hai.
22	<th>Fee</th>	HTML element/component: page ka structure ya UI control banata hai.

Line	Original code	Simple explanation
23	<code><th>Date</th></code>	HTML element/component: page ka structure ya UI control banata hai.
24	<code><th>Status</th></code>	HTML element/component: page ka structure ya UI control banata hai.
25	<code><th>Actions</th></code>	HTML element/component: page ka structure ya UI control banata hai.
26	<code></tr></code>	HTML element/component: page ka structure ya UI control banata hai.
27	<code></thead></code>	HTML element/component: page ka structure ya UI control banata hai.
28	<code><tbody></code>	HTML element/component: page ka structure ya UI control banata hai.
29	<code>@for (e of enrollments(); track e.id) {</code>	Angular loop: array ke har item ke liye HTML repeat karta hai.
30	<code><tr></code>	HTML element/component: page ka structure ya UI control banata hai.
31	<code><td>E{{ e.id }}</td></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
32	<code><td class="name-cell">{{ e.student_name }}</td></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
33	<code><td>{{ e.course_name }}</td></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
34	<code><td>Rs. {{ e.course_fee }}</td></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
35	<code><td>{{ e.enroll_date }}</td></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
36	<code><td></code>	HTML element/component: page ka structure ya UI control banata hai.
37	<code></code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
38	<code>{{ e.status }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
39	<code></code>	HTML element/component: page ka structure ya UI control banata hai.
40	<code></td></code>	HTML element/component: page ka structure ya UI control banata hai.
41	<code><td></code>	HTML element/component: page ka structure ya UI control banata hai.
42	<code><div class="actions"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
43	<code><button class="edit-btn" (click)="editEnrollment(e)">Edit</button></code>	User click par specified TypeScript method/expression chalti hai.
44	<code><button class="delete-btn" (click)="deleteEnrollment(e.id)">Delete</button></code>	User click par specified TypeScript method/expression chalti hai.
45	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
46	<code></td></code>	HTML element/component: page ka structure ya UI control banata hai.
47	<code></tr></code>	HTML element/component: page ka structure ya UI control banata hai.
48	<code>}</code>	Pichla block/function/object close karta hai.
49		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
50	<code>@if (enrollments().length === 0) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
51	<code><tr></code>	HTML element/component: page ka structure ya UI control banata hai.
52	<code><td colspan="7" class="empty-state"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
53	<code><div class="empty-content"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
54	<code><div class="empty-icon">■</div></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
55	<code><h3>No Enrollments Yet</h3></code>	HTML element/component: page ka structure ya UI control banata hai.
56	<code><p>Enroll a student to a course</p></code>	HTML element/component: page ka structure ya UI control banata hai.

Line	Original code	Simple explanation
57	<code><button class="add-first-btn" (click)="openModal()">+ New Enrollment</button></code>	User click par specified TypeScript method/expression chalti hai.
58	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
59	<code></td></code>	HTML element/component: page ka structure ya UI control banata hai.
60	<code></tr></code>	HTML element/component: page ka structure ya UI control banata hai.
61	<code>}</code>	Pichla block/function/object close karta hai.
62	<code></tbody></code>	HTML element/component: page ka structure ya UI control banata hai.
63	<code></table></code>	HTML element/component: page ka structure ya UI control banata hai.
64	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
65		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
66	<code>@if (showModal()) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
67	<code><div class="modal-overlay" (click)="closeModal()"></code>	User click par specified TypeScript method/expression chalti hai.
68	<code><div class="modal-box" (click)="\$event.stopPropagation()"></code>	User click par specified TypeScript method/expression chalti hai.
69		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
70	<code><div class="modal-header"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
71	<code><h3>{{ selectedEnrollmentId() === null ? 'New Enrollment' : 'Update Enrollment' }}</h3></code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
72	<code><button class="modal-close" (click)="closeModal()">X</button></code>	User click par specified TypeScript method/expression chalti hai.
73	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
74		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
75	<code><form [formGroup]="form" (ngSubmit)="saveEnrollment()" class="modal-form"></code>	Form submit par specified handler chalata hai.
76		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
77	<code><!-- Student --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
78	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
79	<code><label>Student *</label></code>	HTML element/component: page ka structure ya UI control banata hai.
80		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
81	<code><select</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
82	<code>formControlName="student_id"</code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
83	<code>class="form-select"</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
84	<code>[class.error]="isValid('student_id')"</code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
85	<code>(change)="onStudentChange(\$event)"</code>	Input/select ki value change par handler chalta hai.
86	<code>></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
87	<code><option value="">Select Student</option></code>	HTML element/component: page ka structure ya UI control banata hai.
88		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
89	<code>@for (student of students(); track student.id) {</code>	Angular loop: array ke har item ke liye HTML repeat karta hai.
90	<code><option [value]="student.id"></code>	Property/attribute binding: HTML property ko TS value se connect karta hai.

Line	Original code	Simple explanation
91	<code>{{ student.name }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
92	<code></option></code>	HTML element/component: page ka structure ya UI control banata hai.
93	<code>}</code>	Pichla block/function/object close karta hai.
94	<code></select></code>	HTML element/component: page ka structure ya UI control banata hai.
95		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
96	<code>@if (isValid('student_id')) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
97	<code>■ Student required</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
98	<code>}</code>	Pichla block/function/object close karta hai.
99	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
100		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
101	<code><!-- Course --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
102	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
103	<code><label>Course *</label></code>	HTML element/component: page ka structure ya UI control banata hai.
104		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
105	<code><select</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
106	<code>formControlName="course_id"</code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
107	<code>class="form-select"</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
108	<code>[class.error]="isValid('course_id')"</code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
109	<code>(change)="onCourseChange(\$event)"</code>	Input/select ki value change par handler chalta hai.
110	<code>></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
111	<code><option value="">Select Course</option></code>	HTML element/component: page ka structure ya UI control banata hai.
112		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
113	<code>@for (course of courses(); track course.id) {</code>	Angular loop: array ke har item ke liye HTML repeat karta hai.
114	<code><option [value]="course.id"></code>	Property/attribute binding: HTML property ko TS value se connect karta hai.
115	<code>{{ course.course_name }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
116	<code></option></code>	HTML element/component: page ka structure ya UI control banata hai.
117	<code>}</code>	Pichla block/function/object close karta hai.
118	<code></select></code>	HTML element/component: page ka structure ya UI control banata hai.
119		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
120	<code>@if (isValid('course_id')) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
121	<code>■ Course required</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
122	<code>}</code>	Pichla block/function/object close karta hai.
123	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
124		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
125	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.

Line	Original code	Simple explanation
126	<label>Course Fee</label>	HTML element/component: page ka structure ya UI control banata hai.
127	<input formControlName="course_fee" readonly />	HTML input ko TS reactive form ke same-named control se connect karta hai.
128	</div>	HTML element/component: page ka structure ya UI control banata hai.
129		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
130	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
131	<label>Enrollment Date *</label>	HTML element/component: page ka structure ya UI control banata hai.
132	<input type="date" formControlName="enroll_date"	HTML input ko TS reactive form ke same-named control se connect karta hai.
133	[class.error]="isValid('enroll_date')"/>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
134	</div>	HTML element/component: page ka structure ya UI control banata hai.
135		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
136	<div class="form-group full-width">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
137	<label>Status *</label>	HTML element/component: page ka structure ya UI control banata hai.
138	<select formControlName="status" class="status-select">	HTML input ko TS reactive form ke same-named control se connect karta hai.
139	<option value="Active">Active</option>	HTML element/component: page ka structure ya UI control banata hai.
140	<option value="Inactive">Inactive</option>	HTML element/component: page ka structure ya UI control banata hai.
141	<option value="Completed">Completed</option>	HTML element/component: page ka structure ya UI control banata hai.
142	</select>	HTML element/component: page ka structure ya UI control banata hai.
143	</div>	HTML element/component: page ka structure ya UI control banata hai.
144		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
145	<div class="modal-actions">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
146	<button type="button" class="cancel-btn" (click)="closeModal()">Cancel</button>	User click par specified TypeScript method/expression chalti hai.
147	<button type="submit" class="save-btn">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
148	{{ selectedEnrollmentId() === null ? 'Enroll' : 'Update' }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
149	</button>	HTML element/component: page ka structure ya UI control banata hai.
150	</div>	HTML element/component: page ka structure ya UI control banata hai.
151		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
152	</form>	HTML element/component: page ka structure ya UI control banata hai.
153	</div>	HTML element/component: page ka structure ya UI control banata hai.
154	</div>	HTML element/component: page ka structure ya UI control banata hai.
155	}	Pichla block/function/object close karta hai.
156		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
157	</div>	HTML element/component: page ka structure ya UI control banata hai.

[src/app/pages/enrollment/enrollment.ts](#)

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	<code>import { Component, OnInit, signal, inject } from '@angular/core';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
2	<code>import { CommonModule } from '@angular/common';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
3	<code>import { HttpClient } from '@angular/common/http';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
4	<code>import { FormBuilder, ReactiveFormsModule, Validators } from '@angular/forms';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
5	<code>import { ThemeService } from '../services/theme';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
6		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
7	<code>@Component({</code>	Angular decorator: neech wali class ko Angular component declare karta hai.
8	<code>selector: 'app-enrollment',</code>	Component ka HTML tag name; parent template mein isi selector se component lagta hai.
9	<code>standalone: true,</code>	Component standalone hai; NgModule mein declare karna zaroori nahi.
10	<code>imports: [CommonModule, ReactiveFormsModule],</code>	Template mein use hone wale Angular directives/forms/router features import karta hai.
11	<code>templateUrl: './enrollment.html',</code>	Is component ki HTML template file ka path.
12	<code>styleUrl: './enrollment.css'</code>	Is component ki CSS file ka path.
13	<code>})</code>	Pichla block/function/object close karta hai.
14	<code>export class Enrollment implements OnInit {</code>	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
15		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
16	<code>private http = inject(HttpClient);</code>	Angular se HttpClient service leta hai; API request bhejne ke liye.
17	<code>private fb = inject(FormBuilder);</code>	FormBuilder leta hai taake reactive form aur controls bana sake.
18	<code>public themeService = inject(ThemeService);</code>	Shared ThemeService use hoti hai taake theme state/component access milay.
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	<code>showModal = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
21	<code>enrollments = signal<any[]>([]);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
22	<code>students = signal<any[]>([]);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
23	<code>courses = signal<any[]>([]);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
24	<code>selectedEnrollmentId = signal<number null>(null);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
25		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
26	<code>// Toast</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
27	<code>showToast = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
28	<code>toastMessage = signal<string>('');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
29	<code>toastType = signal<'error' 'success'>('error');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
30		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
31	<code>// Dropdowns</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
32	<code>showStudentDropdown = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
33	<code>selectedStudentLabel = signal<string>('');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
34	<code>showCourseDropdown = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().

Line	Original code	Simple explanation
35	<code>selectedCourseLabel = signal<string>('');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
36		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
37	<code>form = this.fb.group({</code>	Reactive form group banata hai; fields, current values aur validation ek jagah manage hotay hain.
38	<code> student_id: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
39	<code> student_name: ['',</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
40	<code> course_id: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
41	<code> course_name: ['',</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
42	<code> course_fee: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
43	<code> enroll_date: [new Date().toISOString().slice(0, 10), Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
44	<code> status: ['Active', Validators.required]</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
45	<code>});</code>	Pichla block/function/object close karta hai.
46		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
47	<code>ngOnInit() {</code>	Component initialize hone ke baad Angular is method ko automatically call karta hai.
48	<code> this.loadEnrollments();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
49	<code> this.loadStudents();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
50	<code> this.loadCourses();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
51	<code>}</code>	Pichla block/function/object close karta hai.
52		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
53	<code>showToastMsg(msg: string, type: 'error' 'success' = 'error') {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
54	<code> this.toastMessage.set(msg);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
55	<code> this.toastType.set(type);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
56	<code> this.showToast.set(true);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
57	<code> setTimeout(() => this.showToast.set(false), 3000);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
58	<code>}</code>	Pichla block/function/object close karta hai.
59		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
60	<code>loadEnrollments() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
61	<code> this.http.get<any[]>('http://localhost:3000/enrollments')</code>	HTTP GET request: backend se data read karta hai.
62	<code> .subscribe(data => this.enrollments.set(data));</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
63	<code>}</code>	Pichla block/function/object close karta hai.
64		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
65	<code>loadStudents() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
66	<code> this.http.get<any[]>('http://localhost:3000/students')</code>	HTTP GET request: backend se data read karta hai.
67	<code> .subscribe(data => this.students.set(data));</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
68	<code>}</code>	Pichla block/function/object close karta hai.
69	<code>onStudentChange(event: Event) {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
70	<code> const select = event.target as HTMLSelectElement;</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.

Line	Original code	Simple explanation
71	<code>const studentId = select.value; // ■ Number() HATAO</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
72		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
73	<code>console.log('■ Student ID:', studentId);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
74		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
75	<code>const student = this.students().find(s => String(s.id) === studentId);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
76		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
77	<code>if (student) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
78	<code>this.form.patchValue({</code>	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
79	<code>student_id: student.id,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
80	<code>student_name: student.name</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
81	<code>});</code>	Pichla block/function/object close karta hai.
82	<code>console.log('■ Student set:', student.name);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
83	<code>} else {</code>	Alternative branch: pehli condition false ho to yeh code chalta hai.
84	<code>console.log('■ Student not found:', studentId);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
85	<code>}</code>	Pichla block/function/object close karta hai.
86	<code>}</code>	Pichla block/function/object close karta hai.
87		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
88	<code>loadCourses() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
89	<code>this.http.get<any[]>('http://localhost:3000/courses')</code>	HTTP GET request: backend se data read karta hai.
90	<code>.subscribe(data => this.courses.set(data));</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
91	<code>}</code>	Pichla block/function/object close karta hai.
92	<code>onCourseChange(event: Event) {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
93	<code>const select = event.target as HTMLSelectElement;</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
94	<code>const courseId = select.value; // ■ Number() HATAO</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
95		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
96	<code>console.log('■ Course ID:', courseId);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
97	<code>console.log('■ Courses:', this.courses());</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
98		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
99	<code>const course = this.courses().find(c => String(c.id) === courseId);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
100		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
101	<code>if (course) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
102	<code>this.form.patchValue({</code>	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
103	<code>course_id: course.id,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
104	<code>course_name: course.course_name,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
105	<code>course_fee: course.course_fee</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
106	});	Pichla block/function/object close karta hai.
107	console.log('■ Fee set:', course.course_fee);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
108	} else {	Alternative branch: pehli condition false ho to yeh code chalta hai.
109	console.log('■ Course not found:', courseId);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
110	}	Pichla block/function/object close karta hai.
111	}	Pichla block/function/object close karta hai.
112	toggleStudentDropdown() { this.showStudentDropdown.update(v => !v); }	Arrow function: callback/function ko short syntax mein define karta hai.
113	toggleCourseDropdown() { this.showCourseDropdown.update(v => !v); }	Arrow function: callback/function ko short syntax mein define karta hai.
114		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
115	selectStudent(s: any) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
116	this.selectedStudentLabel.set(s.name);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
117	this.form.patchValue({	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
118	student_id: s.id,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
119	student_name: s.name	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
120	});	Pichla block/function/object close karta hai.
121	this.showStudentDropdown.set(false);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
122	}	Pichla block/function/object close karta hai.
123		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
124	selectCourse(c: any) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
125	this.selectedCourseLabel.set(c.course_name);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
126	this.form.patchValue({	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
127	course_id: c.id,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
128	course_name: c.course_name,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
129	course_fee: c.course_fee	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
130	});	Pichla block/function/object close karta hai.
131	this.showCourseDropdown.set(false);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
132	}	Pichla block/function/object close karta hai.
133		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
134	openModal() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
135	this.selectedEnrollmentId.set(null);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
136	this.form.reset({	Form ko clear/default values par reset karta hai.
137	status: 'Active',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
138	enroll_date: new Date().toISOString().slice(0, 10)	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
139	});	Pichla block/function/object close karta hai.
140	this.selectedStudentLabel.set('');	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
141	this.selectedCourseLabel.set('');	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
142	this.showModal.set(true);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
143	}	Pichla block/function/object close karta hai.
144		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
145	closeModal() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
146	this.showModal.set(false);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
147	this.selectedEnrollmentId.set(null);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
148	this.form.reset();	Form ko clear/default values par reset karta hai.
149	this.selectedStudentLabel.set('');	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
150	this.selectedCourseLabel.set('');	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
151	}	Pichla block/function/object close karta hai.
152		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
153	saveEnrollment() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
154	if (this.form.invalid) {	Check karta hai ke form/field validation pass kar raha hai ya nahi.
155	this.form.markAllAsTouched();	Validation messages dikhane ke liye sab controls ko touched mark karta hai.
156	this.showToastMsg('■ Please fill all required fields', 'error');	User ko success/error ka short notification dikhata hai.
157	return;	Current function ko yahin stop karta hai.
158	}	Pichla block/function/object close karta hai.
159		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
160	const url = 'http://localhost:3000/enrollments';	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
161		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
162	if (this.selectedEnrollmentId() === null) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
163	this.http.post(url, this.form.value).subscribe(() => {	HTTP POST request: backend mein naya record create karta hai.
164	this.showToastMsg('■ Enrollment created!', 'success');	User ko success/error ka short notification dikhata hai.
165	this.closeModal();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
166	this.loadEnrollments();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
167	});	Pichla block/function/object close karta hai.
168	} else {	Alternative branch: pehli condition false ho to yeh code chalta hai.
169	this.http.put(`\${url}/\${this.selectedEnrollmentId()}`, this.form.value)	HTTP PUT request: selected record ko update/replace karta hai.
170	.subscribe(() => {	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
171	this.showToastMsg('■ Enrollment updated!', 'success');	User ko success/error ka short notification dikhata hai.
172	this.closeModal();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
173	this.loadEnrollments();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
174	});	Pichla block/function/object close karta hai.
175	}	Pichla block/function/object close karta hai.
176	}	Pichla block/function/object close karta hai.
177		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
178	editEnrollment(e: any) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
179	this.selectedEnrollmentId.set(e.id);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
180	<code>this.form.setValue({</code>	Form ke tamam controls ko values deta hai; required controls miss hon to error ho sakta hai.
181	<code>student_id: e.student_id,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
182	<code>student_name: e.student_name,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
183	<code>course_id: e.course_id,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
184	<code>course_name: e.course_name,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
185	<code>course_fee: e.course_fee,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
186	<code>enroll_date: e.enroll_date,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
187	<code>status: e.status</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
188	<code>});</code>	Pichla block/function/object close karta hai.
189	<code>this.selectedStudentLabel.set(e.student_name);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
190	<code>this.selectedCourseLabel.set(e.course_name);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
191	<code>this.showModal.set(true);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
192	<code>}</code>	Pichla block/function/object close karta hai.
193		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
194	<code>deleteEnrollment(id: number) {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
195	<code>this.http.delete('http://localhost:3000/enrollments/\${id}')</code>	HTTP DELETE request: selected record remove karta hai.
196	<code>.subscribe(() => {</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
197	<code>this.showToastMsg('■ Enrollment deleted', 'success');</code>	User ko success/error ka short notification dikhata hai.
198	<code>this.loadEnrollments();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
199	<code>});</code>	Pichla block/function/object close karta hai.
200	<code>}</code>	Pichla block/function/object close karta hai.
201		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
202	<code>isValid(field: string): boolean {</code>	Code block start hota hai; andar related statements group hote hain.
203	<code>const c = this.form.get(field);</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
204	<code>return !(c && c.invalid && (c.dirty c.touched));</code>	Check karta hai ke form/field validation pass kar raha hai ya nahi.
205	<code>}</code>	Pichla block/function/object close karta hai.
206	<code>}</code>	Pichla block/function/object close karta hai.

src/app/pages/fees/fees.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
2	<code>Fees Page</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
3	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
4	<code>.fees-page {</code>	Code block start hota hai; andar related statements group hote hain.
5	<code>padding: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
6	<code>background: #f5f6fa;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
7	<code>min-height: 100vh;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
8	<code>font-family: 'Inter', sans-serif;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
9	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
10	<code>transition: background 0.3s, color 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
11	<code>}</code>	Pichla block/function/object close karta hai.
12		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
13	<code>.fees-page.dark {</code>	Code block start hota hai; andar related statements group hote hain.
14	<code>background: #0f0f1a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
15	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
16	<code>}</code>	Pichla block/function/object close karta hai.
17		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
18	<code>/* Page Header */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
19	<code>.page-header {</code>	Code block start hota hai; andar related statements group hote hain.
20	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
21	<code>justify-content: space-between;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
22	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
23	<code>margin-bottom: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
24	<code>}</code>	Pichla block/function/object close karta hai.
25		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
26	<code>.page-header h2 {</code>	Code block start hota hai; andar related statements group hote hain.
27	<code>font-size: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
28	<code>font-weight: 700;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
29	<code>margin: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
30	<code>}</code>	Pichla block/function/object close karta hai.
31		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
32	<code>.fees-page.dark .page-header h2 { color: #ffffff; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
33		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
34	<code>.add-new-btn {</code>	Code block start hota hai; andar related statements group hote hain.
35	<code>background: #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
36	<code>color: #fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
37	<code>border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
38	<code>padding: 10px 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
39	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
40	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
41	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Lin e	Original code	Simple explanation
42	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
43	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
44	<code>box-shadow: 0 4px 12px rgba(16, 185, 129, 0.3);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
45	<code>}</code>	Pichla block/function/object close karta hai.
46		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
47	<code>.add-new-btn:hover {</code>	Code block start hota hai; andar related statements group hote hain.
48	<code>background: #059669;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
49	<code>transform: translateY(-1px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
50	<code>}</code>	Pichla block/function/object close karta hai.
51		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
52	<code>/* Table */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
53	<code>.table-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
54	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
55	<code>border-radius: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
56	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
57	<code>box-shadow: 0 2px 8px rgba(0, 0, 0, 0.04);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
58	<code>overflow: hidden;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
59	<code>}</code>	Pichla block/function/object close karta hai.
60		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
61	<code>.fees-page.dark .table-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
62	<code>background: #14141f;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
63	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
64	<code>}</code>	Pichla block/function/object close karta hai.
65		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
66	<code>table { width: 100%; border-collapse: collapse; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
67		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
68	<code>thead { background: #f9fafb; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
69	<code>.fees-page.dark thead { background: #1a1a28; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
70		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
71	<code>th {</code>	Code block start hota hai; andar related statements group hote hain.
72	<code>text-align: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
73	<code>padding: 14px 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
74	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
75	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
76	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
77	<code>text-transform: uppercase;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
78	<code>letter-spacing: 0.5px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
79	<code>border-bottom: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
80	<code>}</code>	Pichla block/function/object close karta hai.
81		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
82	<code>.fees-page.dark th {</code>	Code block start hota hai; andar related statements group hote hain.
83	<code>color: #9ca3af;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
84	<code>border-bottom-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
85	<code>}</code>	Pichla block/function/object close karta hai.
86		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
87	<code>td {</code>	Code block start hota hai; andar related statements group hote hain.
88	<code>padding: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
89	<code>text-align: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
90	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
91	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
92	<code>border-bottom: 1px solid #f3f4f6;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
93	<code>}</code>	Pichla block/function/object close karta hai.
94		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
95	<code>.fees-page.dark td {</code>	Code block start hota hai; andar related statements group hote hain.
96	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
97	<code>border-bottom-color: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
98	<code>}</code>	Pichla block/function/object close karta hai.
99		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
100	<code>tbody tr:last-child td { border-bottom: none; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
101	<code>tbody tr:hover td { background: #f9fafb; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
102	<code>.fees-page.dark tbody tr:hover td { background: #1a1a28; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
103		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
104	<code>.name-cell { font-weight: 600; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
105	<code>.fees-page.dark .name-cell { color: #ffffff; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
106		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
107	<code>/* Status Badges */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
108	<code>.status-badge {</code>	Code block start hota hai; andar related statements group hote hain.
109	<code>display: inline-block;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
110	<code>padding: 4px 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
111	<code>border-radius: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
112	font-size: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
113	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
114	}	Pichla block/function/object close karta hai.
115		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
116	.status-badge.paid {	Code block start hota hai; andar related statements group hote hain.
117	background: #d1fae5;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
118	color: #065f46;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
119	}	Pichla block/function/object close karta hai.
120	.fees-page.dark .status-badge.paid {	Code block start hota hai; andar related statements group hote hain.
121	background: #064e3b;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
122	color: #6ee7b7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
123	}	Pichla block/function/object close karta hai.
124		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
125	.status-badge.partial {	Code block start hota hai; andar related statements group hote hain.
126	background: #fef3c7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
127	color: #92400e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
128	}	Pichla block/function/object close karta hai.
129	.fees-page.dark .status-badge.partial {	Code block start hota hai; andar related statements group hote hain.
130	background: #78350f;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
131	color: #fcd34d;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
132	}	Pichla block/function/object close karta hai.
133		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
134	.status-badge.unpaid {	Code block start hota hai; andar related statements group hote hain.
135	background: #fee2e2;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
136	color: #991b1b;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
137	}	Pichla block/function/object close karta hai.
138	.fees-page.dark .status-badge.unpaid {	Code block start hota hai; andar related statements group hote hain.
139	background: #7f1d1d;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
140	color: #fca5a5;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
141	}	Pichla block/function/object close karta hai.
142		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
143	/* Buttons */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
144	.actions { display: flex; gap: 6px; justify-content: center; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
145		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
146	.edit-btn {	Code block start hota hai; andar related statements group hote hain.
147	background: #465fff; color: #fff; border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
148	padding: 7px 14px; border-radius: 6px; font-size: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
149	font-weight: 600; cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
150	}	Pichla block/function/object close karta hai.
151	.edit-btn:hover { background: #3548d4; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
152		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
153	.delete-btn {	Code block start hota hai; andar related statements group hote hain.
154	background: #ef4444; color: #fff; border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
155	padding: 7px 14px; border-radius: 6px; font-size: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
156	font-weight: 600; cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
157	}	Pichla block/function/object close karta hai.
158	.delete-btn:hover { background: #dc2626; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
159		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
160	/* Empty */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
161	.empty-state { padding: 60px 20px !important; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
162	.empty-content { display: flex; flex-direction: column; align-items: center; gap: 10px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
163	.empty-icon { font-size: 48px; opacity: 0.6; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
164	.empty-content h3 { font-size: 20px; font-weight: 600; margin: 0; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
165	.fees-page.dark .empty-content h3 { color: #ffffff; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
166	.empty-content p { font-size: 14px; color: #6b7280; margin: 0 0 16px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
167	.fees-page.dark .empty-content p { color: #9ca3af; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
168	.add-first-btn {	Code block start hota hai; andar related statements group hote hain.
169	background: #10b981; color: #fff; border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
170	padding: 10px 24px; border-radius: 8px; font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
171	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
172	}	Pichla block/function/object close karta hai.
173	.add-first-btn:hover { background: #059669; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
174		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
175	/* Modal */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
176	.modal-overlay {	Code block start hota hai; andar related statements group hote hain.
177	position: fixed; inset: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
178	background: rgba(0, 0, 0, 0.6);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
179	backdrop-filter: blur(4px);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
180	display: flex; align-items: center; justify-content: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
181	z-index: 1000; padding: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
182	}	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
183		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
184	<code>.modal-box {</code>	Code block start hota hai; andar related statements group hote hain.
185	<code>background: #ffffff; border-radius: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
186	<code>max-width: 700px; width: 100%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
187	<code>max-height: 90vh; overflow-y: auto;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
188	<code>box-shadow: 0 20px 60px rgba(0, 0, 0, 0.3);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
189	<code>}</code>	Pichla block/function/object close karta hai.
190		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
191	<code>.fees-page.dark .modal-box { background: #14141f; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
192		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
193	<code>.modal-header {</code>	Code block start hota hai; andar related statements group hote hain.
194	<code>display: flex; justify-content: space-between; align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
195	<code>padding: 20px 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
196	<code>border-bottom: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
197	<code>}</code>	Pichla block/function/object close karta hai.
198		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
199	<code>.fees-page.dark .modal-header { border-bottom-color: #2a2a3a; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
200		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
201	<code>.modal-header h3 { font-size: 18px; font-weight: 700; margin: 0; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
202	<code>.fees-page.dark .modal-header h3 { color: #ffffff; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
203		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
204	<code>.modal-close {</code>	Code block start hota hai; andar related statements group hote hain.
205	<code>background: transparent; border: none; font-size: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
206	<code>color: #6b7280; cursor: pointer; padding: 4px 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
207	<code>border-radius: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
208	<code>}</code>	Pichla block/function/object close karta hai.
209	<code>.modal-close:hover { background: #f3f4f6; color: #ef4444; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
210		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
211	<code>/* Modal Form */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
212	<code>.modal-form {</code>	Code block start hota hai; andar related statements group hote hain.
213	<code>padding: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
214	<code>display: grid;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
215	<code>grid-template-columns: 1fr 1fr;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
216	<code>gap: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
217	<code>}</code>	Pichla block/function/object close karta hai.

Lin e	Original code	Simple explanation
218		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
219	<code>.form-group { display: flex; flex-direction: column; gap: 6px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
220	<code>.form-group.full-width { grid-column: 1 / -1; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
221		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
222	<code>.form-group label {</code>	Code block start hota hai; andar related statements group hote hain.
223	<code>font-size: 13px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
224	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
225	<code>color: #374151;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
226	<code>}</code>	Pichla block/function/object close karta hai.
227		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
228	<code>.fees-page.dark .form-group label { color: #d1d5db; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
229		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
230	<code>.form-group input,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
231	<code>.form-group select {</code>	Code block start hota hai; andar related statements group hote hain.
232	<code>padding: 10px 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
233	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
234	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
235	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
236	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
237	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
238	<code>outline: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
239	<code>}</code>	Pichla block/function/object close karta hai.
240		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
241	<code>.fees-page.dark .form-group input,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
242	<code>.fees-page.dark .form-group select {</code>	Code block start hota hai; andar related statements group hote hain.
243	<code>background: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
244	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
245	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
246	<code>}</code>	Pichla block/function/object close karta hai.
247		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
248	<code>.form-group input:focus,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
249	<code>.form-group select:focus {</code>	Code block start hota hai; andar related statements group hote hain.
250	<code>border-color: #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
251	<code>box-shadow: 0 0 0 3px rgba(16, 185, 129, 0.15);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
252	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
253		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
254	<code>.form-group input[readonly] {</code>	Code block start hota hai; andar related statements group hote hain.
255	<code>background: #f9fafb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
256	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
257	<code>cursor: not-allowed;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
258	<code>}</code>	Pichla block/function/object close karta hai.
259		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
260	<code>.fees-page.dark .form-group input[readonly] {</code>	Code block start hota hai; andar related statements group hote hain.
261	<code>background: #1a1a28;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
262	<code>color: #9ca3af;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
263	<code>}</code>	Pichla block/function/object close karta hai.
264		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
265	<code>/* Custom Dropdown */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
266	<code>.custom-dropdown { position: relative; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
267		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
268	<code>.dropdown-btn {</code>	Code block start hota hai; andar related statements group hote hain.
269	<code>width: 100%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
270	<code>display: flex; justify-content: space-between; align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
271	<code>padding: 10px 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
272	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
273	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
274	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
275	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
276	<code>font-size: 14px; cursor: pointer; text-align: left;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
277	<code>}</code>	Pichla block/function/object close karta hai.
278		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
279	<code>.fees-page.dark .dropdown-btn {</code>	Code block start hota hai; andar related statements group hote hain.
280	<code>background: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
281	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
282	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
283	<code>}</code>	Pichla block/function/object close karta hai.
284		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
285	<code>.dropdown-btn:hover,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
286	<code>.dropdown-btn.active {</code>	Code block start hota hai; andar related statements group hote hain.
287	<code>border-color: #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
288	box-shadow: 0 0 0 3px rgba(16, 185, 129, 0.15);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
289	}	Pichla block/function/object close karta hai.
290		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
291	.dropdown-btn.error {	Code block start hota hai; andar related statements group hote hain.
292	border-color: #ef4444 !important;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
293	box-shadow: 0 0 0 3px rgba(239, 68, 68, 0.15) !important;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
294	}	Pichla block/function/object close karta hai.
295		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
296	.dropdown-btn .arrow { font-size: 10px; color: #6b7280; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
297		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
298	.dropdown-menu {	Code block start hota hai; andar related statements group hote hain.
299	position: absolute; top: calc(100% + 6px); left: 0; right: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
300	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
301	border: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
302	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
303	box-shadow: 0 10px 30px rgba(0, 0, 0, 0.1);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
304	max-height: 200px; overflow-y: auto;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
305	z-index: 50;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
306	}	Pichla block/function/object close karta hai.
307		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
308	.fees-page.dark .dropdown-menu {	Code block start hota hai; andar related statements group hote hain.
309	background: #1f1f2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
310	border-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
311	}	Pichla block/function/object close karta hai.
312		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
313	.dropdown-item {	Code block start hota hai; andar related statements group hote hain.
314	padding: 10px 14px; cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
315	font-size: 14px; color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
316	}	Pichla block/function/object close karta hai.
317		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
318	.fees-page.dark .dropdown-item { color: #e4e4e7; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
319	.dropdown-item:hover { background: #f3f4f6; color: #10b981; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
320	.fees-page.dark .dropdown-item:hover { background: #2a2a3a; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
321		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
322	/* Error */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.

Line	Original code	Simple explanation
323	<code>.error-msg {</code>	Code block start hota hai; andar related statements group hote hain.
324	<code>font-size: 12px; color: #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
325	<code>margin-top: 4px; font-weight: 500;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
326	<code>}</code>	Pichla block/function/object close karta hai.
327		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
328	<code>/* Modal Actions */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
329	<code>.modal-actions {</code>	Code block start hota hai; andar related statements group hote hain.
330	<code>grid-column: 1 / -1;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
331	<code>display: flex; gap: 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
332	<code>justify-content: flex-end;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
333	<code>margin-top: 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
334	<code>padding-top: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
335	<code>border-top: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
336	<code>}</code>	Pichla block/function/object close karta hai.
337		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
338	<code>.fees-page.dark .modal-actions { border-top-color: #2a2a3a; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
339		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
340	<code>.save-btn {</code>	Code block start hota hai; andar related statements group hote hain.
341	<code>background: #10b981; color: #fff; border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
342	<code>padding: 10px 24px; border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
343	<code>font-size: 14px; font-weight: 600; cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
344	<code>}</code>	Pichla block/function/object close karta hai.
345	<code>.save-btn:hover { background: #059669; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
346		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
347	<code>.cancel-btn {</code>	Code block start hota hai; andar related statements group hote hain.
348	<code>background: #6b7280; color: #fff; border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
349	<code>padding: 10px 20px; border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
350	<code>font-size: 14px; font-weight: 600; cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
351	<code>}</code>	Pichla block/function/object close karta hai.
352	<code>.cancel-btn:hover { background: #4b5563; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
353		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
354	<code>/* Toast */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
355	<code>.toast-popup {</code>	Code block start hota hai; andar related statements group hote hain.
356	<code>position: fixed; top: 90px; right: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
357	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
358	<code>border-left: 4px solid #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
359	<code>border-radius: 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
360	<code>padding: 14px 18px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
361	<code>display: flex; align-items: center; gap: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
362	<code>box-shadow: 0 10px 30px rgba(0, 0, 0, 0.15);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
363	<code>z-index: 2000;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
364	<code>min-width: 280px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
365	<code>}</code>	Pichla block/function/object close karta hai.
366		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
367	<code>.toast-popup.success { border-left-color: #10b981; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
368	<code>.fees-page.dark .toast-popup { background: #1f1f2e; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
369		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
370	<code>.toast-text { flex: 1; font-size: 14px; font-weight: 500; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
371	<code>.fees-page.dark .toast-text { color: #e4e4e7; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
372		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
373	<code>.toast-close {</code>	Code block start hota hai; andar related statements group hote hain.
374	<code>background: transparent; border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
375	<code>color: #9ca3af; font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
376	<code>cursor: pointer; padding: 2px 8px; border-radius: 4px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
377	<code>}</code>	Pichla block/function/object close karta hai.
378	<code>.toast-close:hover { background: #f3f4f6; color: #ef4444; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
379		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
380	<code>/* Responsive */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
381	<code>@media (max-width: 768px) {</code>	Responsive CSS rule: screen width/device condition par styles apply karta hai.
382	<code>.fees-page { padding: 16px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
383	<code>.page-header { flex-direction: column; gap: 12px; align-items: stretch; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
384	<code>.table-wrapper { overflow-x: auto; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
385	<code>table { min-width: 900px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
386	<code>.modal-form { grid-template-columns: 1fr; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
387	<code>}</code>	Pichla block/function/object close karta hai.

src/app/pages/fees/fees.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Line	Original code	Simple explanation
1	<code><div class="fees-page" [class.dark]="themeService.isDarkMode()"></code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.

Line	Original code	Simple explanation
2		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
3	@if (showToast()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
4	<div class="toast-popup" [class.success]="toastType() === 'success'">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
5	{{ toastMessage() }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
6	<button class="toast-close" (click)="showToast.set(false)">X</button>	User click par specified TypeScript method/expression chalti hai.
7	</div>	HTML element/component: page ka structure ya UI control banata hai.
8	}	Pichla block/function/object close karta hai.
9		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
10	<div class="page-header">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
11	Fees & Payments	HTML element/component: page ka structure ya UI control banata hai.
12	<button class="add-new-btn" (click)="openModal()">+ Submit Payment</button>	User click par specified TypeScript method/expression chalti hai.
13	</div>	HTML element/component: page ka structure ya UI control banata hai.
14		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
15	<div class="table-wrapper">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
16	<table>	HTML element/component: page ka structure ya UI control banata hai.
17	<thead>	HTML element/component: page ka structure ya UI control banata hai.
18	<tr>	HTML element/component: page ka structure ya UI control banata hai.
19	<th>Payment ID</th>	HTML element/component: page ka structure ya UI control banata hai.
20	<th>Student</th>	HTML element/component: page ka structure ya UI control banata hai.
21	<th>Course</th>	HTML element/component: page ka structure ya UI control banata hai.
22	<th>Paid Amount</th>	HTML element/component: page ka structure ya UI control banata hai.
23	<th>Date</th>	HTML element/component: page ka structure ya UI control banata hai.
24	<th>Method</th>	HTML element/component: page ka structure ya UI control banata hai.
25	<th>Status</th>	HTML element/component: page ka structure ya UI control banata hai.
26	<th>Actions</th>	HTML element/component: page ka structure ya UI control banata hai.
27	</tr>	HTML element/component: page ka structure ya UI control banata hai.
28	</thead>	HTML element/component: page ka structure ya UI control banata hai.
29	<tbody>	HTML element/component: page ka structure ya UI control banata hai.
30	@for (p of payments(); track p.id) {	Angular loop: array ke har item ke liye HTML repeat karta hai.
31	<tr>	HTML element/component: page ka structure ya UI control banata hai.
32	<td>P{{ p.id }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
33	<td class="name-cell">{{ p.student_name p.enrollment_id }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
34	<td>{{ p.course_name '-' }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
35	<td>Rs. {{ p.paid_amount }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.

Line	Original code	Simple explanation
36	<td>{{ p.payment_date }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
37	<td>{{ p.payment_method }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
38	<td>	HTML element/component: page ka structure ya UI control banata hai.
39	<span class="status-badge"	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
40	[class.paid]="getStatus(p.paid_amount, p.total_fee) === 'Paid'"	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
41	[class.partial]="getStatus(p.paid_amount, p.total_fee) === 'Partial'"	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
42	[class.unpaid]="getStatus(p.paid_amount, p.total_fee) === 'Unpaid'">	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
43	{{ getStatus(p.paid_amount, p.total_fee) }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
44		HTML element/component: page ka structure ya UI control banata hai.
45	</td>	HTML element/component: page ka structure ya UI control banata hai.
46	<td>	HTML element/component: page ka structure ya UI control banata hai.
47	<div class="actions">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
48	<button class="edit-btn" (click)="editPayment(p)">Edit</button>	User click par specified TypeScript method/expression chalti hai.
49	<button class="delete-btn" (click)="deletePayment(p.id)">Delete</button>	User click par specified TypeScript method/expression chalti hai.
50	</div>	HTML element/component: page ka structure ya UI control banata hai.
51	</td>	HTML element/component: page ka structure ya UI control banata hai.
52	</tr>	HTML element/component: page ka structure ya UI control banata hai.
53	}	Pichla block/function/object close karta hai.
54		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
55	@if (payments().length === 0) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
56	<tr>	HTML element/component: page ka structure ya UI control banata hai.
57	<td colspan="8" class="empty-state">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
58	<div class="empty-content">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
59	<div class="empty-icon">■</div>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
60	No Payments Yet	HTML element/component: page ka structure ya UI control banata hai.
61	Submit a payment to see it here	HTML element/component: page ka structure ya UI control banata hai.
62	<button class="add-first-btn" (click)="openModal()">	User click par specified TypeScript method/expression chalti hai.
63	+ Submit Payment	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
64	</button>	HTML element/component: page ka structure ya UI control banata hai.
65	</div>	HTML element/component: page ka structure ya UI control banata hai.
66	</td>	HTML element/component: page ka structure ya UI control banata hai.
67	</tr>	HTML element/component: page ka structure ya UI control banata hai.
68	}	Pichla block/function/object close karta hai.
69	</tbody>	HTML element/component: page ka structure ya UI control banata hai.

Line	Original code	Simple explanation
70	</table>	HTML element/component: page ka structure ya UI control banata hai.
71	</div>	HTML element/component: page ka structure ya UI control banata hai.
72		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
73	@if (showModal()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
74	<div class="modal-overlay" (click)="closeModal()">	User click par specified TypeScript method/expression chalti hai.
75	<div class="modal-box" (click)="\$event.stopPropagation()">	User click par specified TypeScript method/expression chalti hai.
76		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
77	<div class="modal-header">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
78	{{ selectedPaymentId() === null ? 'Submit Fee Payment' : 'Update Payment' }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
79	<button class="modal-close" (click)="closeModal()">X</button>	User click par specified TypeScript method/expression chalti hai.
80	</div>	HTML element/component: page ka structure ya UI control banata hai.
81		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
82	<form [formGroup]="form" (ngSubmit)="savePayment()" class="modal-form">	Form submit par specified handler chalata hai.
83		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
84	<!-- Enrollment Dropdown -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
85	<div class="form-group full-width">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
86		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
87	<div class="form-group full-width">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
88	<label>Enrollment *</label>	HTML element/component: page ka structure ya UI control banata hai.
89		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
90	<select	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
91	formControlName="enrollment_id"	HTML input ko TS reactive form ke same-named control se connect karta hai.
92	class="form-select"	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
93	[class.error]="isValid('enrollment_id')"	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
94	(change)="onEnrollmentChange(\$event)"	Input/select ki value change par handler chalta hai.
95	>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
96	<option value="">Select Enrollment</option>	HTML element/component: page ka structure ya UI control banata hai.
97		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
98	@for (e of enrollments(); track e.id) {	Angular loop: array ke har item ke liye HTML repeat karta hai.
99	<option [value]="e.id">	Property/attribute binding: HTML property ko TS value se connect karta hai.
100	E{{ e.id }} – {{ e.student_name }} ({{ e.course_name }}) – Rs. {{ e.course_fee }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
101	</option>	HTML element/component: page ka structure ya UI control banata hai.
102	}	Pichla block/function/object close karta hai.
103	</select>	HTML element/component: page ka structure ya UI control banata hai.

Line	Original code	Simple explanation
104		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
105	@if (isInvalid('enrollment_id')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
106		CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
107	■ Enrollment required	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
108		HTML element/component: page ka structure ya UI control banata hai.
109	}	Pichla block/function/object close karta hai.
110	</div>	HTML element/component: page ka structure ya UI control banata hai.
111	@if (isInvalid('enrollment_id')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
112	■ Enrollment required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
113	}	Pichla block/function/object close karta hai.
114	</div>	HTML element/component: page ka structure ya UI control banata hai.
115		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
116	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
117	<label>Total Course Fee</label>	HTML element/component: page ka structure ya UI control banata hai.
118	<input formControlName="total_fee" readonly />	HTML input ko TS reactive form ke same-named control se connect karta hai.
119	</div>	HTML element/component: page ka structure ya UI control banata hai.
120		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
121	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
122	<label>Previously Paid</label>	HTML element/component: page ka structure ya UI control banata hai.
123	<input [value]="previouslyPaid()" readonly />	Property/attribute binding: HTML property ko TS value se connect karta hai.
124	</div>	HTML element/component: page ka structure ya UI control banata hai.
125		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
126	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
127	<label>Payment Amount *</label>	HTML element/component: page ka structure ya UI control banata hai.
128	<input type="number" formControlName="paid_amount"	HTML input ko TS reactive form ke same-named control se connect karta hai.
129	(input)="onPaidChange(\$event)"	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
130	[class.error]="isInvalid('paid_amount')" />	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
131	@if (isInvalid('paid_amount')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
132	■ Payment amount required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
133	}	Pichla block/function/object close karta hai.
134	</div>	HTML element/component: page ka structure ya UI control banata hai.
135		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
136	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
137	<label>Remaining Balance</label>	HTML element/component: page ka structure ya UI control banata hai.

Line	Original code	Simple explanation
138	<input [value]="remainingBalance()" readonly />	Property/attribute binding: HTML property ko TS value se connect karta hai.
139	</div>	HTML element/component: page ka structure ya UI control banata hai.
140		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
141	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
142	<label>Payment Method *</label>	HTML element/component: page ka structure ya UI control banata hai.
143	<select formControlName="payment_method" class="status-select">	HTML input ko TS reactive form ke same-named control se connect karta hai.
144	@for (m of paymentMethods; track m) {	Angular loop: array ke har item ke liye HTML repeat karta hai.
145	<option [value]="m">{{ m }}</option>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
146	}	Pichla block/function/object close karta hai.
147	</select>	HTML element/component: page ka structure ya UI control banata hai.
148	</div>	HTML element/component: page ka structure ya UI control banata hai.
149		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
150	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
151	<label>Payment Date *</label>	HTML element/component: page ka structure ya UI control banata hai.
152	<input type="date" formControlName="payment_date"	HTML input ko TS reactive form ke same-named control se connect karta hai.
153	[class.error]="isValid('payment_date')" />	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
154	@if (isValid('payment_date')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
155	■ Date required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
156	}	Pichla block/function/object close karta hai.
157	</div>	HTML element/component: page ka structure ya UI control banata hai.
158		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
159	<div class="modal-actions">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
160	<button type="button" class="cancel-btn" (click)="closeModal()">Cancel</button>	User click par specified TypeScript method/expression chalti hai.
161	<button type="submit" class="save-btn">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
162	{{ selectedPaymentId() === null ? 'Submit Payment' : 'Update' }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
163	</button>	HTML element/component: page ka structure ya UI control banata hai.
164	</div>	HTML element/component: page ka structure ya UI control banata hai.
165		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
166	</form>	HTML element/component: page ka structure ya UI control banata hai.
167	</div>	HTML element/component: page ka structure ya UI control banata hai.
168	</div>	HTML element/component: page ka structure ya UI control banata hai.
169	}	Pichla block/function/object close karta hai.
170		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
171	</div>	HTML element/component: page ka structure ya UI control banata hai.

src/app/pages/fees/fees.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	<code>import { Component, OnInit, signal, inject } from '@angular/core';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
2	<code>import { CommonModule } from '@angular/common';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
3	<code>import { HttpClient } from '@angular/common/http';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
4	<code>import { FormBuilder, ReactiveFormsModule, Validators } from '@angular/forms';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
5	<code>import { ThemeService } from '../services/theme';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
6		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
7	<code>@Component({</code>	Angular decorator: neech wali class ko Angular component declare karta hai.
8	<code>selector: 'app-fees',</code>	Component ka HTML tag name; parent template mein isi selector se component lagta hai.
9	<code>standalone: true,</code>	Component standalone hai; NgModule mein declare karna zaroori nahi.
10	<code>imports: [CommonModule, ReactiveFormsModule],</code>	Template mein use hone wale Angular directives/forms/router features import karta hai.
11	<code>templateUrl: './fees.html',</code>	Is component ki HTML template file ka path.
12	<code>styleUrl: './fees.css'</code>	Is component ki CSS file ka path.
13	<code>})</code>	Pichla block/function/object close karta hai.
14	<code>export class Fees implements OnInit {</code>	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
15		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
16	<code>private http = inject(HttpClient);</code>	Angular se HttpClient service leta hai; API request bhejne ke liye.
17	<code>private fb = inject(FormBuilder);</code>	FormBuilder leta hai taake reactive form aur controls bana sake.
18	<code>public themeService = inject(ThemeService);</code>	Shared ThemeService use hoti hai taake theme state/component access milay.
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	<code>showModal = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
21	<code>payments = signal<any[]>([]);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
22	<code>enrollments = signal<any[]>([]);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
23	<code>selectedPaymentId = signal<number null>(null);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
24		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
25	<code>// Toast</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
26	<code>showToast = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
27	<code>toastMessage = signal<string>('');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
28	<code>toastType = signal<'error' 'success'>('error');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
29		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
30	<code>// Enrollment Dropdown</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
31	<code>showEnrollDropdown = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
32	<code>selectedEnrollLabel = signal<string>('');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
33		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
34	// Auto calculation signals	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
35	totalFee = signal<number>(0);	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
36	previouslyPaid = signal<number>(0);	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
37	remainingBalance = signal<number>(0);	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
38		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
39	// Payment methods	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
40	paymentMethods = ['Cash', 'Bank Transfer', 'Online', 'Cheque'];	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
41		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
42	form = this.fb.group({	Reactive form group banata hai; fields, current values aur validation ek jagah manage hotay hain.
43	enrollment_id: ['', Validators.required],	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
44	student_name: ['',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
45	course_name: ['',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
46	total_fee: ['', Validators.required],	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
47	paid_amount: ['', Validators.required],	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
48	payment_method: ['Cash', Validators.required],	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
49	payment_date: [new Date().toISOString().slice(0, 10), Validators.required]	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
50	});	Pichla block/function/object close karta hai.
51		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
52	ngOnInit() {	Component initialize hone ke baad Angular is method ko automatically call karta hai.
53	this.loadPayments();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
54	this.loadEnrollments();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
55	}	Pichla block/function/object close karta hai.
56		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
57	showToastMsg(msg: string, type: 'error' 'success' = 'error') {	Function/method define karta hai; call hone par is block ka kaam hota hai.
58	this.toastMessage.set(msg);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
59	this.toastType.set(type);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
60	this.showToast.set(true);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
61	setTimeout(() => this.showToast.set(false), 3000);	Arrow function: callback/function ko short syntax mein define karta hai.
62	}	Pichla block/function/object close karta hai.
63		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
64	loadPayments() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
65	this.http.get<any[]>('http://localhost:3000/payments')	HTTP GET request: backend se data read karta hai.
66	.subscribe(data => this.payments.set(data));	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
67	}	Pichla block/function/object close karta hai.
68		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
69	loadEnrollments() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
70	this.http.get<any[]>('http://localhost:3000/enrollments')	HTTP GET request: backend se data read karta hai.
71	.subscribe(data => this.enrollments.set(data));	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
72	}	Pichla block/function/object close karta hai.
73	onEnrollmentChange(event: Event) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
74	// Select se selected ID lena	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
75	const select = event.target as HTMLSelectElement;	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
76		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
77	// ID ko string mein lena	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
78	const enrollmentId = select.value;	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
79		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
80	// Agar kuch select nahi kiya	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
81	if (!enrollmentId) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
82	return;	Current function ko yahin stop karta hai.
83	}	Pichla block/function/object close karta hai.
84		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
85	// Selected enrollment find karna	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
86	const enrollment = this.enrollments()	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
87	.find(e => String(e.id) === enrollmentId);	Arrow function: callback/function ko short syntax mein define karta hai.
88		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
89	// Agar enrollment nahi mila	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
90	if (!enrollment) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
91	return;	Current function ko yahin stop karta hai.
92	}	Pichla block/function/object close karta hai.
93		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
94	// Form mein enrollment ki information set karna	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
95	this.form.patchValue({	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
96	enrollment_id: enrollment.id,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
97	student_name: enrollment.student_name '',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
98	course_name: enrollment.course_name '',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
99	total_fee: enrollment.course_fee	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
100	});	Pichla block/function/object close karta hai.
101		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
102	// Total fee set karna	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
103	const total = Number(enrollment.course_fee) 0;	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
104	this.totalFee.set(total);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
105		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
106	// Pehle kitni payment hui hai wo calculate karna	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
107	const paid = this.payments()	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
108	.filter(p =>	Arrow function: callback/function ko short syntax mein define karta hai.
109	String(p.enrollment_id) === String(enrollment.id)	Function/method define karta hai; call hone par is block ka kaam hota hai.
110	)	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
111	.reduce(	Array ke items ko combine karke aik result banata hai, jaise total fee.
112	(sum, p) => sum + Number(p.paid_amount 0),	Arrow function: callback/function ko short syntax mein define karta hai.
113	0	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
114	);	Pichla block/function/object close karta hai.
115		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
116	// Previously paid set	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
117	this.previouslyPaid.set(paid);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
118		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
119	// Remaining balance calculate	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
120	this.remainingBalance.set(total - paid);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
121	}	Pichla block/function/object close karta hai.
122		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
123	toggleEnrollDropdown() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
124	this.showEnrollDropdown.update(v => !v);	Arrow function: callback/function ko short syntax mein define karta hai.
125	}	Pichla block/function/object close karta hai.
126		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
127	selectEnrollment(e: any) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
128	this.selectedEnrollLabel.set(`E\${e.id} - \${e.student_name    ''} (\${e.course_name    ''})`);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
129		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
130	this.form.patchValue({	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
131	enrollment_id: e.id,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
132	student_name: e.student_name '',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
133	course_name: e.course_name '',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
134	total_fee: e.course_fee	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
135	});	Pichla block/function/object close karta hai.
136		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
137	this.totalFee.set(Number(e.course_fee) 0);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
138		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
139	const paid = this.payments()	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.

Line	Original code	Simple explanation
140	<code>.filter(p => p.enrollment_id === e.id)</code>	Arrow function: callback/function ko short syntax mein define karta hai.
141	<code>.reduce((sum, p) => sum + Number(p.paid_amount 0), 0);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
142	<code>this.previouslyPaid.set(paid);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
143	<code>this.remainingBalance.set((Number(e.course_fee) 0) - paid);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
144		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
145	<code>this.showEnrollDropdown.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
146	<code>}</code>	Pichla block/function/object close karta hai.
147		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
148	<code>onPaidChange(event: Event) {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
149	<code>const val = Number((event.target as HTMLInputElement).value) 0;</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
150	<code>const remaining = this.totalFee() - this.previouslyPaid() - val;</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
151	<code>this.remainingBalance.set(remaining);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
152	<code>}</code>	Pichla block/function/object close karta hai.
153		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
154	<code>openModal() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
155	<code>this.selectedPaymentId.set(null);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
156	<code>this.form.reset({</code>	Form ko clear/default values par reset karta hai.
157	<code>payment_method: 'Cash',</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
158	<code>payment_date: new Date().toISOString().slice(0, 10)</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
159	<code>});</code>	Pichla block/function/object close karta hai.
160	<code>this.selectedEnrollLabel.set('');</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
161	<code>this.totalFee.set(0);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
162	<code>this.previouslyPaid.set(0);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
163	<code>this.remainingBalance.set(0);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
164	<code>this.showModal.set(true);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
165	<code>}</code>	Pichla block/function/object close karta hai.
166		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
167	<code>closeModal() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
168	<code>this.showModal.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
169	<code>this.selectedPaymentId.set(null);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
170	<code>this.form.reset();</code>	Form ko clear/default values par reset karta hai.
171	<code>this.selectedEnrollLabel.set('');</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
172	<code>this.totalFee.set(0);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
173	<code>this.previouslyPaid.set(0);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
174	<code>this.remainingBalance.set(0);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
175	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
176		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
177	savePayment() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
178	if (this.form.invalid) {	Check karta hai ke form/field validation pass kar raha hai ya nahi.
179	this.form.markAllAsTouched();	Validation messages dikhane ke liye sab controls ko touched mark karta hai.
180	this.showToastMsg('■ Please fill all required fields', 'error');	User ko success/error ka short notification dikhata hai.
181	return;	Current function ko yahin stop karta hai.
182	}	Pichla block/function/object close karta hai.
183		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
184	const url = 'http://localhost:3000/payments';	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
185		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
186	if (this.selectedPaymentId() === null) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
187	this.http.post(url, this.form.value).subscribe(() => {	HTTP POST request: backend mein naya record create karta hai.
188	this.showToastMsg('■ Payment submitted successfully!', 'success');	User ko success/error ka short notification dikhata hai.
189	this.closeModal();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
190	this.loadPayments();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
191	});	Pichla block/function/object close karta hai.
192	} else {	Alternative branch: pehli condition false ho to yeh code chalta hai.
193	this.http.put(`\${url}/\${this.selectedPaymentId()}`, this.form.value)	HTTP PUT request: selected record ko update/replace karta hai.
194	.subscribe(() => {	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
195	this.showToastMsg('■ Payment updated!', 'success');	User ko success/error ka short notification dikhata hai.
196	this.closeModal();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
197	this.loadPayments();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
198	});	Pichla block/function/object close karta hai.
199	}	Pichla block/function/object close karta hai.
200	}	Pichla block/function/object close karta hai.
201		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
202	editPayment(p: any) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
203	this.selectedPaymentId.set(p.id);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
204	this.form.setValue({	Form ke tamam controls ko values deta hai; required controls miss hon to error ho sakta hai.
205	enrollment_id: p.enrollment_id,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
206	student_name: p.student_name '',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
207	course_name: p.course_name '',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
208	total_fee: p.total_fee,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
209	paid_amount: p.paid_amount,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
210	payment_method: p.payment_method,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
211	payment_date: p.payment_date	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
212	});	Pichla block/function/object close karta hai.
213	this.selectedEnrollLabel.set(`E\${p.enrollment_id}`);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
214	this.totalFee.set(p.total_fee 0);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
215	this.previouslyPaid.set(0);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
216	this.remainingBalance.set((p.total_fee 0) - (p.paid_amount 0));	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
217	this.showModal.set(true);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
218	}	Pichla block/function/object close karta hai.
219		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
220	deletePayment(id: number) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
221	if (!confirm('Delete this payment?')) return;	Condition: andar ka code sirf tab chalta hai jab condition true ho.
222		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
223	this.http.delete(`http://localhost:3000/payments/\${id}`)	HTTP DELETE request: selected record remove karta hai.
224	.subscribe(() => {	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
225	this.showToastMsg('■ Payment deleted', 'success');	User ko success/error ka short notification dikhata hai.
226	this.loadPayments();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
227	});	Pichla block/function/object close karta hai.
228	}	Pichla block/function/object close karta hai.
229		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
230	getStatus(paid: number, total: number): string {	Code block start hota hai; andar related statements group hote hain.
231	const p = Number(paid) 0;	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
232	const t = Number(total) 0;	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
233	if (p <= 0) return 'Unpaid';	Condition: andar ka code sirf tab chalta hai jab condition true ho.
234	if (p >= t) return 'Paid';	Condition: andar ka code sirf tab chalta hai jab condition true ho.
235	return 'Partial';	Function ka result return karta hai.
236	}	Pichla block/function/object close karta hai.
237		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
238	isInvalid(field: string): boolean {	Code block start hota hai; andar related statements group hote hain.
239	const c = this.form.get(field);	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
240	return !(c && c.invalid && (c.dirty c.touched));	Check karta hai ke form/field validation pass kar raha hai ya nahi.
241	}	Pichla block/function/object close karta hai.
242	}	Pichla block/function/object close karta hai.

src/app/pages/setting/setting.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	/* Settings Page */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
2	.settings-page {	Code block start hota hai; andar related statements group hote hain.
3	padding: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
4	background: #f5f6fa;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
5	min-height: 100vh;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
6	font-family: 'Inter', sans-serif;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
7	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
8	transition: background 0.3s, color 0.3s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
9	}	Pichla block/function/object close karta hai.
10	.settings-page.dark { background: #0f0f1a; color: #e4e4e7; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
11		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
12	.page-header { margin-bottom: 24px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
13	.page-header h2 { font-size: 24px; font-weight: 700; margin: 0; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
14	.settings-page.dark .page-header h2 { color: #ffffff; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
15		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
16	/* Tabs */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
17	.tabs {	Code block start hota hai; andar related statements group hote hain.
18	display: flex; gap: 8px; margin-bottom: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
19	border-bottom: 1px solid #e5e7eb; padding-bottom: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
20	flex-wrap: wrap;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
21	}	Pichla block/function/object close karta hai.
22	.settings-page.dark .tabs { border-bottom-color: #2a2a3a; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
23		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
24	.tab-btn {	Code block start hota hai; andar related statements group hote hain.
25	background: transparent; border: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
26	padding: 10px 20px; border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
27	font-size: 14px; font-weight: 600; color: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
28	cursor: pointer; transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
29	}	Pichla block/function/object close karta hai.
30	.settings-page.dark .tab-btn { border-color: #2a2a3a; color: #9ca3af; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
31	.tab-btn:hover { border-color: #10b981; color: #10b981; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
32	.tab-btn.active {	Code block start hota hai; andar related statements group hote hain.
33	background: #10b981; color: white; border-color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
34	box-shadow: 0 4px 12px rgba(16, 185, 129, 0.3);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
35	}	Pichla block/function/object close karta hai.
36		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
37	/* Settings Card */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
38	.settings-card {	Code block start hota hai; andar related statements group hote hain.

Lin e	Original code	Simple explanation
39	background: #ffffff; border-radius: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
40	border: 1px solid #e5e7eb; padding: 28px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
41	box-shadow: 0 2px 8px rgba(0, 0, 0, 0.04);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
42	max-width: 700px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
43	}	Pichla block/function/object close karta hai.
44	.settings-page.dark .settings-card {	Code block start hota hai; andar related statements group hote hain.
45	background: #14141f; border-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
46	}	Pichla block/function/object close karta hai.
47	.settings-card h3 {	Code block start hota hai; andar related statements group hote hain.
48	font-size: 18px; font-weight: 700; margin: 0 0 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
49	}	Pichla block/function/object close karta hai.
50	.settings-page.dark .settings-card h3 { color: #ffffff; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
51		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
52	/* Form */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
53	.settings-form { display: flex; flex-direction: column; gap: 16px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
54	.form-group { display: flex; flex-direction: column; gap: 6px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
55	.form-group label { font-size: 13px; font-weight: 600; color: #374151; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
56	.settings-page.dark .form-group label { color: #d1d5db; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
57		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
58	.form-group input {	Code block start hota hai; andar related statements group hote hain.
59	padding: 10px 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
60	border: 1px solid #e5e7eb; border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
61	background: #ffffff; color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
62	font-size: 14px; outline: none; transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
63	}	Pichla block/function/object close karta hai.
64	.settings-page.dark .form-group input {	Code block start hota hai; andar related statements group hote hain.
65	background: #1f1f2e; border-color: #2a2a3a; color: #e4e4e7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
66	}	Pichla block/function/object close karta hai.
67	.form-group input:focus {	Code block start hota hai; andar related statements group hote hain.
68	border-color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
69	box-shadow: 0 0 3px rgba(16, 185, 129, 0.15);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
70	}	Pichla block/function/object close karta hai.
71	.form-group input.error {	Code block start hota hai; andar related statements group hote hain.
72	border-color: #ef4444 !important;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
73	box-shadow: 0 0 3px rgba(239, 68, 68, 0.15) !important;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
74	}	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
75		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
76	<code>.error-msg { font-size: 12px; color: #ef4444; font-weight: 500; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
77		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
78	<code>.form-actions { display: flex; justify-content: flex-end; margin-top: 12px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
79	<code>.save-btn {</code>	Code block start hota hai; andar related statements group hote hain.
80	<code>background: #10b981; color: white; border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
81	<code>padding: 10px 24px; border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
82	<code>font-size: 14px; font-weight: 600; cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
83	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
84	<code>}</code>	Pichla block/function/object close karta hai.
85	<code>.save-btn:hover { background: #059669; transform: translateY(-1px); }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
86		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
87	<code>/* Appearance */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
88	<code>.appearance-section { padding: 12px 0; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
89	<code>.label { font-size: 13px; font-weight: 600; color: #374151; margin-bottom: 12px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
90	<code>.settings-page.dark .label { color: #d1d5db; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
91	<code>.color-picker { display: flex; gap: 12px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
92	<code>.color-dot {</code>	Code block start hota hai; andar related statements group hote hain.
93	<code>width: 40px; height: 40px; border-radius: 50%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
94	<code>cursor: pointer; border: 3px solid transparent;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
95	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
96	<code>}</code>	Pichla block/function/object close karta hai.
97	<code>.color-dot:hover { transform: scale(1.15); border-color: #ffffff; box-shadow: 0 4px 12px rgba(0,0,0,0.2); }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
98		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
99	<code>/* Data */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
100	<code>.data-section {</code>	Code block start hota hai; andar related statements group hote hain.
101	<code>padding: 16px 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
102	<code>border-bottom: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
103	<code>}</code>	Pichla block/function/object close karta hai.
104	<code>.settings-page.dark .data-section { border-bottom-color: #2a2a3a; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
105	<code>.data-section:last-child { border-bottom: none; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
106	<code>.data-section p { font-size: 14px; color: #6b7280; margin: 0 0 12px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
107	<code>.settings-page.dark .data-section p { color: #9ca3af; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
108	<code>.data-section.danger p { color: #ef4444; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
109		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
110	<code>.export-btn, .danger-btn {</code>	Code block start hota hai; andar related statements group hote hain.
111	<code>padding: 10px 20px; border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
112	<code>font-size: 14px; font-weight: 600; cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
113	<code>border: none; transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
114	<code>}</code>	Pichla block/function/object close karta hai.
115	<code>.export-btn { background: #465fff; color: white; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
116	<code>.export-btn:hover { background: #3548d4; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
117	<code>.danger-btn { background: #ef4444; color: white; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
118	<code>.danger-btn:hover { background: #dc2626; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
119		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
120	<code>/* Toast */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
121	<code>.toast-popup {</code>	Code block start hota hai; andar related statements group hote hain.
122	<code>position: fixed; top: 90px; right: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
123	<code>background: #ffffff; border-left: 4px solid #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
124	<code>border-radius: 10px; padding: 14px 18px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
125	<code>display: flex; align-items: center; gap: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
126	<code>box-shadow: 0 10px 30px rgba(0,0,0,0.15);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
127	<code>z-index: 2000; min-width: 280px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
128	<code>}</code>	Pichla block/function/object close karta hai.
129	<code>.toast-popup.success { border-left-color: #10b981; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
130	<code>.settings-page.dark .toast-popup { background: #1f1f2e; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
131	<code>.toast-text { flex: 1; font-size: 14px; font-weight: 500; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
132	<code>.toast-close { background: transparent; border: none; color: #9ca3af; cursor: pointer; padding: 2px 8px; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

src/app/pages/setting/setting.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Line	Original code	Simple explanation
1	<code><div class="settings-page" [class.dark]="themeService.isDarkMode()"></code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
2		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
3	<code>@if (showToast()) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
4	<code><div class="toast-popup" [class.success]="toastType() === 'success'"></code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
5	<code>{{ toastMessage() }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
6	<code><button class="toast-close" (click)="showToast.set(false)">X</button></code>	User click par specified TypeScript method/expression chalti hai.
7	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.

Line	Original code	Simple explanation
8	}	Pichla block/function/object close karta hai.
9		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
10	<div class="page-header">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
11	Settings	HTML element/component: page ka structure ya UI control banata hai.
12	</div>	HTML element/component: page ka structure ya UI control banata hai.
13		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
14	<!-- Tabs -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
15	<div class="tabs">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
16	<button class="tab-btn" [class.active]="activeTab() === 'profile'" (click)="setTab('profile')">	User click par specified TypeScript method/expression chalti hai.
17	■ Profile	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
18	</button>	HTML element/component: page ka structure ya UI control banata hai.
19	<button class="tab-btn" [class.active]="activeTab() === 'password'" (click)="setTab('password')">	User click par specified TypeScript method/expression chalti hai.
20	■ Password	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
21	</button>	HTML element/component: page ka structure ya UI control banata hai.
22	<button class="tab-btn" [class.active]="activeTab() === 'appearance'" (click)="setTab('appearance')">	User click par specified TypeScript method/expression chalti hai.
23	■ Appearance	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
24	</button>	HTML element/component: page ka structure ya UI control banata hai.
25	<button class="tab-btn" [class.active]="activeTab() === 'data'" (click)="setTab('data')">	User click par specified TypeScript method/expression chalti hai.
26	■ Data	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
27	</button>	HTML element/component: page ka structure ya UI control banata hai.
28	</div>	HTML element/component: page ka structure ya UI control banata hai.
29		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
30	<!-- PROFILE TAB -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
31	@if (activeTab() === 'profile') {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
32	<div class="settings-card">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
33	Profile Information	HTML element/component: page ka structure ya UI control banata hai.
34	<form [formGroup]="profileForm" (ngSubmit)="saveProfile()" class="settings-form">	Form submit par specified handler chalata hai.
35	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
36	<label>Name *</label>	HTML element/component: page ka structure ya UI control banata hai.
37	<input formControlName="name" [class.error]="isInvalid(profileForm, 'name')" />	HTML input ko TS reactive form ke same-named control se connect karta hai.
38	@if (isInvalid(profileForm, 'name')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
39	■ Name required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
40	}	Pichla block/function/object close karta hai.
41	</div>	HTML element/component: page ka structure ya UI control banata hai.

Lin e	Original code	Simple explanation
42	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
43	<label>Email *</label>	HTML element/component: page ka structure ya UI control banata hai.
44	<input type="email" formControlName="email" [class.error]="isValid(profileForm, 'email')" />	HTML input ko TS reactive form ke same-named control se connect karta hai.
45	@if (isValid(profileForm, 'email')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
46	■ Valid email required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
47	}	Pichla block/function/object close karta hai.
48	</div>	HTML element/component: page ka structure ya UI control banata hai.
49	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
50	<label>Phone</label>	HTML element/component: page ka structure ya UI control banata hai.
51	<input formControlName="phone" />	HTML input ko TS reactive form ke same-named control se connect karta hai.
52	</div>	HTML element/component: page ka structure ya UI control banata hai.
53	<div class="form-actions">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
54	<button type="submit" class="save-btn">■ Save Changes</button>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
55	</div>	HTML element/component: page ka structure ya UI control banata hai.
56	</form>	HTML element/component: page ka structure ya UI control banata hai.
57	</div>	HTML element/component: page ka structure ya UI control banata hai.
58	}	Pichla block/function/object close karta hai.
59		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
60	<!-- PASSWORD TAB -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
61	@if (activeTab() === 'password') {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
62	<div class="settings-card">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
63	Change Password	HTML element/component: page ka structure ya UI control banata hai.
64	<form [formGroup]="passwordForm" (ngSubmit)="changePassword()" class="settings-form">	Form submit par specified handler chalata hai.
65	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
66	<label>Current Password *</label>	HTML element/component: page ka structure ya UI control banata hai.
67	<input type="password" formControlName="current" [class.error]="isValid(passwordForm, 'current')" />	HTML input ko TS reactive form ke same-named control se connect karta hai.
68	@if (isValid(passwordForm, 'current')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
69	■ Current password required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
70	}	Pichla block/function/object close karta hai.
71	</div>	HTML element/component: page ka structure ya UI control banata hai.
72	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
73	<label>New Password *</label>	HTML element/component: page ka structure ya UI control banata hai.
74	<input type="password" formControlName="newPass" [class.error]="isValid(passwordForm, 'newPass')" />	HTML input ko TS reactive form ke same-named control se connect karta hai.
75	@if (isValid(passwordForm, 'newPass')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.

Line	Original code	Simple explanation
76	<code>■ Min 6 characters</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
77	<code>}</code>	Pichla block/function/object close karta hai.
78	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
79	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
80	<code><label>Confirm Password *</label></code>	HTML element/component: page ka structure ya UI control banata hai.
81	<code><input type="password" formControlName="confirm" [class.error]="isValid(passwordForm, 'confirm')" /></code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
82	<code>@if (isValid(passwordForm, 'confirm')) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
83	<code>■ Confirm password required</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
84	<code>}</code>	Pichla block/function/object close karta hai.
85	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
86	<code><div class="form-actions"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
87	<code><button type="submit" class="save-btn">■ Update Password</button></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
88	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
89	<code></form></code>	HTML element/component: page ka structure ya UI control banata hai.
90	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
91	<code>}</code>	Pichla block/function/object close karta hai.
92		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
93	<code><!-- APPEARANCE TAB --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
94	<code>@if (activeTab() === 'appearance') {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
95	<code><div class="settings-card"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
96	<code><h3>Appearance</h3></code>	HTML element/component: page ka structure ya UI control banata hai.
97	<code><div class="appearance-section"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
98	<code><p class="label">Accent Color</p></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
99	<code><div class="color-picker"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
100	<code><div class="color-dot" style="background: #10b981;" (click)="setAccent('#10b981')"></div></code>	User click par specified TypeScript method/expression chalti hai.
101	<code><div class="color-dot" style="background: #465fff;" (click)="setAccent('#465fff')"></div></code>	User click par specified TypeScript method/expression chalti hai.
102	<code><div class="color-dot" style="background: #a855f7;" (click)="setAccent('#a855f7')"></div></code>	User click par specified TypeScript method/expression chalti hai.
103	<code><div class="color-dot" style="background: #ef4444;" (click)="setAccent('#ef4444')"></div></code>	User click par specified TypeScript method/expression chalti hai.
104	<code><div class="color-dot" style="background: #f59e0b;" (click)="setAccent('#f59e0b')"></div></code>	User click par specified TypeScript method/expression chalti hai.
105	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
106	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
107	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
108	<code>}</code>	Pichla block/function/object close karta hai.
109		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
110	<code><!-- DATA TAB --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.

Line	Original code	Simple explanation
111	@if (activeTab() === 'data') {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
112	<div class="settings-card">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
113	Data Management	HTML element/component: page ka structure ya UI control banata hai.
114	<div class="data-section">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
115	Export all your data as a JSON file for backup.	HTML element/component: page ka structure ya UI control banata hai.
116	<button class="export-btn" (click)="exportJSON()">■ Export JSON</button>	User click par specified TypeScript method/expression chalti hai.
117	</div>	HTML element/component: page ka structure ya UI control banata hai.
118	<div class="data-section danger">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
119	■ This will delete all students, courses, enrollments and payments. This cannot be undone.	HTML element/component: page ka structure ya UI control banata hai.
120	<button class="danger-btn" (click)="resetAllData()">■ Reset All Data</button>	User click par specified TypeScript method/expression chalti hai.
121	</div>	HTML element/component: page ka structure ya UI control banata hai.
122	</div>	HTML element/component: page ka structure ya UI control banata hai.
123	}	Pichla block/function/object close karta hai.
124		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
125	</div>	HTML element/component: page ka structure ya UI control banata hai.

src/app/pages/setting/setting.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	import { Component, OnInit, signal, inject } from '@angular/core';	Import: doosri library/file se required class ya function is file mein available karta hai.
2	import { CommonModule } from '@angular/common';	Import: doosri library/file se required class ya function is file mein available karta hai.
3	import { HttpClient } from '@angular/common/http';	Import: doosri library/file se required class ya function is file mein available karta hai.
4	import { FormBuilder, ReactiveFormsModule, Validators } from '@angular/forms';	Import: doosri library/file se required class ya function is file mein available karta hai.
5	import { ThemeService } from '../services/theme';	Import: doosri library/file se required class ya function is file mein available karta hai.
6		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
7	@Component({	Angular decorator: neech wali class ko Angular component declare karta hai.
8	selector: 'app-settings',	Component ka HTML tag name; parent template mein isi selector se component lagta hai.
9	standalone: true,	Component standalone hai; NgModule mein declare karna zaroori nahi.
10	imports: [CommonModule, ReactiveFormsModule],	Template mein use hone wale Angular directives/forms/router features import karta hai.
11	templateUrl: './setting.html',	Is component ki HTML template file ka path.
12	styleUrl: './setting.css'	Is component ki CSS file ka path.
13	})	Pichla block/function/object close karta hai.
14	export class Setting implements OnInit {	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
15		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
16	private http = inject(HttpClient);	Angular se HttpClient service leta hai; API request bhejne ke liye.

Lin e	Original code	Simple explanation
17	<code>private fb = inject(FormBuilder);</code>	FormBuilder leta hai taake reactive form aur controls bana sake.
18	<code>public themeService = inject(ThemeService);</code>	Shared ThemeService use hoti hai taake theme state/component access milay.
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	<code>// Toast</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
21	<code>showToast = signal<boolean>(false);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
22	<code>toastMessage = signal<string>('');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
23	<code>toastType = signal<'error' 'success'>('error');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
24		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
25	<code>// Active tab</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
26	<code>activeTab = signal<'profile' 'password' 'appearance' 'data'>('profile');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
27		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
28	<code>// Profile Form</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
29	<code>profileForm = this.fb.group({</code>	Reactive form group banata hai; fields, current values aur validation ek jagah manage hotay hain.
30	<code>name: ['Admin', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
31	<code>email: ['admin@cms.com', [Validators.required, Validators.email]],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
32	<code>phone: ['03001234567']</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
33	<code>});</code>	Pichla block/function/object close karta hai.
34		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
35	<code>// Password Form</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
36	<code>passwordForm = this.fb.group({</code>	Reactive form group banata hai; fields, current values aur validation ek jagah manage hotay hain.
37	<code>current: ['', Validators.required],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
38	<code>newPass: ['', [Validators.required, Validators.minLength(6)]],</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
39	<code>confirm: ['', Validators.required]</code>	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
40	<code>});</code>	Pichla block/function/object close karta hai.
41		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
42	<code>// Appearance</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
43	<code>accentColor = signal<string>('#10b981');</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
44		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
45	<code>ngOnInit() {</code>	Component initialize hone ke baad Angular is method ko automatically call karta hai.
46	<code>// Load saved profile from localStorage</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
47	<code>const saved = localStorage.getItem('admin_profile');</code>	Browser ke localStorage se saved string read karta hai.
48	<code>if (saved) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
49	<code>this.profileForm.patchValue(JSON.parse(saved));</code>	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
50	<code>}</code>	Pichla block/function/object close karta hai.
51	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
52		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
53	<code>showToastMsg(msg: string, type: 'error' 'success' = 'error') {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
54	<code> this.toastMessage.set(msg);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
55	<code> this.toastType.set(type);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
56	<code> this.showToast.set(true);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
57	<code> setTimeout(() => this.showToast.set(false), 3000);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
58	<code>}</code>	Pichla block/function/object close karta hai.
59		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
60	<code>setTab(tab: 'profile' 'password' 'appearance' 'data') {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
61	<code> this.activeTab.set(tab);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
62	<code>}</code>	Pichla block/function/object close karta hai.
63		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
64	<code>// ■ Save Profile</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
65	<code>saveProfile() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
66	<code> if (this.profileForm.invalid) {</code>	Check karta hai ke form/field validation pass kar raha hai ya nahi.
67	<code> this.profileForm.markAllAsTouched();</code>	Validation messages dikhane ke liye sab controls ko touched mark karta hai.
68	<code> this.showToastMsg('■ Please fill all fields correctly', 'error');</code>	User ko success/error ka short notification dikhata hai.
69	<code> return;</code>	Current function ko yahin stop karta hai.
70	<code>}</code>	Pichla block/function/object close karta hai.
71		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
72	<code>localStorage.setItem('admin_profile', JSON.stringify(this.profileForm.value));</code>	Browser localStorage mein string save karta hai; browser-side persistence.
73	<code> this.showToastMsg('■ Profile updated successfully!', 'success');</code>	User ko success/error ka short notification dikhata hai.
74	<code>}</code>	Pichla block/function/object close karta hai.
75		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
76	<code>// ■ Change Password</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
77	<code>changePassword() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
78	<code> if (this.passwordForm.invalid) {</code>	Check karta hai ke form/field validation pass kar raha hai ya nahi.
79	<code> this.passwordForm.markAllAsTouched();</code>	Validation messages dikhane ke liye sab controls ko touched mark karta hai.
80	<code> this.showToastMsg('■ Please fill all fields', 'error');</code>	User ko success/error ka short notification dikhata hai.
81	<code> return;</code>	Current function ko yahin stop karta hai.
82	<code>}</code>	Pichla block/function/object close karta hai.
83		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
84	<code>const { current, newPass, confirm } = this.passwordForm.value;</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
85		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
86	<code>if (newPass !== confirm) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
87	<code> this.showToastMsg('■ Passwords do not match', 'error');</code>	User ko success/error ka short notification dikhata hai.
88	<code> return;</code>	Current function ko yahin stop karta hai.
89	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
90		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
91	<code>this.showToastMsg('■ Password changed successfully!', 'success');</code>	User ko success/error ka short notification dikhata hai.
92	<code>this.passwordForm.reset();</code>	Form ko clear/default values par reset karta hai.
93	<code>}</code>	Pichla block/function/object close karta hai.
94		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
95	<code>// ■ Set Accent Color</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
96	<code>setAccent(color: string) {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
97	<code>this.accentColor.set(color);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
98	<code>document.documentElement.style.setProperty('--accent-color', color);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
99	<code>localStorage.setItem('accent_color', color);</code>	Browser localStorage mein string save karta hai; browser-side persistence.
100	<code>this.showToastMsg('■ Accent color updated!', 'success');</code>	User ko success/error ka short notification dikhata hai.
101	<code>}</code>	Pichla block/function/object close karta hai.
102		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
103	<code>// ■ Export JSON</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
104	<code>exportJSON() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
105	<code>this.http.get<any>('http://localhost:3000/students').subscribe(students => {</code>	HTTP GET request: backend se data read karta hai.
106	<code>this.http.get<any>('http://localhost:3000/courses').subscribe(courses => {</code>	HTTP GET request: backend se data read karta hai.
107	<code>this.http.get<any>('http://localhost:3000/enrollments').subscribe(enrollments => {</code>	HTTP GET request: backend se data read karta hai.
108	<code>this.http.get<any>('http://localhost:3000/payments').subscribe(payments => {</code>	HTTP GET request: backend se data read karta hai.
109	<code>const data = { students, courses, enrollments, payments };</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
110	<code>const blob = new Blob([JSON.stringify(data, null, 2)], { type: 'application/json' });</code>	JavaScript value ko JSON text mein convert karta hai.
111	<code>const url = window.URL.createObjectURL(blob);</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
112	<code>const a = document.createElement('a');</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
113	<code>a.href = url;</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
114	<code>a.download = `cms-backup-\${new Date().toISOString().slice(0, 10)}.json`;</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
115	<code>a.click();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
116	<code>this.showToastMsg('■ Data exported successfully!', 'success');</code>	User ko success/error ka short notification dikhata hai.
117	<code>});</code>	Pichla block/function/object close karta hai.
118	<code>});</code>	Pichla block/function/object close karta hai.
119	<code>});</code>	Pichla block/function/object close karta hai.
120	<code>});</code>	Pichla block/function/object close karta hai.
121	<code>}</code>	Pichla block/function/object close karta hai.
122		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
123	<code>// ■ Reset All Data</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
124	<code>resetAllData() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
125	<code>if (!confirm('■ Are you sure? This will DELETE all students, courses, enrollments and payments!')) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
126	<code>return;</code>	Current function ko yahin stop karta hai.
127	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
128		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
129	<code>this.http.put('http://localhost:3000/students', []).subscribe();</code>	HTTP PUT request: selected record ko update/replace karta hai.
130	<code>this.http.put('http://localhost:3000/courses', []).subscribe();</code>	HTTP PUT request: selected record ko update/replace karta hai.
131	<code>this.http.put('http://localhost:3000/enrollments', []).subscribe();</code>	HTTP PUT request: selected record ko update/replace karta hai.
132	<code>this.http.put('http://localhost:3000/payments', []).subscribe(() => {</code>	HTTP PUT request: selected record ko update/replace karta hai.
133	<code>this.showToastMsg('■ All data reset successfully!', 'success');</code>	User ko success/error ka short notification dikhata hai.
134	<code>});</code>	Pichla block/function/object close karta hai.
135	<code>}</code>	Pichla block/function/object close karta hai.
136		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
137	<code>isValid(form: any, field: string): boolean {</code>	Code block start hota hai; andar related statements group hote hain.
138	<code>const c = form.get(field);</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
139	<code>return !(c && c.invalid && (c.dirty c.touched));</code>	Check karta hai ke form/field validation pass kar raha hai ya nahi.
140	<code>}</code>	Pichla block/function/object close karta hai.
141	<code>}</code>	Pichla block/function/object close karta hai.

src/app/pages/student/student.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
2	<code>Students Page – Light (default)</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
3	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
4	<code>.students-page {</code>	Code block start hota hai; andar related statements group hote hain.
5	<code>padding: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
6	<code>background: #f5f6fa;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
7	<code>min-height: 100vh;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
8	<code>font-family: 'Inter', sans-serif;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
9	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
10	<code>transition: background 0.3s, color 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
11	<code>}</code>	Pichla block/function/object close karta hai.
12		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
13	<code>.students-page.dark {</code>	Code block start hota hai; andar related statements group hote hain.
14	<code>background: #0f0f1a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
15	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
16	<code>}</code>	Pichla block/function/object close karta hai.
17		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
18	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.

Lin e	Original code	Simple explanation
19	Page Header	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
20	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
21	.page-header {	Code block start hota hai; andar related statements group hote hain.
22	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
23	justify-content: space-between;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
24	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
25	margin-bottom: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
26	}	Pichla block/function/object close karta hai.
27		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
28	.page-header h2 {	Code block start hota hai; andar related statements group hote hain.
29	font-size: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
30	font-weight: 700;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
31	margin: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
32	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
33	}	Pichla block/function/object close karta hai.
34		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
35	.students-page.dark .page-header h2 {	Code block start hota hai; andar related statements group hote hain.
36	color: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
37	}	Pichla block/function/object close karta hai.
38		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
39	.add-new-btn {	Code block start hota hai; andar related statements group hote hain.
40	background: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
41	color: #fff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
42	border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
43	padding: 10px 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
44	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
45	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
46	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
47	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
48	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
49	box-shadow: 0 4px 12px rgba(16, 185, 129, 0.3);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
50	}	Pichla block/function/object close karta hai.
51		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
52	.add-new-btn:hover {	Code block start hota hai; andar related statements group hote hain.
53	background: #059669;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
54	<code>transform: translateY(-1px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
55	<code>}</code>	Pichla block/function/object close karta hai.
56		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
57	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
58	<code>Table</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
59	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
60	<code>.table-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
61	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
62	<code>border-radius: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
63	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
64	<code>box-shadow: 0 2px 8px rgba(0, 0, 0, 0.04);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
65	<code>overflow: hidden;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
66	<code>transition: background 0.3s, border-color 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
67	<code>}</code>	Pichla block/function/object close karta hai.
68		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
69	<code>.students-page.dark .table-wrapper {</code>	Code block start hota hai; andar related statements group hote hain.
70	<code>background: #14141f;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
71	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
72	<code>}</code>	Pichla block/function/object close karta hai.
73		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
74	<code>table {</code>	Code block start hota hai; andar related statements group hote hain.
75	<code>width: 100%;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
76	<code>border-collapse: collapse;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
77	<code>}</code>	Pichla block/function/object close karta hai.
78		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
79	<code>thead {</code>	Code block start hota hai; andar related statements group hote hain.
80	<code>background: #f9fafb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
81	<code>transition: background 0.3s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
82	<code>}</code>	Pichla block/function/object close karta hai.
83		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
84	<code>.students-page.dark thead {</code>	Code block start hota hai; andar related statements group hote hain.
85	<code>background: #1a1a28;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
86	<code>}</code>	Pichla block/function/object close karta hai.
87		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
88	<code>th {</code>	Code block start hota hai; andar related statements group hote hain.

Line	Original code	Simple explanation
89	<code>text-align: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
90	<code>padding: 14px 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
91	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
92	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
93	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
94	<code>text-transform: uppercase;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
95	<code>letter-spacing: 0.5px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
96	<code>border-bottom: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
97	<code>}</code>	Pichla block/function/object close karta hai.
98		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
99	<code>.students-page.dark th {</code>	Code block start hota hai; andar related statements group hote hain.
100	<code>color: #9ca3af;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
101	<code>border-bottom-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
102	<code>}</code>	Pichla block/function/object close karta hai.
103		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
104	<code>td {</code>	Code block start hota hai; andar related statements group hote hain.
105	<code>padding: 16px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
106	<code>text-align: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
107	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
108	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
109	<code>border-bottom: 1px solid #f3f4f6;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
110	<code>transition: background 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
111	<code>}</code>	Pichla block/function/object close karta hai.
112		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
113	<code>.students-page.dark td {</code>	Code block start hota hai; andar related statements group hote hain.
114	<code>color: #e4e4e7;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
115	<code>border-bottom-color: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
116	<code>}</code>	Pichla block/function/object close karta hai.
117		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
118	<code>tbody tr:last-child td {</code>	Code block start hota hai; andar related statements group hote hain.
119	<code>border-bottom: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
120	<code>}</code>	Pichla block/function/object close karta hai.
121		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
122	<code>tbody tr:hover td {</code>	Code block start hota hai; andar related statements group hote hain.
123	<code>background: #f9fafb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
124	}	Pichla block/function/object close karta hai.
125		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
126	.students-page.dark tbody tr:hover td {	Code block start hota hai; andar related statements group hote hain.
127	background: #1a1a28;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
128	}	Pichla block/function/object close karta hai.
129		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
130	.name-cell {	Code block start hota hai; andar related statements group hote hain.
131	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
132	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
133	}	Pichla block/function/object close karta hai.
134		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
135	.students-page.dark .name-cell {	Code block start hota hai; andar related statements group hote hain.
136	color: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
137	}	Pichla block/function/object close karta hai.
138		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
139	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
140	Action Buttons	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
141	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
142	.actions {	Code block start hota hai; andar related statements group hote hain.
143	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
144	gap: 6px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
145	justify-content: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
146	}	Pichla block/function/object close karta hai.
147		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
148	.edit-btn {	Code block start hota hai; andar related statements group hote hain.
149	background: #465fff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
150	color: #fff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
151	border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
152	padding: 7px 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
153	border-radius: 6px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
154	font-size: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
155	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
156	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
157	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
158	}	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
159		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
160	<code>.edit-btn:hover {</code>	Code block start hota hai; andar related statements group hote hain.
161	<code>background: #3548d4;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
162	<code>transform: translateY(-1px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
163	<code>}</code>	Pichla block/function/object close karta hai.
164		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
165	<code>.delete-btn {</code>	Code block start hota hai; andar related statements group hote hain.
166	<code>background: #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
167	<code>color: #fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
168	<code>border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
169	<code>padding: 7px 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
170	<code>border-radius: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
171	<code>font-size: 12px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
172	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
173	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
174	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
175	<code>}</code>	Pichla block/function/object close karta hai.
176		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
177	<code>.delete-btn:hover {</code>	Code block start hota hai; andar related statements group hote hain.
178	<code>background: #dc2626;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
179	<code>transform: translateY(-1px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
180	<code>}</code>	Pichla block/function/object close karta hai.
181		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
182	<code>/* =====</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
183	<code>Empty State</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
184	<code>===== */</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
185	<code>.empty-state {</code>	Code block start hota hai; andar related statements group hote hain.
186	<code>padding: 60px 20px !important;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
187	<code>}</code>	Pichla block/function/object close karta hai.
188		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
189	<code>.empty-content {</code>	Code block start hota hai; andar related statements group hote hain.
190	<code>display: flex;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
191	<code>flex-direction: column;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
192	<code>align-items: center;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
193	<code>gap: 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
194	}	Pichla block/function/object close karta hai.
195		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
196	.empty-icon {	Code block start hota hai; andar related statements group hote hain.
197	font-size: 48px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
198	margin-bottom: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
199	opacity: 0.6;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
200	}	Pichla block/function/object close karta hai.
201		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
202	.empty-content h3 {	Code block start hota hai; andar related statements group hote hain.
203	font-size: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
204	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
205	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
206	margin: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
207	}	Pichla block/function/object close karta hai.
208		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
209	.students-page.dark .empty-content h3 {	Code block start hota hai; andar related statements group hote hain.
210	color: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
211	}	Pichla block/function/object close karta hai.
212		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
213	.empty-content p {	Code block start hota hai; andar related statements group hote hain.
214	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
215	color: #6b7280;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
216	margin: 0 0 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
217	}	Pichla block/function/object close karta hai.
218		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
219	.students-page.dark .empty-content p {	Code block start hota hai; andar related statements group hote hain.
220	color: #9ca3af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
221	}	Pichla block/function/object close karta hai.
222		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
223	.add-first-btn {	Code block start hota hai; andar related statements group hote hain.
224	background: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
225	color: #fff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
226	border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
227	padding: 10px 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
228	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
229	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
230	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
231	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
232	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
233	box-shadow: 0 4px 12px rgba(16, 185, 129, 0.3);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
234	}	Pichla block/function/object close karta hai.
235		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
236	.add-first-btn:hover {	Code block start hota hai; andar related statements group hote hain.
237	background: #059669;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
238	transform: translateY(-2px);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
239	}	Pichla block/function/object close karta hai.
240		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
241	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
242	Modal Popup	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
243	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
244	.modal-overlay {	Code block start hota hai; andar related statements group hote hain.
245	position: fixed;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
246	top: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
247	left: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
248	right: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
249	bottom: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
250	background: rgba(0, 0, 0, 0.6);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
251	backdrop-filter: blur(4px);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
252	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
253	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
254	justify-content: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
255	z-index: 1000;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
256	padding: 20px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
257	animation: fadeIn 0.2s ease;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
258	}	Pichla block/function/object close karta hai.
259		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
260	.modal-box {	Code block start hota hai; andar related statements group hote hain.
261	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
262	border-radius: 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Lin e	Original code	Simple explanation
263	padding: 0;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
264	max-width: 700px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
265	width: 100%;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
266	max-height: 90vh;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
267	overflow-y: auto;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
268	box-shadow: 0 20px 60px rgba(0, 0, 0, 0.3);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
269	animation: slideUp 0.3s ease;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
270	}	Pichla block/function/object close karta hai.
271		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
272	.students-page.dark .modal-box {	Code block start hota hai; andar related statements group hote hain.
273	background: #14141f;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
274	box-shadow: 0 20px 60px rgba(0, 0, 0, 0.8);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
275	}	Pichla block/function/object close karta hai.
276		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
277	@keyframes fadeIn {	Code block start hota hai; andar related statements group hote hain.
278	from { opacity: 0; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
279	to { opacity: 1; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
280	}	Pichla block/function/object close karta hai.
281		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
282	@keyframes slideUp {	Code block start hota hai; andar related statements group hote hain.
283	from { opacity: 0; transform: translateY(20px); }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
284	to { opacity: 1; transform: translateY(0); }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
285	}	Pichla block/function/object close karta hai.
286		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
287	.modal-header {	Code block start hota hai; andar related statements group hote hain.
288	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
289	justify-content: space-between;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
290	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
291	padding: 20px 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
292	border-bottom: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
293	}	Pichla block/function/object close karta hai.
294		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
295	.students-page.dark .modal-header {	Code block start hota hai; andar related statements group hote hain.
296	border-bottom-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
297	}	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
298		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
299	<code>.modal-header h3 {</code>	Code block start hota hai; andar related statements group hote hain.
300	<code>font-size: 18px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
301	<code>font-weight: 700;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
302	<code>margin: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
303	<code>color: #1a1a2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
304	<code>}</code>	Pichla block/function/object close karta hai.
305		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
306	<code>.students-page.dark .modal-header h3 {</code>	Code block start hota hai; andar related statements group hote hain.
307	<code>color: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
308	<code>}</code>	Pichla block/function/object close karta hai.
309		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
310	<code>.modal-close {</code>	Code block start hota hai; andar related statements group hote hain.
311	<code>background: transparent;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
312	<code>border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
313	<code>font-size: 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
314	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
315	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
316	<code>padding: 4px 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
317	<code>border-radius: 6px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
318	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
319	<code>}</code>	Pichla block/function/object close karta hai.
320		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
321	<code>.modal-close:hover {</code>	Code block start hota hai; andar related statements group hote hain.
322	<code>background: #f3f4f6;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
323	<code>color: #ef4444;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
324	<code>}</code>	Pichla block/function/object close karta hai.
325		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
326	<code>.students-page.dark .modal-close:hover {</code>	Code block start hota hai; andar related statements group hote hain.
327	<code>background: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
328	<code>}</code>	Pichla block/function/object close karta hai.
329		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
330	<code>/* Modal Form */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
331	<code>.modal-form {</code>	Code block start hota hai; andar related statements group hote hain.
332	<code>padding: 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
333	display: grid;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
334	grid-template-columns: 1fr 1fr;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
335	gap: 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
336	}	Pichla block/function/object close karta hai.
337		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
338	.modal-form .form-group {	Code block start hota hai; andar related statements group hote hain.
339	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
340	flex-direction: column;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
341	gap: 6px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
342	}	Pichla block/function/object close karta hai.
343		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
344	.modal-form .form-group label {	Code block start hota hai; andar related statements group hote hain.
345	font-size: 13px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
346	font-weight: 600;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
347	color: #374151;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
348	}	Pichla block/function/object close karta hai.
349		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
350	.students-page.dark .modal-form .form-group label {	Code block start hota hai; andar related statements group hote hain.
351	color: #d1d5db;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
352	}	Pichla block/function/object close karta hai.
353		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
354	.modal-form input {	Code block start hota hai; andar related statements group hote hain.
355	padding: 10px 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
356	border: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
357	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
358	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
359	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
360	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
361	outline: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
362	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
363	}	Pichla block/function/object close karta hai.
364		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
365	.students-page.dark .modal-form input {	Code block start hota hai; andar related statements group hote hain.
366	background: #1f1f2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
367	border-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
368	color: #e4e4e7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
369	}	Pichla block/function/object close karta hai.
370		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
371	.modal-form input:focus {	Code block start hota hai; andar related statements group hote hain.
372	border-color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
373	box-shadow: 0 0 0 3px rgba(16, 185, 129, 0.15);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
374	}	Pichla block/function/object close karta hai.
375		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
376	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
377	Field Errors	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
378	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
379	.modal-form input.error {	Code block start hota hai; andar related statements group hote hain.
380	border-color: #ef4444 !important;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
381	box-shadow: 0 0 0 3px rgba(239, 68, 68, 0.15) !important;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
382	}	Pichla block/function/object close karta hai.
383		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
384	.error-msg {	Code block start hota hai; andar related statements group hote hain.
385	display: block;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
386	font-size: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
387	color: #ef4444;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
388	margin-top: 4px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
389	font-weight: 500;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
390	}	Pichla block/function/object close karta hai.
391		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
392	/* Modal Actions */	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
393	.modal-actions {	Code block start hota hai; andar related statements group hote hain.
394	grid-column: 1 / -1;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
395	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
396	gap: 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
397	justify-content: flex-end;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
398	margin-top: 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
399	padding-top: 16px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
400	border-top: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
401	}	Pichla block/function/object close karta hai.
402		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
403	<code>.students-page.dark .modal-actions {</code>	Code block start hota hai; andar related statements group hote hain.
404	<code>border-top-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
405	<code>}</code>	Pichla block/function/object close karta hai.
406		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
407	<code>.save-btn {</code>	Code block start hota hai; andar related statements group hote hain.
408	<code>background: #10b981;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
409	<code>color: #fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
410	<code>border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
411	<code>padding: 10px 24px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
412	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
413	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
414	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
415	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
416	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
417	<code>}</code>	Pichla block/function/object close karta hai.
418		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
419	<code>.save-btn:hover {</code>	Code block start hota hai; andar related statements group hote hain.
420	<code>background: #059669;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
421	<code>transform: translateY(-1px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
422	<code>}</code>	Pichla block/function/object close karta hai.
423		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
424	<code>.cancel-btn {</code>	Code block start hota hai; andar related statements group hote hain.
425	<code>background: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
426	<code>color: #fff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
427	<code>border: none;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
428	<code>padding: 10px 20px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
429	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
430	<code>font-size: 14px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
431	<code>font-weight: 600;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
432	<code>cursor: pointer;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
433	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
434	<code>}</code>	Pichla block/function/object close karta hai.
435		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
436	<code>.cancel-btn:hover {</code>	Code block start hota hai; andar related statements group hote hain.
437	<code>background: #4b5563;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
438	}	Pichla block/function/object close karta hai.
439		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
440	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
441	Toast Popup	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
442	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
443	.toast-popup {	Code block start hota hai; andar related statements group hote hain.
444	position: fixed;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
445	top: 90px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
446	right: 24px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
447	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
448	border-left: 4px solid #ef4444;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
449	border-radius: 10px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
450	padding: 14px 18px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
451	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
452	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
453	gap: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
454	box-shadow: 0 10px 30px rgba(0, 0, 0, 0.15);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
455	z-index: 2000;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
456	min-width: 280px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
457	max-width: 400px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
458	animation: slideInRight 0.3s ease;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
459	}	Pichla block/function/object close karta hai.
460		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
461	.toast-popup.success {	Code block start hota hai; andar related statements group hote hain.
462	border-left-color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
463	}	Pichla block/function/object close karta hai.
464		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
465	.students-page.dark .toast-popup {	Code block start hota hai; andar related statements group hote hain.
466	background: #1f1f2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
467	box-shadow: 0 10px 30px rgba(0, 0, 0, 0.5);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
468	}	Pichla block/function/object close karta hai.
469		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
470	@keyframes slideInRight {	Code block start hota hai; andar related statements group hote hain.
471	from { opacity: 0; transform: translateX(100%); }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
472	to { opacity: 1; transform: translateX(0); }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
473	}	Pichla block/function/object close karta hai.
474		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
475	.toast-text {	Code block start hota hai; andar related statements group hote hain.
476	flex: 1;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
477	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
478	font-weight: 500;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
479	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
480	}	Pichla block/function/object close karta hai.
481		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
482	.students-page.dark .toast-text {	Code block start hota hai; andar related statements group hote hain.
483	color: #e4e4e7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
484	}	Pichla block/function/object close karta hai.
485		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
486	.toast-close {	Code block start hota hai; andar related statements group hote hain.
487	background: transparent;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
488	border: none;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
489	color: #9ca3af;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
490	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
491	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
492	padding: 2px 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
493	border-radius: 4px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
494	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
495	}	Pichla block/function/object close karta hai.
496		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
497	.toast-close:hover {	Code block start hota hai; andar related statements group hote hain.
498	background: #f3f4f6;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
499	color: #ef4444;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
500	}	Pichla block/function/object close karta hai.
501		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
502	.students-page.dark .toast-close:hover {	Code block start hota hai; andar related statements group hote hain.
503	background: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
504	}	Pichla block/function/object close karta hai.
505		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
506	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
507	Department Dropdown	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
508	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
509	.dept-dropdown {	Code block start hota hai; andar related statements group hote hain.
510	position: relative;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
511	}	Pichla block/function/object close karta hai.
512		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
513	.dept-btn {	Code block start hota hai; andar related statements group hote hain.
514	width: 100%;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
515	display: flex;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
516	justify-content: space-between;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
517	align-items: center;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
518	padding: 10px 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
519	border: 1px solid #e5e7eb;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
520	border-radius: 8px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
521	background: #ffffff;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
522	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
523	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
524	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
525	transition: 0.2s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
526	text-align: left;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
527	}	Pichla block/function/object close karta hai.
528		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
529	.students-page.dark .dept-btn {	Code block start hota hai; andar related statements group hote hain.
530	background: #1f1f2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
531	border-color: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
532	color: #e4e4e7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
533	}	Pichla block/function/object close karta hai.
534		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
535	.dept-btn:hover,	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
536	.dept-btn.active {	Code block start hota hai; andar related statements group hote hain.
537	border-color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
538	box-shadow: 0 0 3px rgba(16, 185, 129, 0.15);	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
539	}	Pichla block/function/object close karta hai.
540		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
541	.dept-btn.error {	Code block start hota hai; andar related statements group hote hain.
542	border-color: #ef4444 !important;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.

Line	Original code	Simple explanation
543	<code>box-shadow: 0 0 0 3px rgba(239, 68, 68, 0.15) !important;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
544	<code>}</code>	Pichla block/function/object close karta hai.
545		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
546	<code>.dept-btn .arrow {</code>	Code block start hota hai; andar related statements group hote hain.
547	<code>font-size: 10px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
548	<code>color: #6b7280;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
549	<code>transition: 0.2s;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
550	<code>}</code>	Pichla block/function/object close karta hai.
551		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
552	<code>.dept-btn.active .arrow {</code>	Code block start hota hai; andar related statements group hote hain.
553	<code>transform: rotate(180deg);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
554	<code>}</code>	Pichla block/function/object close karta hai.
555		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
556	<code>.dept-menu {</code>	Code block start hota hai; andar related statements group hote hain.
557	<code>position: absolute;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
558	<code>top: calc(100% + 6px);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
559	<code>left: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
560	<code>right: 0;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
561	<code>background: #ffffff;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
562	<code>border: 1px solid #e5e7eb;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
563	<code>border-radius: 8px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
564	<code>box-shadow: 0 10px 30px rgba(0, 0, 0, 0.1);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
565	<code>max-height: 220px;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
566	<code>overflow-y: auto;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
567	<code>z-index: 50;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
568	<code>animation: fadeIn 0.15s ease;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
569	<code>}</code>	Pichla block/function/object close karta hai.
570		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
571	<code>.students-page.dark .dept-menu {</code>	Code block start hota hai; andar related statements group hote hain.
572	<code>background: #1f1f2e;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
573	<code>border-color: #2a2a3a;</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
574	<code>box-shadow: 0 10px 30px rgba(0, 0, 0, 0.5);</code>	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
575	<code>}</code>	Pichla block/function/object close karta hai.
576		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
577	<code>.dept-item {</code>	Code block start hota hai; andar related statements group hote hain.

Line	Original code	Simple explanation
578	padding: 10px 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
579	cursor: pointer;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
580	font-size: 14px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
581	color: #1a1a2e;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
582	transition: 0.15s;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
583	}	Pichla block/function/object close karta hai.
584		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
585	.students-page.dark .dept-item {	Code block start hota hai; andar related statements group hote hain.
586	color: #e4e4e7;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
587	}	Pichla block/function/object close karta hai.
588		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
589	.dept-item:hover {	Code block start hota hai; andar related statements group hote hain.
590	background: #f3f4f6;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
591	color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
592	}	Pichla block/function/object close karta hai.
593		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
594	.students-page.dark .dept-item:hover {	Code block start hota hai; andar related statements group hote hain.
595	background: #2a2a3a;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
596	color: #10b981;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
597	}	Pichla block/function/object close karta hai.
598		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
599	/* =====	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
600	Responsive	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
601	===== */	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
602	@media (max-width: 768px) {	Responsive CSS rule: screen width/device condition par styles apply karta hai.
603	.students-page { padding: 16px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
604		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
605	.page-header {	Code block start hota hai; andar related statements group hote hain.
606	flex-direction: column;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
607	gap: 12px;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
608	align-items: stretch;	CSS property set karta hai: property design/layout ka ek pehlu control karti hai.
609	}	Pichla block/function/object close karta hai.
610		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
611	.table-wrapper { overflow-x: auto; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
612	table { min-width: 800px; }	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
613		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
614	<code>.modal-form { grid-template-columns: 1fr; }</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
615	<code>}</code>	Pichla block/function/object close karta hai.

src/app/pages/student/student.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Line	Original code	Simple explanation
1	<code><div class="students-page" [class.dark]="themeService.isDarkMode()"></code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
2		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
3	<code><!-- ■ Toast Notification --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
4	<code>@if (showToast()) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
5	<code><div class="toast-popup" [class.success]="toastType() === 'success'"></code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
6	<code>{{ toastMessage() }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
7	<code><button class="toast-close" (click)="showToast.set(false)">X</button></code>	User click par specified TypeScript method/expression chalti hai.
8	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
9	<code>}</code>	Pichla block/function/object close karta hai.
10		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
11	<code><!-- Page Header --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
12	<code><div class="page-header"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karta hai.
13	<code><h2>Student List</h2></code>	HTML element/component: page ka structure ya UI control banata hai.
14		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
15	<code><button class="add-new-btn" (click)="openModal()"></code>	User click par specified TypeScript method/expression chalti hai.
16	<code>+ Add Student</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
17	<code></button></code>	HTML element/component: page ka structure ya UI control banata hai.
18	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	<code><!-- Table --></code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
21	<code><div class="table-wrapper"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karta hai.
22	<code><table></code>	HTML element/component: page ka structure ya UI control banata hai.
23	<code><thead></code>	HTML element/component: page ka structure ya UI control banata hai.
24	<code><tr></code>	HTML element/component: page ka structure ya UI control banata hai.
25	<code><th>Name</th></code>	HTML element/component: page ka structure ya UI control banata hai.
26	<code><th>Father Name</th></code>	HTML element/component: page ka structure ya UI control banata hai.
27	<code><th>Email</th></code>	HTML element/component: page ka structure ya UI control banata hai.
28	<code><th>Phone</th></code>	HTML element/component: page ka structure ya UI control banata hai.

Line	Original code	Simple explanation
29	<th>Address</th>	HTML element/component: page ka structure ya UI control banata hai.
30	<th>Department</th>	HTML element/component: page ka structure ya UI control banata hai.
31	<th>Actions</th>	HTML element/component: page ka structure ya UI control banata hai.
32	</tr>	HTML element/component: page ka structure ya UI control banata hai.
33	</thead>	HTML element/component: page ka structure ya UI control banata hai.
34	<tbody>	HTML element/component: page ka structure ya UI control banata hai.
35	@for (student of students()); track student.id) {	Angular loop: array ke har item ke liye HTML repeat karta hai.
36	<tr>	HTML element/component: page ka structure ya UI control banata hai.
37	<td class="name-cell">{{ student.name }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
38	<td>{{ student.father_name }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
39	<td>{{ student.email }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
40	<td>{{ student.phone }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
41	<td>{{ student.address }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
42	<td>{{ student.department }}</td>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
43	<td>	HTML element/component: page ka structure ya UI control banata hai.
44	<div class="actions">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
45	<button class="edit-btn" (click)="editStudent(student)">Edit</button>	User click par specified TypeScript method/expression chalti hai.
46	<button class="delete-btn" (click)="deleteStudent(student.id)">Delete</button>	User click par specified TypeScript method/expression chalti hai.
47	</div>	HTML element/component: page ka structure ya UI control banata hai.
48	</td>	HTML element/component: page ka structure ya UI control banata hai.
49	</tr>	HTML element/component: page ka structure ya UI control banata hai.
50	}	Pichla block/function/object close karta hai.
51		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
52	@if (students().length === 0) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
53	<tr>	HTML element/component: page ka structure ya UI control banata hai.
54	<td colspan="7" class="empty-state">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
55	<div class="empty-content">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
56	<div class="empty-icon">■■■</div>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
57	No Students Yet	HTML element/component: page ka structure ya UI control banata hai.
58	Click the button below to add your first student	HTML element/component: page ka structure ya UI control banata hai.
59	<button class="add-first-btn" (click)="openModal()">	User click par specified TypeScript method/expression chalti hai.
60	+ Add First Student	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
61	</button>	HTML element/component: page ka structure ya UI control banata hai.
62	</div>	HTML element/component: page ka structure ya UI control banata hai.

Line	Original code	Simple explanation
63	</td>	HTML element/component: page ka structure ya UI control banata hai.
64	</tr>	HTML element/component: page ka structure ya UI control banata hai.
65	}	Pichla block/function/object close karta hai.
66	</tbody>	HTML element/component: page ka structure ya UI control banata hai.
67	</table>	HTML element/component: page ka structure ya UI control banata hai.
68	</div>	HTML element/component: page ka structure ya UI control banata hai.
69		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
70	<!-- ■ Modal Popup -->	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
71	@if (showModal()) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
72	<div class="modal-overlay" (click)="closeModal()">	User click par specified TypeScript method/expression chalti hai.
73	<div class="modal-box" (click)="\$event.stopPropagation()">	User click par specified TypeScript method/expression chalti hai.
74		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
75	<div class="modal-header">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
76	{{ selectedStudentId() === null ? 'Add New Student' : 'Update Student' }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
77	<button class="modal-close" (click)="closeModal()">X</button>	User click par specified TypeScript method/expression chalti hai.
78	</div>	HTML element/component: page ka structure ya UI control banata hai.
79		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
80	<form [formGroup]="studentForm" (ngSubmit)="saveStudent()" class="modal-form">	Form submit par specified handler chalata hai.
81		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
82	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
83	<label>Name *</label>	HTML element/component: page ka structure ya UI control banata hai.
84	<input	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
85	placeholder="Enter student name"	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
86	formControlName="name"	HTML input ko TS reactive form ke same-named control se connect karta hai.
87	[class.error]="isInvalid('name')"	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
88	/>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
89	@if (isInvalid('name')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
90	■ Name is required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
91	}	Pichla block/function/object close karta hai.
92	</div>	HTML element/component: page ka structure ya UI control banata hai.
93		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
94	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
95	<label>Father Name *</label>	HTML element/component: page ka structure ya UI control banata hai.
96	<input	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
97	placeholder="Enter father name"	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
98	formControlName="father_name"	HTML input ko TS reactive form ke same-named control se connect karta hai.
99	[class.error]="isInvalid('father_name')"	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
100	/>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
101	@if (isInvalid('father_name')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
102	■ Father name is required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
103	}	Pichla block/function/object close karta hai.
104	</div>	HTML element/component: page ka structure ya UI control banata hai.
105		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
106	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
107	<label>Email *</label>	HTML element/component: page ka structure ya UI control banata hai.
108	<input	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
109	type="email"	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
110	placeholder="Enter email"	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
111	formControlName="email"	HTML input ko TS reactive form ke same-named control se connect karta hai.
112	[class.error]="isInvalid('email')"	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
113	/>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
114	@if (isInvalid('email')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
115	■ Email is required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
116	}	Pichla block/function/object close karta hai.
117	</div>	HTML element/component: page ka structure ya UI control banata hai.
118		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
119	<div class="form-group">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
120	<label>Phone *</label>	HTML element/component: page ka structure ya UI control banata hai.
121	<input	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
122	placeholder="Enter phone"	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
123	formControlName="phone"	HTML input ko TS reactive form ke same-named control se connect karta hai.
124	[class.error]="isInvalid('phone')"	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
125	/>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
126	@if (isInvalid('phone')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
127	■ Phone is required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
128	}	Pichla block/function/object close karta hai.
129	</div>	HTML element/component: page ka structure ya UI control banata hai.
130		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
131	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
132	<code><label>Address *</label></code>	HTML element/component: page ka structure ya UI control banata hai.
133	<code><input</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
134	<code>placeholder="Enter address"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
135	<code>formControlName="address"</code>	HTML input ko TS reactive form ke same-named control se connect karta hai.
136	<code>[class.error]="isInvalid('address')"</code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
137	<code>/></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
138	<code>@if (isInvalid('address')) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
139	<code>■ Address is required</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
140	<code>}</code>	Pichla block/function/object close karta hai.
141	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
142		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
143	<code><div class="form-group"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
144	<code><label>Department *</label></code>	HTML element/component: page ka structure ya UI control banata hai.
145		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
146	<code><div class="dept-dropdown"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
147	<code><button</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
148	<code>type="button"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
149	<code>class="dept-btn"</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
150	<code>(click)="toggleDeptDropdown()"</code>	User click par specified TypeScript method/expression chalti hai.
151	<code>[class.active]="showDeptDropdown()"</code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
152	<code>[class.error]="isInvalid('department')"</code>	Property/class binding: state ke mutabiq CSS class dynamically lagata/hatata hai.
153	<code>></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
154	<code>{{ selectedDeptLabel() 'Select Department' }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
155	<code>▼</code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
156	<code></button></code>	HTML element/component: page ka structure ya UI control banata hai.
157		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
158	<code>@if (showDeptDropdown()) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
159	<code><div class="dept-menu"></code>	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
160	<code>@for (dept of departmentOptions; track dept) {</code>	Angular loop: array ke har item ke liye HTML repeat karta hai.
161	<code><div class="dept-item" (click)="selectDept(dept)"></code>	User click par specified TypeScript method/expression chalti hai.
162	<code>{{ dept }}</code>	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
163	<code></div></code>	HTML element/component: page ka structure ya UI control banata hai.
164	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
165	</div>	HTML element/component: page ka structure ya UI control banata hai.
166	}	Pichla block/function/object close karta hai.
167	</div>	HTML element/component: page ka structure ya UI control banata hai.
168		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
169	@if (isValid('department')) {	Condition: andar ka code sirf tab chalta hai jab condition true ho.
170	■ Department is required	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
171	}	Pichla block/function/object close karta hai.
172	</div>	HTML element/component: page ka structure ya UI control banata hai.
173		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
174	<div class="modal-actions">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
175	<button type="button" class="cancel-btn" (click)="closeModal()">Cancel</button>	User click par specified TypeScript method/expression chalti hai.
176	<button type="submit" class="save-btn">	CSS class attach karta hai; styling file is class ke zariye design apply karti hai.
177	{{ selectedStudentId() === null ? 'Add Student' : 'Update Student' }}	Interpolation: TypeScript value ko screen par text ki form mein dikhata hai.
178	</button>	HTML element/component: page ka structure ya UI control banata hai.
179	</div>	HTML element/component: page ka structure ya UI control banata hai.
180		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
181	</form>	HTML element/component: page ka structure ya UI control banata hai.
182	</div>	HTML element/component: page ka structure ya UI control banata hai.
183	</div>	HTML element/component: page ka structure ya UI control banata hai.
184	}	Pichla block/function/object close karta hai.
185		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
186	</div>	HTML element/component: page ka structure ya UI control banata hai.

src/app/pages/student/student.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	import { Component, OnInit, signal, inject } from '@angular/core';	Import: doosri library/file se required class ya function is file mein available karta hai.
2	import { CommonModule } from '@angular/common';	Import: doosri library/file se required class ya function is file mein available karta hai.
3	import { HttpClient } from '@angular/common/http';	Import: doosri library/file se required class ya function is file mein available karta hai.
4	import { FormBuilder, ReactiveFormsModule, Validators } from '@angular/forms';	Import: doosri library/file se required class ya function is file mein available karta hai.
5	import { ThemeService } from '../services/theme';	Import: doosri library/file se required class ya function is file mein available karta hai.
6		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
7	@Component({	Angular decorator: neech wali class ko Angular component declare karta hai.
8	selector: 'app-students',	Component ka HTML tag name; parent template mein isi selector se component lagta hai.
9	standalone: true,	Component standalone hai; NgModule mein declare karna zaroori nahi.

Line	Original code	Simple explanation
10	imports: [CommonModule, ReactiveFormsModule],	Template mein use hone wale Angular directives/forms/router features import karta hai.
11	templateUrl: './student.html',	Is component ki HTML template file ka path.
12	styleUrl: './student.css'	Is component ki CSS file ka path.
13	})	Pichla block/function/object close karta hai.
14	export class Student implements OnInit {	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
15		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
16	private http = inject(HttpClient);	Angular se HttpClient service leta hai; API request bhejne ke liye.
17	private fb = inject(FormBuilder);	FormBuilder leta hai taake reactive form aur controls bana sake.
18	public themeService = inject(ThemeService);	Shared ThemeService use hoti hai taake theme state/component access milay.
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	// ■ Modal state	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
21	showModal = signal<boolean>(false);	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
22		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
23	students = signal<any[]>([]);	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
24	selectedStudentId = signal<number null>(null);	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
25		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
26	// ■ Toast signals	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
27	showToast = signal<boolean>(false);	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
28	toastMessage = signal<string>('');	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
29	toastType = signal<'error' 'success'>('error');	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
30		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
31	// ■ Department dropdown signals	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
32	showDeptDropdown = signal<boolean>(false);	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
33	selectedDeptLabel = signal<string>('');	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
34		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
35	// ■ Department options	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
36	departmentOptions = [	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
37	'Computer Science',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
38	'Software Engineering',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
39	'Information Technology',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
40	'Artificial Intelligence',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
41	'Data Science',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
42	'Cyber Security',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
43	'Electrical Engineering',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
44	'Mechanical Engineering',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
45	'Business Administration',	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
46	'Mathematics'	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
47	];	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
48		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
49	// ■ Sab fields required	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
50	studentForm = this.fb.group({	Reactive form group banata hai; fields, current values aur validation ek jagah manage hotay hain.
51	name: ['', Validators.required],	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
52	father_name: ['', Validators.required],	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
53	email: ['', Validators.required],	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
54	phone: ['', Validators.required],	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
55	address: ['', Validators.required],	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
56	department: ['', Validators.required]	Validation: field empty na ho. Invalid ho to form submit rok sakta hai.
57	});	Pichla block/function/object close karta hai.
58		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
59	ngOnInit() {	Component initialize hone ke baad Angular is method ko automatically call karta hai.
60	this.loadStudents();	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
61	}	Pichla block/function/object close karta hai.
62		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
63	// ■ Toast helper	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
64	showToastMsg(msg: string, type: 'error' 'success' = 'error') {	Function/method define karta hai; call hone par is block ka kaam hota hai.
65	this.toastMessage.set(msg);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
66	this.toastType.set(type);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
67	this.showToast.set(true);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
68	setTimeout(() => this.showToast.set(false), 3000);	Arrow function: callback/function ko short syntax mein define karta hai.
69	}	Pichla block/function/object close karta hai.
70		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
71	// ■ Department dropdown toggle	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
72	toggleDeptDropdown() {	Function/method define karta hai; call hone par is block ka kaam hota hai.
73	this.showDeptDropdown.update(v => !v);	Arrow function: callback/function ko short syntax mein define karta hai.
74	}	Pichla block/function/object close karta hai.
75		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
76	// ■ Select department	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
77	selectDept(dept: string) {	Function/method define karta hai; call hone par is block ka kaam hota hai.
78	this.selectedDeptLabel.set(dept);	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.

Line	Original code	Simple explanation
79	<code>this.studentForm.patchValue({</code>	Form ke kuch selected fields update karta hai; har field dena zaroori nahi.
80	<code>department: dept</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
81	<code>});</code>	Pichla block/function/object close karta hai.
82	<code>this.showDeptDropdown.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
83	<code>}</code>	Pichla block/function/object close karta hai.
84		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
85	<code>// ■ Modal open</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
86	<code>openModal() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
87	<code>this.selectedStudentId.set(null);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
88	<code>this.studentForm.reset();</code>	Form ko clear/default values par reset karta hai.
89	<code>this.selectedDeptLabel.set('');</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
90	<code>this.showDeptDropdown.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
91	<code>this.showModal.set(true);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
92	<code>}</code>	Pichla block/function/object close karta hai.
93		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
94	<code>// ■ Modal close</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
95	<code>closeModal() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
96	<code>this.showModal.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
97	<code>this.selectedStudentId.set(null);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
98	<code>this.studentForm.reset();</code>	Form ko clear/default values par reset karta hai.
99	<code>this.selectedDeptLabel.set('');</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
100	<code>this.showDeptDropdown.set(false);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
101	<code>}</code>	Pichla block/function/object close karta hai.
102		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
103	<code>loadStudents() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
104	<code>this.http.get<any[]>('http://localhost:3000/students')</code>	HTTP GET request: backend se data read karta hai.
105	<code>.subscribe(data => this.students.set(data));</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
106	<code>}</code>	Pichla block/function/object close karta hai.
107		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
108	<code>// ■ Save with validation</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
109	<code>saveStudent() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
110	<code>if (this.studentForm.invalid) {</code>	Check karta hai ke form/field validation pass kar raha hai ya nahi.
111	<code>this.studentForm.markAllAsTouched();</code>	Validation messages dikhane ke liye sab controls ko touched mark karta hai.
112	<code>this.showToastMsg('■ Please fill all required fields', 'error');</code>	User ko success/error ka short notification dikhata hai.
113	<code>return;</code>	Current function ko yahin stop karta hai.
114	<code>}</code>	Pichla block/function/object close karta hai.
115		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.

Line	Original code	Simple explanation
116	<code>const url = 'http://localhost:3000/students';</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
117		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
118	<code>if (this.selectedStudentId() === null) {</code>	Condition: andar ka code sirf tab chalta hai jab condition true ho.
119	<code>this.http.post(url, this.studentForm.value).subscribe(() => {</code>	HTTP POST request: backend mein naya record create karta hai.
120	<code>this.showToastMsg('■ Student added successfully!', 'success');</code>	User ko success/error ka short notification dikhata hai.
121	<code>this.closeModal();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
122	<code>this.loadStudents();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
123	<code>});</code>	Pichla block/function/object close karta hai.
124	<code>} else {</code>	Alternative branch: pehli condition false ho to yeh code chalta hai.
125	<code>this.http.put(`\${url}/\${this.selectedStudentId()}`, this.studentForm.value)</code>	HTTP PUT request: selected record ko update/replace karta hai.
126	<code>.subscribe(() => {</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
127	<code>this.showToastMsg('■ Student updated successfully!', 'success');</code>	User ko success/error ka short notification dikhata hai.
128	<code>this.closeModal();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
129	<code>this.loadStudents();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
130	<code>});</code>	Pichla block/function/object close karta hai.
131	<code>}</code>	Pichla block/function/object close karta hai.
132	<code>}</code>	Pichla block/function/object close karta hai.
133		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
134	<code>editStudent(student: any) {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
135	<code>this.selectedStudentId.set(student.id);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
136	<code>this.studentForm.setValue({</code>	Form ke tamam controls ko values deta hai; required controls miss hon to error ho sakta hai.
137	<code>name: student.name,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
138	<code>father_name: student.father_name,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
139	<code>email: student.email,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
140	<code>phone: student.phone,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
141	<code>address: student.address,</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
142	<code>department: student.department</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
143	<code>});</code>	Pichla block/function/object close karta hai.
144	<code>this.selectedDeptLabel.set(student.department); // ■ dropdown label set</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
145	<code>this.showModal.set(true);</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
146	<code>}</code>	Pichla block/function/object close karta hai.
147		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
148	<code>deleteStudent(id: number) {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
149	<code>this.http.delete(`http://localhost:3000/students/\${id}`)</code>	HTTP DELETE request: selected record remove karta hai.
150	<code>.subscribe(() => {</code>	Observable ko subscribe karta hai; asynchronous HTTP response milne par callback chalta hai.
151	<code>this.showToastMsg('■ Student deleted', 'success');</code>	User ko success/error ka short notification dikhata hai.

Line	Original code	Simple explanation
152	<code>this.loadStudents();</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
153	<code>});</code>	Pichla block/function/object close karta hai.
154	<code>}</code>	Pichla block/function/object close karta hai.
155		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
156	<code>// ■ Field invalid check</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.
157	<code>isValid(field: string): boolean {</code>	Code block start hota hai; andar related statements group hote hain.
158	<code>const control = this.studentForm.get(field);</code>	Constant variable declare karta hai; is binding ko dobara assign nahi kiya jata.
159	<code>return !(control && control.invalid && (control.dirty control.touched));</code>	Check karta hai ke form/field validation pass kar raha hai ya nahi.
160	<code>}</code>	Pichla block/function/object close karta hai.
161	<code>}</code>	Pichla block/function/object close karta hai.

src/app/services/courses.service.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	<code>import { Injectable, inject } from '@angular/core';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
2	<code>import { HttpClient } from '@angular/common/http';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
3	<code>import { Observable } from 'rxjs';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
4		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
5	<code>@Injectable({</code>	Decorator: service ko Angular dependency injection ke liye register/prepare karta hai.
6	<code>providedIn: 'root'</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
7	<code>})</code>	Pichla block/function/object close karta hai.
8	<code>export class CoursesService {</code>	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
9		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
10	<code>private http = inject(HttpClient);</code>	Angular se HttpClient service leta hai; API request bhejne ke liye.
11	<code>private apiUrl = 'http://localhost:3000/courses';</code>	Class property/method ko class ke andar se access karne ke liye mark karta hai.
12		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
13	<code>getCourses(): Observable<any[]> {</code>	Code block start hota hai; andar related statements group hote hain.
14	<code>return this.http.get<any[]>(this.apiUrl);</code>	HTTP GET request: backend se data read karta hai.
15	<code>}</code>	Pichla block/function/object close karta hai.
16		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
17	<code>addCourse(course: any): Observable<any> {</code>	Code block start hota hai; andar related statements group hote hain.
18	<code>return this.http.post(this.apiUrl, course);</code>	HTTP POST request: backend mein naya record create karta hai.
19	<code>}</code>	Pichla block/function/object close karta hai.
20		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
21	<code>updateCourse(id: number, course: any): Observable<any> {</code>	Code block start hota hai; andar related statements group hote hain.
22	<code>return this.http.put(`\${this.apiUrl}/\${id}`, course);</code>	HTTP PUT request: selected record ko update/replace karta hai.
23	<code>}</code>	Pichla block/function/object close karta hai.

Line	Original code	Simple explanation
24		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
25	<code>deleteCourse(id: number): Observable<any> {</code>	Code block start hota hai; andar related statements group hote hain.
26	<code>return this.http.delete(`\${this.apiUrl}/\${id}`);</code>	HTTP DELETE request: selected record remove karta hai.
27	<code>}</code>	Pichla block/function/object close karta hai.
28	<code>}</code>	Pichla block/function/object close karta hai.

src/app/services/students.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	<code>import { HttpClient } from '@angular/common/http';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
2	<code>import { inject, Injectable } from '@angular/core';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
3		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
4	<code>@Injectable({</code>	Decorator: service ko Angular dependency injection ke liye register/prepare karta hai.
5	<code>providedIn: 'root'</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
6	<code>})</code>	Pichla block/function/object close karta hai.
7	<code>export class Students {</code>	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
8	<code>private http = inject(HttpClient);</code>	Angular se HttpClient service leta hai; API request bhejne ke liye.
9		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
10	<code>private apiUrl = 'http://localhost:3000/students';</code>	Class property/method ko class ke andar se access karne ke liye mark karta hai.
11		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
12	<code>getStudents() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
13	<code>return this.http.get<any[]>(this.apiUrl);</code>	HTTP GET request: backend se data read karta hai.
14	<code>}</code>	Pichla block/function/object close karta hai.
15		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
16	<code>addStudent(data: any) {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
17	<code>return this.http.post<any>(this.apiUrl, data)</code>	HTTP POST request: backend mein naya record create karta hai.
18	<code>}</code>	Pichla block/function/object close karta hai.
19		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
20	<code>updateStudent(id: number, data: any) {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
21	<code>return this.http.put<any>(`\${this.apiUrl}/\${id}`, data);</code>	HTTP PUT request: selected record ko update/replace karta hai.
22	<code>}</code>	Pichla block/function/object close karta hai.
23		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
24	<code>deleteStudent(id: number) {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
25	<code>return this.http.delete<any>(`\${this.apiUrl}/\${id}`);</code>	HTTP DELETE request: selected record remove karta hai.
26	<code>}</code>	Pichla block/function/object close karta hai.
27	<code>}</code>	Pichla block/function/object close karta hai.

src/app/services/theme.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	<code>import { Injectable, signal } from '@angular/core';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
2		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
3	<code>@Injectable({</code>	Decorator: service ko Angular dependency injection ke liye register/prepare karta hai.
4	<code>providedIn: 'root'</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
5	<code>})</code>	Pichla block/function/object close karta hai.
6	<code>export class ThemeService {</code>	Class define/export hoti hai; doosri files is component/class ko import kar sakti hain.
7	<code>isDarkMode = signal<boolean>(true);</code>	Signal state banata hai; value read karne ke liye name() aur update ke liye .set()/.update().
8		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
9	<code>toggleTheme() {</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
10	<code>this.isDarkMode.update(v => !v);</code>	Arrow function: callback/function ko short syntax mein define karta hai.
11	<code>}</code>	Pichla block/function/object close karta hai.
12	<code>}</code>	Pichla block/function/object close karta hai.

src/index.html

HTML template: page structure, binding, buttons, forms, conditions and loops.

Line	Original code	Simple explanation
1	<code><!doctype html></code>	HTML element/component: page ka structure ya UI control banata hai.
2	<code><html lang="en"></code>	HTML element/component: page ka structure ya UI control banata hai.
3	<code><head></code>	HTML element/component: page ka structure ya UI control banata hai.
4	<code><meta charset="utf-8" /></code>	HTML element/component: page ka structure ya UI control banata hai.
5	<code><title>CourseManagementSystem</title></code>	HTML element/component: page ka structure ya UI control banata hai.
6	<code><base href="/" /></code>	HTML element/component: page ka structure ya UI control banata hai.
7	<code><meta name="viewport" content="width=device-width, initial-scale=1" /></code>	HTML element/component: page ka structure ya UI control banata hai.
8	<code><link</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
9	<code>rel="stylesheet"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
10	<code>href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/7.3.1/css/all.min.css"</code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
11	<code>/></code>	Is line ka role: is file ke surrounding code ke saath mil kar component ka data, logic, UI ya styling configure karta hai.
12	<code></head></code>	HTML element/component: page ka structure ya UI control banata hai.
13	<code><body></code>	HTML element/component: page ka structure ya UI control banata hai.
14	<code><app-root></app-root></code>	HTML element/component: page ka structure ya UI control banata hai.
15	<code></body></code>	HTML element/component: page ka structure ya UI control banata hai.
16	<code></html></code>	HTML element/component: page ka structure ya UI control banata hai.

src/main.ts

TypeScript: component/service logic, state, forms, events, lifecycle or API connections.

Line	Original code	Simple explanation
1	<code>import { bootstrapApplication } from '@angular/platform-browser';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
2	<code>import { appConfig } from './app/app.config';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
3	<code>import { App } from './app/app';</code>	Import: doosri library/file se required class ya function is file mein available karta hai.
4		Blank line: code ko readable groups mein separate karti hai; program par direct effect nahi.
5	<code>bootstrapApplication(App, appConfig)</code>	Function/method define karta hai; call hone par is block ka kaam hota hai.
6	<code>.catch((err) => console.error(err));</code>	Arrow function: callback/function ko short syntax mein define karta hai.

src/styles.css

CSS/SCSS: visual styling, layout, colors, spacing and responsive behavior.

Line	Original code	Simple explanation
1	<code>/* You can add global styles to this file, and also import other style files */</code>	Comment: developer ko code ka purpose samjhata hai; khud execute nahi hota.

9. API cheat sheet

HTTP method	Endpoint	Purpose	Example
GET	/students	Student list read karna	loadStudents()
POST	/students	Naya student add	saveStudent() add branch
PUT	/students/{id}	Student update	saveStudent() edit branch
DELETE	/students/{id}	Student delete	deleteStudent(id)
GET/POST/PUT/DELETE	/courses	Course CRUD	CoursesService
GET/POST/PUT/DELETE	/enrollments	Enrollment CRUD	Enrollment component
GET/POST/PUT/DELETE	/payments	Payment CRUD	Fees component

10. Viva / interview questions with short answers

Q. What is this project?

An Angular-based Course Management System UI that manages students, courses, enrollments and fee payments using json-server as a development REST API.

Q. Why Angular?

Component-based structure, routing, forms, dependency injection and reactive state make the app organized and maintainable.

Q. What is a component?

A reusable UI unit usually made from a TypeScript class, HTML template and CSS file.

Q. What is standalone: true?

Component can import its own dependencies without being declared in an NgModule.

Q. What is a service?

A reusable class for shared logic/API calls. CoursesService keeps course HTTP requests out of the page component.

Q. What is dependency injection?

Angular supplies a required object/service, for example inject(HttpClient).

Q. What does HttpClient do?

Sends HTTP requests to the backend API.

Q. GET vs POST vs PUT vs DELETE?

GET reads; POST creates; PUT updates/replaces; DELETE removes.

Q. What is a signal?

Angular reactive state container; read with signalName(), update with set/update.

Q. Why subscribe?

HTTP requests are asynchronous; subscribe handles the response when it arrives.

Q. What is ReactiveFormsModule?

Angular module that enables reactive forms in the component template.

Q. What does Validators.required do?

Marks a control invalid if it has no acceptable value.

Q. What is form.patchValue?

Updates selected form fields without needing to provide every control.

Q. What is routing?

Maps a URL such as /students to the Student component.

Q. What is json-server?

A quick mock REST API that reads/writes a JSON file; useful for learning, not a production database.

Q. What is localStorage used for here?

Browser-local profile/theme-related values and exported UI settings; it is not server-side storage.

Q. How is enrollment related to student/course?

Enrollment stores student_id and course_id to connect one student with one course in a specific enrollment record.

Q. How is remaining fee calculated?

Total fee minus previously paid minus the payment amount currently being entered.

Q. What happens if validation is omitted?

Empty/incorrect data may be sent to the API and create inconsistent records.

Q. Does password change really update a password?

No. Current code validates fields and shows a toast, but does not call a secure authentication backend.

Q. Does logout securely end a server session?

No server session handling is visible; current logout clears localStorage and navigates to root.

Q. Are monthly chart figures real?

No. salesData is hard-coded demonstration data.

Q. What is a known architecture improvement?

Use Students service inside Student component too, strongly type the data models, and centralize API base URL.

Q. What security changes are needed for production?

Real authentication/authorization, server-side validation, secure password hashing, database constraints, environment configuration and protected API endpoints.

11. Important improvements / limitations

- Use TypeScript interfaces instead of any[] / any, for example Student, Course, Enrollment and Payment models.
- Move Student component API operations to Students service, so the project uses a consistent service layer.
- Keep API base URL in environment configuration instead of repeating http://localhost:3000.
- Add Validators.email for student email; add positive-number validators for course_fee and paid_amount; prevent overpayment if business rules require it.
- Compute payment status using the sum of all payments for an enrollment, not only one payment row.
- Handle HTTP errors for all create/update/delete requests and show a clear error toast.
- Use string | number for IDs where json-server records may have alphanumeric IDs; compare normalized strings.
- Settings reset should wait for all requests to finish and confirm the backend operation; collection-wide PUT may not behave like deleting every record in every json-server version.
- Password change must be connected to a real authentication backend. Never store plain-text passwords in localStorage.
- Logout should invalidate server-side/session tokens if authentication is added.
- Admin profile and photo are stored only in browser localStorage; other browsers/devices will not see them.
- Add tests for form validation, CRUD success/failure, fee calculations and routing.

12. Short project explanation to memorize

"Mera project Course Management System hai. Iska frontend Angular mein bana hai. Application main.ts se start hoti hai. app.config.ts Router aur HttpClient provide karta hai. app.routes.ts URL ko relevant standalone page component se connect karta hai. Har page ki TypeScript file state, forms aur business logic handle karti hai; HTML template data bindings aur user interactions show karti hai; CSS UI ko style karti hai. Students, courses, enrollments aur payments json-server ke REST endpoints par db.json mein save hotay hain. CRUD ke liye GET, POST, PUT aur DELETE requests use hoti hain. Signals UI state ko reactive rakhte hain aur Reactive Forms required-field validation karte hain. Courses ke liye separate CoursesService use hoti hai. ThemeService shared dark-mode state manage karta hai. Kuch features, jaise password change aur monthly chart, current version mein demo/UI-level hain, production security nahi."

13. Run the project locally

1) Node.js install ho. 2) Project folder mein terminal kholen. 3) npm install run karein. 4) npm start ya package.json mein defined Angular script se frontend start karein. 5) Doosre terminal mein json-server ko db.json par port 3000 par run karein, project ke package scripts ko follow karte hue. 6) Browser mein Angular dev-server URL open karein. Agar API data nahi aata, check

karein ke json-server port 3000 par chal raha hai aur db.json mein required collections hain.

Project scripts

```
{
  "ng": "ng",
  "start": "ng serve",
  "build": "ng build",
  "watch": "ng build --watch --configuration development",
  "test": "ng test"
}
```

Final reminder

Yeh document uploaded source ke mutabiq banaya gaya hai. Code ke exact behavior ko confirm karne ke liye apne local app mein form submissions, browser console aur Network tab test karein. Viva mein feature aur uski limitation dono honestly explain karna strong understanding ki nishani hai.